Project Data Page:

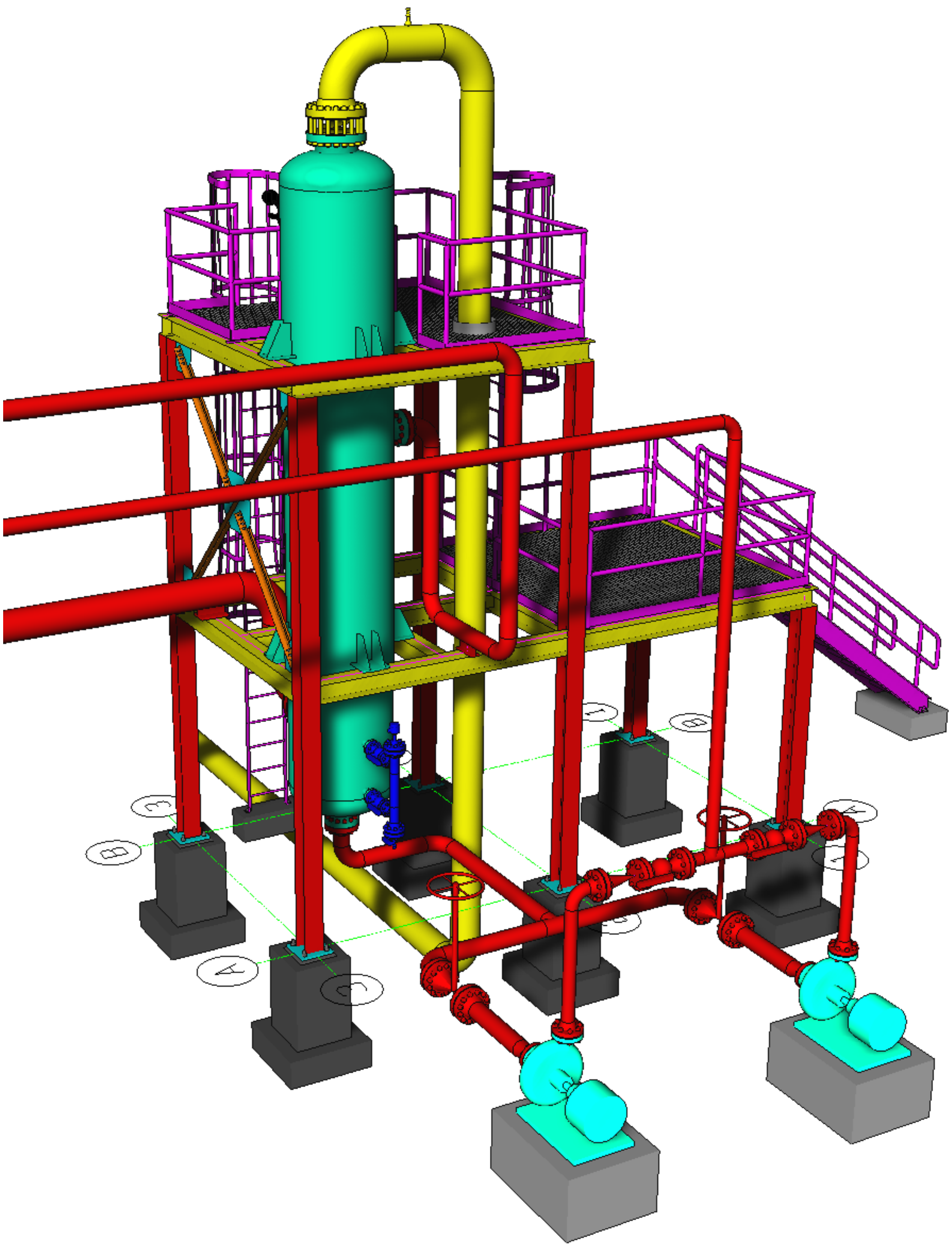

Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	9
Warnings and Errors:	11
Input Echo:	13
XY Coordinate Calculations:	31
Flg Calc [Int P]: FLANGE-1	32
Flg Calc [Int P]: FLANGE-2	40
Flg Calc [Int P]: FLANGE-3	48
Flg Calc [Int P]: FLANGE-4	56
Flg Calc [Int P]: FLANGE-5	64
Flg Calc [Int P]: FLANGE-6	72
Flg Calc [Int P]: FLANGE-7	80
Flg Calc [Int P]: FLANGE-8	88
Internal Pressure Calculations:	96
Element: Bottom head	97
Element: Shell-1	99
Element: Shell-2	101
Element: Shell-3	103
Element: Shell-4	105
Element: Shell-5	107
Element: Shell-6	109
Element: Top head	111
Shop Pressure Test Results:	113
Field Pressure Test Results:	115
External Pressure Calculations:	117
Element and Detail Weights:	119
Nozzle Flange MAWP:	123
Natural Frequency Calculation:	124
Forces/Moments Applied to Vessel	125
Wind Load Calculation:	126
Earthquake Load Calculation:	129
User Force/Moment Shear and Bendi	131
Wind/Earthquake Shear, Bending:	132
Wind/Earthquake Deflection:	133
Longitudinal Stress Constants:	135
Longitudinal Allowable Stresses:	136
Longitudinal Stresses due to:	137
Stress due to Combined Loads:	140
Skirt Stress Calculations:	146
Skirt Stress Calculations:	150
Center of Gravity Calculation:	157
Basering Calculations:	158
Nozzle Calcs.: N4_80	162
Nozzle Calcs.: N1_150	169
Nozzle Calcs.: K1_50	179
Nozzle Calcs.: K2_50	182
Nozzle Calcs.: K3_50	185
Nozzle Calcs.: H2_200	188
Nozzle Calcs.: H1_200	193
Nozzle Calcs.: N2_80	198
Nozzle Calcs.: K4_50	208
Nozzle Calcs.: N3_150	211

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : D:\QEUP\Columns\Tag numbers\124-D\Load data\Rev 01\124-DOpeRev 01.pvdb

Date of Analysis : Feb 10,2026 3:16pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

```
Diameter Spec : 762.002 x 826.002 x 762.002 mm ID
Vessel Design Length, Tangent to Tangent          14686.00  mm

Distance of Bottom Tangent above Grade             3500.00  mm
Distance of Base above Grade                        300.00  mm
Specified Datum Line Distance                         0.00  mm

Internal Design Temperature                             82  °C
Internal Design Pressure                             9.403  MPa

External Design Temperature                             82  °C
External Design Pressure                             0.103  MPa

Maximum Allowable Working Pressure                      10  MPa
External Max. Allowable Working Pressure                 4  MPa
Shop Test Pressure                                  13.776  MPa
Field Test Pressure                                 12.224  MPa

Required Minimum Design Metal Temperature            -28.0  °C
Warmest Computed Minimum Design Metal Temperature      4.0  °C
```

Warning: Computed overall MDMT was higher than the required value !

```
Wind Design Code                             ASCE/SEI 7-16
Earthquake Design Code                       ASCE/SEI 7-16
```

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	+IT
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
Bottom head	9.536	0.10	10.000	3.0000	No	No
Shell-1	9.534	0.10	10.000	3.0000	N/A	No
Flange-1	9.515	0.10	10.000	3.0000	N/A	No
Flange-2	9.513	0.10	10.000	3.0000	N/A	No
Shell-2	9.511	0.10	10.000	3.0000	N/A	No
Shell-3	9.484	0.10	10.000	3.0000	N/A	No
Flange-3	9.474	0.10	10.000	3.0000	N/A	No
Flange-4	9.472	0.10	10.000	3.0000	N/A	No
Shell-4	9.470	0.10	10.000	3.0000	N/A	No
Shell-5	9.443	0.10	10.000	3.0000	N/A	No
Flange-5	9.433	0.10	10.000	3.0000	N/A	No

Flange-6	9.431	0.10	10.000	3.0000	N/A	No
Shell-6	9.429	0.10	10.000	3.0000	N/A	No
Flange-7	9.408	0.10	10.000	3.0000	N/A	No
Flange-8	9.406	0.10	11.000	3.0000	N/A	No
Top head	9.405	0.10	10.000	3.0000	No	No

Liquid Level: 14722.53 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.920

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	2200.0	2200.0	16.0	16.0	...	...	0.70	0.70
Skirt	3200.0	1000.0	16.0	16.0	...	...	0.70	0.50
Ellipse	3250.0	50.0	36.0	30.0	29.4	5.1	1.00	1.00
Cylinder	5250.0	2000.0	32.0	32.0	30.7	6.9	1.00	1.00
Body Flg	5490.5	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Body Flg	5726.5	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Cylinder	8726.5	3000.0	32.0	32.0	30.6	8.2	1.00	1.00
Cylinder	9876.5	1150.0	32.0	32.0	30.5	8.2	1.00	1.00
Body Flg	10117.0	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Body Flg	10353.0	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Cylinder	13353.0	3000.0	32.0	32.0	30.6	8.2	1.00	1.00
Cylinder	14481.0	1128.0	32.0	32.0	30.4	8.2	1.00	1.00
Body Flg	14721.5	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Body Flg	14957.5	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Cylinder	17357.5	2400.0	32.0	32.0	30.4	7.1	1.00	1.00
Body Flg	17600.0	238.0	180.0	178.0	166.2	166.2	1.00	1.00
Body Flg	17836.0	236.0	178.0	176.0	158.0	158.0	1.00	1.00
Ellipse	17886.0	50.0	36.0	29.5	29.1	5.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N4_80	WRC-107/537	0.882	n/a	Passed
N1_150	WRC-297	0.949	Shell	Passed
N2_80	WRC-297	0.935	Shell	Passed
N3_150	WRC-107/537	0.886	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1100	13510	25234.8	124878792	325366912
20	30	2700	12106.2	25150.6	96689520	269920480
30	40	3225	11405.3	25092.7	84928960	244788608
40	50	4250	11366.8	25014.9	84359432	243535440
50	60	5372.5	9819.07	24250.9	63120756	194140512
60	70	5608.5	9652.06	23904.6	60822232	188455856
70	80	7226.5	9485.05	23539.3	58563152	182855200
80	90	9301.5	7163.44	21358.6	33580292	115481064
90	100	9999	6273.49	20320.8	25822684	91414232
100	110	10235	6106.48	19583.4	24361258	86703632
110	120	11853	5939.47	18824.7	22939264	82169656
120	130	13917	3617.86	12673.3	8597463	34903448
130	140	14603.5	2744.93	11009.6	4995039	21491276
140	150	14839.5	2577.92	9843.48	4366689	19029614
150	160	16157.5	2410.91	8654.3	3777769	16845996
160	170	17481	553.624	4141.03	216393	1466727
170	180	17718	385.416	2667.09	104603	656235
180	190	17861	219.912	1270.21	33145.2	191446

Abs Max of the all of the Stress Ratio's : 0.8468

Basering Data : Continuous Top Ring W/Gussets

Thickness of Basering	54.0000	mm
Inside Diameter of Basering	629.0000	mm
Outside Diameter of Basering	1149.0000	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	981.0000	mm
Number of Bolts	16	
Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	158.0000	mm
Height of Gussets	252.0000	mm
Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	168.0000	mm

Basering Sketch for Cylindrical Skirts:

Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	53901.	N
Total Earthquake Shear on Support	65625.	N
Wind Moment on Support	481291456.	N-mm
Earthquake Moment on Support	681779584.	N-mm
Maximum Initial Bolt Load	155566.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated	- Bare w/o removable internals	19902.7	kgm
Shop Test	- Fabricated + water (full)	26722.4	kgm
Shipping	- Fab. + rem. intls.+ shipping app.	21952.4	kgm
Erected	- Fab. + rem. intls.+ insul. (etc)	21952.4	kgm
Empty	- Fab. + intls. + details + wghts.	21952.4	kgm
Operating	- Empty + operating liquid (no ca)	28767.2	kgm
Field Test	- Empty weight + test medium (full)	28015.2	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N4_80	10	OK	...	OK	7.80	OK	Passed
N1_150	10	OK	...	OK	9.22	OK	Passed
K1_50	10	...	...	OK	8.26	OK	No Calc[*]
K2_50	10	...	...	OK	8.26	OK	No Calc[*]
K3_50	10	...	...	OK	8.26	OK	No Calc[*]
H2_200	10	OK	...	...		OK	Passed
H1_200	10	OK	...	...		OK	Passed
N2_80	10	OK	...	OK	7.80	OK	Passed
K4_50	10	...	...	OK	8.26	OK	No Calc[*]
N3_150	10	OK	...	OK	9.22	OK	Passed

Nozzle MAWP Summary:

```
Minimum MAWP Nozzles         :        10 Nozzle  : N3150
Minimum MAWP Shells/Flanges  :        10 Element : Top head
Minimum MAPnc Shells/Flanges :        11 Element : Top head

Computed Vessel M.A.W.P.     :        10 MPa
```

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N4_80	0.000	0.000	145.299
40	N1_150	1225.000	180.000	310.145
40	K1_50	250.000	0.000	162.648
40	K2_50	1250.000	0.000	162.648
40	K3_50	600.000	180.000	162.648
70	H2_200	2870.492	0.000	386.231
110	H1_200	7826.240	0.000	377.100
150	N2_80	12502.207	180.000	150.862
150	K4_50	13264.007	0.000	162.648
180	N3_150	0.000	0.000	310.145

The nozzle spacing is computed by the following:

$= \sqrt{(ll^2 + lc^2)}$ where

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Reinforcing Pad Thk mm	Cut Length mm	Flg Class
K1_50	50 mm	Actual	LWN	104.6	26.924	...	...	222	900
K2_50	50 mm	Actual	LWN	104.6	26.924	...	...	222	900
K3_50	50 mm	Actual	LWN	104.6	26.924	...	...	222	900
K4_50	50 mm	Actual	LWN	104.6	26.924	...	...	222	900
N4_80	80 mm	160	WNF	88.9	11.125	...	...	231	900
N2_80	80 mm	160	WNF	88.9	11.125	...	...	234	900
N1_150	150 mm	80	WNF	168.3	10.973	...	...	241	900
N3_150	150 mm	80	WNF	168.3	10.973	...	...	234	900
H2_200	200 mm	120	WNF	219.1	18.263	...	...	248	900
H1_200	200 mm	120	WNF	219.1	18.263	...	...	248	900

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:
Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1_50	SA-105	30.800	10.000	...	...	...
K2_50	SA-105	30.800	10.000	...	...	...
K3_50	SA-105	30.800	10.000	...	...	...
K4_50	SA-105	32.000	10.000	...	...	...
N4_80	SA-105	30.000	10.000	...	...	...
N2_80	SA-105	32.000	10.000	...	...	...
N1_150	SA-105	30.800	10.000	...	...	...
N3_150	SA-105	29.400	10.000	...	...	...
H2_200	SA-105	32.000	10.000	...	...	...
H1_200	SA-105	32.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1_50	3450.000	0.0	187.32	0.00	Shell-1
K2_50	4450.000	0.0	187.32	0.00	Shell-1
K3_50	3800.000	180.0	187.32	0.00	Shell-1
K4_50	16464.008	0.0	187.32	0.00	Shell-6
N4_80	...	0.0	200.00	0.00	Bottom head
N2_80	15702.207	180.0	200.00	0.00	Shell-6
N1_150	4425.000	180.0	200.00	0.00	Shell-1
N3_150	...	0.0	200.00	0.00	Top head
H2_200	6070.491	0.0	200.00	0.00	Shell-2
H1_200	11026.239	0.0	200.00	0.00	Shell-4

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 16.0mm THK X 781.0mm ID X 2200.0mm	SA-516 70
1	SKIRT: 16.0mm THK X 781.0mm ID X 1000.0mm	SA-516 70
2	ELLIPTICAL HEAD: 2.0 X 1, 36.0mm THK X 762.0mm ID X 50.0mm	SA-516 70
6	WELD NECK FLANGE: 178.0mm THK X 1126.0mm OD	SA-266 2
2	CYLINDER: 32.0mm THK X 762.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 32.0mm THK X 762.0mm ID X 1150.0mm	SA-516 70
1	CYLINDER: 32.0mm THK X 762.0mm ID X 1128.0mm	SA-516 70
1	CYLINDER: 32.0mm THK X 762.0mm ID X 2400.0mm	SA-516 70
1	WELD NECK FLANGE: 180.0mm THK X 1128.0mm OD	SA-266 2
1	WELD NECK FLANGE: 178.0mm THK X 1128.0mm OD	SA-266 2
1	INSULATION: 500mm X 80mm THK	
2	INSULATION: 240mm X 80mm THK	
1	INSULATION: 2000mm X 80mm THK	
7	INSULATION: 206mm X 80mm THK	
1	PACKING: 2900.000mm	...
2	INSULATION: 3000mm X 80mm THK	
1	PACKING: 800.000mm	...
1	INSULATION: 1150mm X 80mm THK	
1	PACKING: 2922.000mm	...
1	PACKING: 778.000mm	...
1	INSULATION: 1128mm X 80mm THK	
1	INSULATION: 2400mm X 80mm THK	
1	INSULATION: 194mm X 80mm THK	
2	CLASS 900 GR 1.1, 200.0mm BLIND FLANGE(S)	SA-105
1	BASERING: 1149.0mm OD X 629.0 ID X 54.0mm	SA-516 70
16	BASERING BOLTS: 48.0mm DIA	SA-193 B7
16	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 1149.0mm OD X 813.0 ID X 44.0mm	...
32	BASERING GUSSET PLATES: 158.0mm X 158.0mm X 252.0mm X 16.0mm	...
4	GASKET: 826.0mm OD X 794.0mm ID	...
96	BODY FLANGE BOLTS: 56.0mm DIA	SA-193 B7
192	NUTS FOR BODY FLANGE BOLTS: 56.0mm DIA	...
1	CYLINDRICAL SEGMENT 32.0mm THK X 997.4mm X 2000.0mm	SA-516 70

1	CYLINDRICAL SEGMENT 32.0mm THK X 704.9mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm THK X 691.6mm X 2000.0mm	SA-516 70
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Flange-1	[11]	B	6	0		0.903	32.000	1.00	No
Flange-2	[11]	B	6	0		0.898	32.000	1.00	No
Flange-3	[11]	B	6	-1		0.872	32.000	1.00	No
Flange-4	[11]	B	6	-1		0.872	32.000	1.00	No
Flange-5	[11]	B	6	-2		0.869	32.000	1.00	No
Flange-6	[11]	B	6	-2		0.868	32.000	1.00	No
Flange-7	[11]	B	6	-2		0.866	32.000	1.00	No
Flange-8	[11]	B	-8	-20		0.779	36.000	1.00	Yes
Bottom head	[10]	B	4	4		0.995	30.000	1.00	No
Bottom head	[7]	B	9	1		0.853	36.000	1.00	No
Shell-1	[8]	B	6	4		0.970	32.000	1.00	No
Shell-2	[8]	B	6	4		0.968	32.000	1.00	No
Shell-3	[8]	B	6	4		0.965	32.000	1.00	No
Shell-4	[8]	B	6	4		0.966	32.000	1.00	No
Shell-5	[8]	B	6	4		0.961	32.000	1.00	No
Shell-6	[8]	B	6	3		0.959	32.000	1.00	No
Top head	[10]	B	4	4		1.000	29.500	1.00	No
Top head	[7]	B	9	0		0.841	36.000	1.00	No
N480	[1]	B	-8	-48	-29	0.444	9.735	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N1150	[1]	B	6	3		0.953	32.000	1.00	No
Nozzle Flg	[5]	!	-18	-39					
K150	[1]	B	1	-104		0.085	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-39					
K250	[1]	B	1	-104		0.085	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-39					
K350	[1]	B	1	-104		0.085	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-39					
H2200	[1]	B	6	3		0.952	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
H1200	[1]	B	6	3		0.947	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N280	[1]	B	6	3		0.943	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
K450	[1]	B	1	-104		0.084	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N3150	[1]	B	4	-3		0.888	29.500	1.00	No
Nozzle Flg	[5]	A	-18	-39					
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						
Warmest MDMT:			9	4					

Required Minimum Design Metal Temperature -28.0 °C
Warmest Computed Minimum Design Metal Temperature 4.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:
[!] - This was an impact tested material.
[1] - Governing Nozzle Weld.
[4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
[5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
[6] - MDMT Calculations at the Shell/Head Joint.
[7] - MDMT Calculations for the Straight Flange.
[8] - Cylinder/Cone/Flange Junction MDMT.
[9] - Calculations in the Spherical Portion of the Head.
[10] - Calculations in the Knuckle Portion of the Head.
[11] - Calculated (Body Flange) Flange MDMT.
[12] - Calculated Flat Head MDMT per UCS-66.3
[13] - Tubesheet MDMT, shell side, if applicable

[14] - Tubesheet MDMT, tube side, if applicable
[15] - Nozzle Material
[16] - Shell or Head Material
[17] - Impact Testing required
[18] - Impact Testing not required, see UCS-66(b)(3)
[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
[21] - Body Flange Bolting Material
[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:
Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.
Note	40	50	The wind diameter multiplier is very big.
Note	50	60	The wind diameter multiplier is very big.
Note	60	70	The wind diameter multiplier is very big.
Note	70	80	The wind diameter multiplier is very big.
Note	80	90	The wind diameter multiplier is very big.
Note	90	100	The wind diameter multiplier is very big.
Note	100	110	The wind diameter multiplier is very big.
Note	110	120	The wind diameter multiplier is very big.
Note	120	130	The wind diameter multiplier is very big.
Note	130	140	The wind diameter multiplier is very big.
Note	140	150	The wind diameter multiplier is very big.
Note	150	160	The wind diameter multiplier is very big.
Note	160	170	The wind diameter multiplier is very big.
Note	170	180	The wind diameter multiplier is very big.
Note	180	190	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.

One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.

The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The *static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.*

Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm²
Inertia	in.4	416231.0000	mm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	9.4029	MPa
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-28.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW

```
Load Case 16                                              IP+VE+EW
Load Case 17                                              NP+VO+OW
Load Case 18                                           FS+BS+IP+OW
Load Case 19                                           FS+BS+EP+OW
Load Case 20                                              BL+IP+OW

Wind Design Code                                     ASCE/SEI 7-16
Wind Load Reduction Scale Factor                             1.100
Basic Wind Speed                       [V]                    97.2  km/hr
Surface Roughness Category                         C: Open Terrain
Type of Surface                                  Moderately Smooth
Base Elevation                                                 300  mm
Percent Wind for Hydrotest                                    50.0
Using User defined Wind Press. Vs Elev.                          N
Height of Hill or Escarpment      H or Hh                        0  mm
Distance Upwind of Crest               Lh                        0  mm
Distance from Crest to the Vessel       x                        0  mm
Type of Terrain ( Hill, Escarpment )                          Flat
Damping Factor (Beta) for Wind (Ope)                        0.0100
Damping Factor (Beta) for Wind (Empty)                      0.0000
Damping Factor (Beta) for Wind (Filled)                     0.0000
Site Elevation above Sea Level                                   0  mm

Seismic Design Code                                  ASCE/SEI 7-16
Seismic Load Reduction Scale Factor                          1.000
Importance Factor                                            2.000
Table Value Fa                                               1.300
Table Value Fv                                               1.500
Max. Mapped Res. Acceleration         [Ss]                   0.147
Max. Eff. Ground Acceleration          [S]                   0.065
Force Modification Factor R                                  2.000
Site Class                                                       C
Component Elevation Ratio              z/h                   0.000
Amplification Factor                    Ap                   2.500
Force Factor                                                 0.000
Consider Vertical Acceleration                                 Yes
Minimum Acceleration Multiplier                              0.000
User Value of Sds (used if > 0 )                             0.000
User Value of Sd1 (used if > 0 )                             0.000
Moment Reduction Factor Tau                                  1.000

Design Pressure + Static Head                                    Y
Consider MAP New and Cold in Noz. Design                         N
Consider External Loads for Nozzle Des.                          Y
Use ASME VIII-1 Appendix 1-9                                     N

Perform Blast Load Analysis                                     No
Material Database Year               Current w/Addenda or Code Year
```

Configuration Directives:

```
g0 is taken as min( g0, attached shell thk )               Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37           No
Use Table G instead of exact equation for "A"               Yes
Shell Head Joints are Tapered                               Yes
Compute "K" in corroded condition                           Yes
Use Code Case 2286                                           No
Use the MAWP to compute the MDMT                            Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)   Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)        No

Using Metric Material Databases, ASME II D                  Yes
Calculate B31.3 type stress for Nozzles with Loads          Yes
Reduce the MDMT due to lower membrane stress                Yes
Consider Longitudinal Stress in MDMT Calculations           Yes
```

Complete Listing of Vessel Elements and Details:

```
Element#                                                      1/18
Element From Node                                               10
```

Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	2200	mm
Skirt Outside Diameter	813	mm
Diameter of Skirt at Base	813	mm
Skirt Thickness	16	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing lug	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element#	2/18	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	813	mm
Diameter of Skirt at Base	813	mm
Skirt Thickness	16	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Insulation	
Detail ID	FP	
Dist. from "FROM" Node / Offset dist	500	mm
Height/Length of Insulation	500	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	3/18	
Element From Node	30	
Element To Node	40	

Element Type	Elliptical	
Description	Bottom head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	762	mm
Element Thickness	30	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	36	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	-190.5	mm
Height/Length of Liquid	240.5	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	-190.5	mm
Height/Length of Insulation	240.5	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N480	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	80	mm
Nozzle Schedule	160	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	30	
Detail Type	For./Mom.	
Detail ID	Guide sup later	
Dist. from "FROM" Node / Offset dist	-190	mm
Force in X Direction	-20791	N
Force in Y Direction	0	N
Force in Z Direction	14291	N
Moment about X Axis	0	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	0	N-mm
Force/Moment Combination Method	SRSS	

Element#	4/18	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	2000	mm
Inside Diameter	762	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm

```
Nominal Thickness                                        32  mm
External Corrosion Allowance                              0  mm
Design Internal Pressure                             9.4029  MPa
Design Temperature Internal Pressure                     82  °C
Design External Pressure                             0.1034  MPa
Design Temperature External Pressure                     82  °C
Effective Diameter Multiplier                          1.64
Material Name                                     SA-516 70
Efficiency, Longitudinal Seam                           1.0
Efficiency, Circumferential Seam                        1.0
Weld is pre-Heated                                       No

    Element From Node                                    40
    Detail Type                                      Liquid
    Detail ID                                     Liquid 40
    Dist. from "FROM" Node / Offset dist                  0  mm
    Height/Length of Liquid                            2000  mm
    Liquid Density                                0.0009185  kg/cm³

    Element From Node                                    40
    Detail Type                                  Insulation
    Detail ID                                       Ins: 40
    Dist. from "FROM" Node / Offset dist                  0  mm
    Height/Length of Insulation                        2000  mm
    Thickness of Insulation                              80  mm
    Density                                        0.000352  kg/cm³

    Element From Node                                    40
    Detail Type                                      Nozzle
    Detail ID                                         N1150
    Dist. from "FROM" Node / Offset dist               1175  mm
    Nozzle Diameter                                     150  mm
    Nozzle Schedule                                      80
    Nozzle Class                                        900
    Layout Angle                                      180.0
    Blind Flange (Y/N)                                    N
    Weight of Nozzle ( Used if > 0 )                      0  N
    Grade of Attached Flange                         GR 1.1
    Nozzle Matl                                      SA-105 +IT

    Element From Node                                    40
    Detail Type                                      Nozzle
    Detail ID                                          K150
    Dist. from "FROM" Node / Offset dist                200  mm
    Nozzle Diameter                                    50.8  mm
    Nozzle Schedule                                    None
    Nozzle Class                                        900
    Layout Angle                                        0.0
    Blind Flange (Y/N)                                    N
    Weight of Nozzle ( Used if > 0 )                 231.38  N
    Grade of Attached Flange                         GR 1.1
    Nozzle Matl                                      SA-105 +IT

    Element From Node                                    40
    Detail Type                                      Nozzle
    Detail ID                                          K250
    Dist. from "FROM" Node / Offset dist               1200  mm
    Nozzle Diameter                                    50.8  mm
    Nozzle Schedule                                    None
    Nozzle Class                                        900
    Layout Angle                                        0.0
    Blind Flange (Y/N)                                    N
    Weight of Nozzle ( Used if > 0 )                 231.38  N
    Grade of Attached Flange                         GR 1.1
    Nozzle Matl                                      SA-105 +IT

    Element From Node                                    40
    Detail Type                                      Nozzle
    Detail ID                                          K350
    Dist. from "FROM" Node / Offset dist                550  mm
```

Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	231.38	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	40	
Detail Type	Weight	
Detail ID	N1FY	
Dist. from "FROM" Node / Offset dist	1175	mm
Miscellaneous Weight	2539.8	N
Offset from Element Centerline	1649	mm

Element#	5/18	
Element From Node	50	
Element To Node	60	
Element Type	Flange	
Description	Flange-1	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	220	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	6/18	
Element From Node	60	
Element To Node	70	
Element Type	Flange	
Description	Flange-2	

Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	222	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	7/18	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	762	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Packing	
Detail ID	Bed 2	
Dist. from "FROM" Node / Offset dist	100	mm
Height of Packed Section	2900	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	

Specific Gravity of Packing Liquid	0.92000002	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.0009185	kg/cm³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	H2200	
Dist. from "FROM" Node / Offset dist	343.99	mm
Nozzle Diameter	200	mm
Nozzle Schedule	120	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	70	
Detail Type	Weight	
Detail ID	H2Davit	
Dist. from "FROM" Node / Offset dist	343.99	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	350	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	Bed support 2	
Dist. from "FROM" Node / Offset dist	50	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	Bed sup grid 2	
Dist. from "FROM" Node / Offset dist	80	mm
Miscellaneous Weight	343.21	N
Offset from Element Centerline	0	mm

Element#	8/18	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	1150	mm
Inside Diameter	762	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	

Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Packing	
Detail ID	Bed2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	800	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.92000002	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1150	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1150	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	80	
Detail Type	Weight	
Detail ID	Bed limiter sup	
Dist. from "FROM" Node / Offset dist	820	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Bed limiter	
Dist. from "FROM" Node / Offset dist	850	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element#	9/18	
Element From Node	90	
Element To Node	100	
Element Type	Flange	
Description	Flange-3	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	

Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	216	mm
Liquid Density	0.0009185	kg/cm³
Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	10/18	
Element From Node	100	
Element To Node	110	
Element Type	Flange	
Description	Flange-4	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	100	
Detail Type	Liquid	
Detail ID	Liquid 100	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	216	mm
Liquid Density	0.0009185	kg/cm³
Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 100	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	11/18	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell-4	
Distance "FROM" to "TO"	3000	mm
Element Outside Diameter	826	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa

Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm^3
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Packing	
Detail ID	Bed-1	
Dist. from "FROM" Node / Offset dist	78	mm
Height of Packed Section	2922	mm
Density	0.000224	kg/cm^3
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.92000002	

Element From Node	110	
Detail Type	Liquid	
Detail ID	Liquid 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.0009185	kg/cm^3

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm^3

Element From Node	110	
Detail Type	Nozzle	
Detail ID	H1200	
Dist. from "FROM" Node / Offset dist	673.24	mm
Nozzle Diameter	200	mm
Nozzle Schedule	120	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	110	
Detail Type	Weight	
Detail ID	H1Davit	
Dist. from "FROM" Node / Offset dist	673.24	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	350	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Clip weight	
Dist. from "FROM" Node / Offset dist	665	mm
Miscellaneous Weight	3922.4	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Bed support	
Dist. from "FROM" Node / Offset dist	49.999	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Bed sup grid	
Dist. from "FROM" Node / Offset dist	59.999	mm
Miscellaneous Weight	343.21	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Misc wt 10%	
Dist. from "FROM" Node / Offset dist	606.96	mm
Miscellaneous Weight	17160	N
Offset from Element Centerline	0	mm

Element#	12/18	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	1128	mm
Inside Diameter	762	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	120	
Detail Type	Packing	
Detail ID	Bed1	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	778	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.92000002	

Element From Node	120	
Detail Type	Liquid	
Detail ID	Liquid 120	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1128	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 120	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1128	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	120	
Detail Type	Weight	

Detail ID	Bed limiter sup	
Dist. from "FROM" Node / Offset dist	800	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	120	
Detail Type	Weight	
Detail ID	Bed limiter 1	
Dist. from "FROM" Node / Offset dist	820	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element#	13/18	
Element From Node	130	
Element To Node	140	
Element Type	Flange	
Description	Flange-5	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm^3
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	130	
Detail Type	Liquid	
Detail ID	Liquid 130	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	208	mm
Liquid Density	0.0009185	kg/cm^3

Element From Node	130	
Detail Type	Insulation	
Detail ID	Ins: 130	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm^3

Element#	14/18	
Element From Node	140	
Element To Node	150	
Element Type	Flange	
Description	Flange-6	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm

Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	140	
Detail Type	Liquid	
Detail ID	Liquid 140	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	208	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	140	
Detail Type	Insulation	
Detail ID	Ins: 140	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	15/18	
Element From Node	150	
Element To Node	160	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	2400	mm
Inside Diameter	762	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	150	
Detail Type	Liquid	
Detail ID	Liquid 150	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2400	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	150	
Detail Type	Insulation	
Detail ID	Ins: 150	
Dist. from "FROM" Node / Offset dist	0	mm

Height/Length of Insulation	2400	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	150	
Detail Type	Nozzle	
Detail ID	N280	
Dist. from "FROM" Node / Offset dist	744.71	mm
Nozzle Diameter	80	mm
Nozzle Schedule	160	
Nozzle Class	900	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	150	
Detail Type	Nozzle	
Detail ID	K450	
Dist. from "FROM" Node / Offset dist	1506.5	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	231.98	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	150	
Detail Type	Weight	
Detail ID	N2Pipe	
Dist. from "FROM" Node / Offset dist	744.7	mm
Miscellaneous Weight	49.03	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Demister sup	
Dist. from "FROM" Node / Offset dist	2250	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	196.12	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Lifting lug1	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Lifting lug 2	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	0	mm

Element#	16/18
Element From Node	160
Element To Node	170

Element Type	Flange	
Description	Flange-7	
Distance "FROM" to "TO"	238	mm
Flange Inside Diameter	762	mm
Element Thickness	178	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	180	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	160	
Detail Type	Liquid	
Detail ID	Liquid 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	208	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	160	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	17/18	
Element From Node	170	
Element To Node	180	
Element Type	Flange	
Description	Flange-8	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	170	
Detail Type	Liquid	
Detail ID	Liquid 170	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	65.529	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	170	
Detail Type	Insulation	
Detail ID	Ins: 170	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	194	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	18/18	
Element From Node	180	
Element To Node	190	
Element Type	Elliptical	
Description	Top head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	762	mm
Element Thickness	29.5	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	36	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	180	
Detail Type	Liquid	
Detail ID	Liquid 180	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	240.5	mm
Liquid Density	0.0009185	kg/cm³

Element From Node	180	
Detail Type	Insulation	
Detail ID	Ins: 180	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	240.5	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	180	
Detail Type	Nozzle	
Detail ID	N3150	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	150	mm
Nozzle Schedule	80	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	180	
Detail Type	Weight	
Detail ID	N3Wt	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	3137.9	N
Offset from Element Centerline	1649	mm

Element From Node	180	
Detail Type	For./Mom.	
Detail ID	N3F/M	
Dist. from "FROM" Node / Offset dist	170	mm
Force in X Direction	10721	N
Force in Y Direction	-14845	N
Force in Z Direction	10721	N
Moment about X Axis	4950120	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	4950120	N-mm
Force/Moment Combination Method	SRSS	

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	Skirt-2	...	2200	...	2200
	Skirt-1	...	3200	...	1000
	Bottom head	...	3250	...	50
	Shell-1	...	5250	...	2000
	Flange-1	...	5490.5	...	236
	Flange-2	...	5726.5	...	236
	Shell-2	...	8726.5	...	3000
	Shell-3	...	9876.5	...	1150
	Flange-3	...	10117	...	236
	Flange-4	...	10353	...	236
	Shell-4	...	13353	...	3000
	Shell-5	...	14481	...	1128
	Flange-5	...	14721.5	...	236
	Flange-6	...	14957.5	...	236
	Shell-6	...	17357.5	...	2400
	Flange-7	...	17600	...	238
	Flange-8	...	17836	...	236
	Top head	...	17886	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

Flange Input Data Values **Description: FLANGE-1 :**

Flange-1

```
Flange Type                                 Integral Weld Neck

Design Pressure                           P              9.52  MPa
Design Temperature                                         82  °C
Internal Corrosion Allowance             ci            3.0000  mm
External Corrosion Allowance             ce            0.0000  mm
Use Corrosion Allowance in Thickness Calcs.                No

Flange Inside Diameter                    B           762.002  mm
Flange Outside Diameter                   A          1126.000  mm
Flange Thickness                          t          176.0000  mm
Thickness of Hub at Small End            go           32.0000  mm
Thickness of Hub at Large End            g1           48.0000  mm
Length of Hub                             h           60.0000  mm

Flange Material                                      SA-266 2
Flange Material UNS number                             K03506
Flange Allowable Stress At Temperature  Sfo            138.00  MPa
Flange Allowable Stress At Ambient      Sfa            138.00  MPa

Bolt Material                                       SA-193 B7
Bolt Allowable Stress At Temperature     Sb            172.00  MPa
Bolt Allowable Stress At Ambient         Sa            172.00  MPa

Diameter of Bolt Circle                   C          1004.000  mm
Nominal Bolt Diameter                     a           56.0000  mm
Type of Threads                                   TEMA Metric
Number of Bolts                                            24

Flange Face Outside Diameter            Fod           936.000  mm
Flange Face Inside Diameter             Fid           762.002  mm
Flange Facing Sketch                                Sketch 1a

Gasket Outside Diameter                  Go           826.000  mm
Gasket Inside Diameter                   Gi           794.000  mm
Gasket Factor                             m            3.0000
Gasket Design Seating Stress              y             68.95  MPa

Gasket Thickness                         tg            4.5000  mm
Raised face depth                        Fd            0.0000  mm

Axial Force on Flange                               115933.00  N
Bending Moment on Flange                         175200992.00  N-mm
```

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
= (P•(D/2+Ca))/(S•E-0.6•P) per UG-27 (c)(1)
= (9.52•(762.002/2+3.0))/(138.0•1.0-0.6•9.52)+Ca
= 30.6211 mm

Small End Hub MAWP:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 29.0)/(384.001 + 0.6 • 29.0)
= 9.970 MPa

Hub Small End Hub MAPnc:
= (Sa•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 32.0)/(381.001 + 0.6 • 32.0)
= 11.034 MPa

```
Corroded Flange ID,      Bcor = B+2*ci                768.002 mm
Corroded Large Hub,      g1Cor = g1-ci                 45.000 mm
Corroded Small Hub,      g0Cor = go-ci                 29.000 mm
Code R Dimension,            R = ((C-Bcor)/2)-g1cor    72.999 mm
```

Gasket Contact Width,	N = (Go - Gi)/2	16.000	mm
Basic Gasket Width,	bo = N/2	8.000	mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127	mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
= π/4 • G² • P
= π/4 • 811.7452² • 9.515
= 4924049.500 N

Contact Load on Gasket Surfaces [Hp or Wo,2]:
= 2 • b • π • G • m • P
= 2 • 7.1274 • π • 811.7452 • 3.0 • 9.52
= 1037636.688 N

Hydrostatic End Load at Flange ID [Hd]:
= π/4 • Bcor² • P
= π/4 • 768.002² • 9.5155
= 4407656.500 N

Pressure Force on Flange Face [Ht]:
= H - Hd
= 4924050 - 4407657
= 516393.062 N

Design Bolt Load for the Operating Condition [Wo]:
= max(Wo,1 + Wo,2 + H'p, 0)
= max(4924050 + 1037637 + 0, 0)
= 5961686.000 N

Design Bolt Load for the Gasket Seating Condition [Wg]:
= y • b • π • G + yPart • bPart • lp
= 68.95•7.1274•π•811.745+0.0•0.0•0.0
= 1253136.250 N

Required Bolt Area [Am]:
= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
= max((5961686 + 115933 + 4•.17520E+09/811.75)/172, 1253136/172)
= 40356.480 mm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
= 2a + 6t/(m + 0.5)
= 2 • 56.0 + 6 • 176.0/(3.0 + 0.5)
= 413.714 mm

Actual Circumferential Bolt Spacing [Bs]:
= C • sin(π / n)
= 1004.0 • sin(π/24)
= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
= max($\sqrt{\text{(Bs/(2a + t))}}$, 1)
= max($\sqrt{\text{(131.048/(2 • 56.0 + 176.0))}}$, 1)
= 1.0000

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	40356.480	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
= Ab • Sa/(y • π(Go + Gi))

= 44705.398 • 172.0/(68.95 • π(826.0 + 794.0))
= 21.912 mm[Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

```
Sheet Gaskets including laminated sheet gaskets with or without a metal core:     16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm
```

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
= Sa • Ab
= 172.0 • 44705.3984
= 7688536.50 N

Gasket Load for the Operating Condition [HG]:
= Wo - H
= 5961686 - 4924050
= 1037636.69 N

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
= (C - G) / 2
= (1004.0 - 811.7452)/2
= 96.1274 mm

Distance to Face Pressure Reaction [ht]:
= 0.5((C - B)/2 + hg)
= 0.5((1004.0 - 768.002)/2 + 96.1274)
= 107.0632 mm

Distance to End Pressure Reaction [hd]:
= (C - B - g1)/2
= (1004.0 - 768.002 - 45.0)/2
= 95.4990 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

```
Aa = 179.0, Bb = 176.0, Cc = 60.0 and Ddg = 37.0 mm
```

Compute the value of [Kab]:
= Aa • Bb³ • (1/3 - 0.210 • (Bb/Aa) • (1-1/12 • (Bb/Aa)⁴))
= 179.0 • 176.0³ • (1/3 - 0.210 • (176.0/179.0) • (1-1/12 • (176.0/179.0)⁴))
= 139483776.000 mm⁴

Compute the value of [Kcd]:
= Cc • Ddg³ • (1/3 - 0.105 • (Ddg/Cc) • (1-1/192 • (Ddg/Cc)⁴))
= 60.0•37.0³•(1/3-0.105•(37.0/60.0)•(1-1/192•(37.0/60.0)⁴))
= 816420.500 mm⁴

Flange Inertia Term [Ip,term1]:
= Kab + Kcd = 8511.831 + 49.821
= 140300192.000 mm⁴

Flange Inertia Term [Ip,term2]:
= Ar • t³(1/3 - 0.21(t/AR)(1 - 1/12(t/AR)⁴))
= 179.0 • 176.0³(1/3 - 0.21(176.0/179.0)(1 - 1/12(176.0/7.047)⁴))
= 139483776.000 mm⁴

Flange Inertia [Ip]:
= max(Ip,term1, Ip,term2)
= max(0.14030E+09, 0.13948E+09)
= 140300192.0 mm⁴

Flange Moment of Inertia for Integral Types [I]:
= 0.0874 • L • go² • ho • B / V
= 0.0874 • 3.631 • 29.0² • 149.238 • 768.002/0.327
= 93579992.000 mm⁴

Summary of Moments for Internal Pressure: (N-mm)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	4407656.	95.4990	1.0000	421097440.
Face Pressure,	Mt	516393.	107.0632	1.0000	55309116.
Gasket Load,	Mg	1037637.	96.1274	1.0000	99785800.
Gasket Seating,	Matm	7688536.	96.1274	1.0000	739379008.
Moment Load,	Moe	...	...	...	63318768.

Moment Value, eq. 4.16.16, [Moe]:
= 4Me[I/(0.384 • IP + I))•(hD /(C-2•hD)) + FA • hD
= 4•.17520E+09[93579992/(0.384•.14030E+09+93579992))•(95.5/(1004.0-2•95.5))+
115933•95.499
= 63318768.000 N-mm

```
Total Moment for Operation,      Mo                    639511104. N-mm
Total Moment for Gasket seating, Mg                    739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)      149.238 mm
Hub Ratio,           h/h0 = HL / H0                0.402
Thickness Ratio,    g1/g0 = (g1Cor/goCor)          1.552
```

Flange Factors for Integral Flange:

```
Factor F per 4.16.5                                0.860
Factor V per 4.16.5                                0.327
Factor f per 4.16.5                                1.000
```

Stress factor from Table 4.16.4 [K]:
= A / B
= 1126.0/768.002
= 1.466

Stress factor from Table 4.16.4 [Y]:
= (0.66845 + (5.7169(K^2 • log10(K))) / (K^2 - 1))/(K - 1)
= (0.66845 + (5.7169(1.466^2 • log10(1.466)))/(1.466^2 - 1))/(1.466 - 1)
= 5.245

Stress factor from Table 4.16.4 [T]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K^2) • (K - 1))
= (1.466^2(1 + 8.55246 • log10(1.466)) - 1)/((1.0472 + 1.9448 • 1.466^2) • (1.466 - 1))
= 1.725

Stress factor from Table 4.16.4 [U]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/(1.36136(K^2 - 1) • (K - 1))
= (1.466^2(1 + 8.55246 • log10(1.466)) - 1)/(1.36136(1.466^2 - 1) • (1.466 - 1))
= 5.764

Stress factor from Table 4.16.4 [Z]:
= (K^2 + 1)/(K^2 - 1)
= (1.466^2 + 1)/(1.466^2 - 1)
= 2.740

Stress factor from Table 4.16.4 for integral flanges [d]:
= U • $g0^2$ • h0 / V
= 5.764 • 29.0^2 • 149.238/0.327
= 2213042.750 mm^3

Stress factor from Table 4.16.4 for integral flanges [e]:
= F / h0
= 0.86/149.238
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:
= (t • e + 1) / T + t^3/d
= (176.0 • 0.006 + 1)/1.725 + 176.0^3/2213043
= 3.631

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.0•.63951E+09/768.002)/(3.6309•45.0²)
= 113.22 MPa

Longitudinal Hub Stress, Seating [SHa]:
= (f • Matm / Bcor) / (L • g1²)
= (1.0•0.73938E+09/768.002)/(3.6309•45.0²)
= 130.90 MPa

Radial Flange Stress, Operating [SRo]:
= (β • Mop / Bcor) / (L • t²)
= (2.3524•.63951E+09/768.002)/(3.6309•176.0²)
= 17.41 MPa

Radial Flange Stress, Seating [SRa]:
= (β • Matm/Bcor) / (L • t²)
= (2.3524•.73938E+09/768.002)/(3.6309•176.0²)
= 20.13 MPa

Tangential Flange Stress, Operating [STo]:
= (Y • Mo / (t² • Bcor)) - Z • SRo
= (5.2449•.63951E+09/(176.0²•768.002))-2.7398•17
= 93.25 MPa

Tangential Flange Stress, Seating [STa]:
= (y • Matm / (t² • Bcor)) - Z • SRa
= (5.2449•.73938E+09/(176.0²•768.002))-2.7398•20
= 107.81 MPa

Average Flange Stress, Operating [SAo]:
= (SHo + max(SRo, STo)) / 2
= (113+max(17,93))/2
= 103.23 MPa

Average Flange Stress, Seating [SAa]:
= (SHa + max(SRa, STa)) / 2
= (131+max(20,108))/2
= 119.35 MPa

Bolt Stress, Operating [BSo]:
= (Wm1 + Fa + 4Me/G) / Ab
= 6940599/44705.3984
= 155.27 MPa

Bolt Stress, Seating [BSa]:
= (Wm2 / Ab)
= (1253136/44705.3984)
= 28.03 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	113.22	207.00	130.90	207.00
Radial Flange	17.41	138.00	20.13	138.00
Tangential Flange	93.25	138.00	107.81	138.00
Maximum Average	103.23	138.00	119.35	138.00
Bolting	155.27	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 10 MPa
Estimated Finished Weight of Flange at given Thk. 783.1 kgm
Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

	Operating Actual	Operating Allowed	stress ratio	Gasket Seating Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio			1.000			1.000

Coefficients for f and f2

	f	*f2*	*f3*
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c\left(\frac{g1}{g0}\right) + e\left(\frac{h}{h0}\right) + g\left(\frac{g1}{g0}\right)^2 + i\left(\frac{h}{h0}\right)^2 + k\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}{\left[1 + b\left(\frac{g1}{g0}\right) + d\left(\frac{h}{h0}\right) + f\left(\frac{g1}{g0}\right)^2 + h\left(\frac{h}{h0}\right)^2 + j\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}$$

Hub Stress Correction Factor [f]:
= max(1; F1num / F1den)
= max(1; 1.3704/1.4405)
= 1.0000

Hub Stress Correction Factor [f2]:
= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
= max(0.25; 0.5259/1.4506) if h/ho < 0.35, else max (0.25; 0.2734/0.7571)
= 0.3611

In this case, the hub ratio h/ho, 0.4020 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress Sbmax 507.5 MPa

Minimum permissible bolt stress	Sbmin	290.0	MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4	MPa
Target assembly gasket stress	SgT	172.0	MPa
Maximum permissible gasket stress	Sgmax	380.0	MPa
Minimum gasket operating stress	Sgmin-O	97.0	MPa
Minimum gasket seating stress	Sgmin-S	140.0	MPa
Nut Factor	K	0.200	
Fraction of Gasket Load remaining after relaxation	Phig	0.700	
Root Area of a Single Bolt	Abs	1862.7250	mm²
Max. permissible single flange rotation for gasket	Thetagmax	1.000	deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0	MPa
Flange Material Operating Yield Stress	Syo	230.1	MPa
Single flange rotation at Sfmax	Thetafmax	0.336	deg

Gasket Contact Area [Ag]:
= π/4(GO.D.² - GI.D.²)
= π/4(826.0² - 794.0²)
= 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)
= SgT • Ag / (nb • Ab)
= 172.0 • 40714.922/(24 • 1862.725)
= 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
= min(Sbsel; Sbmax)
= min(156.647; 507.5)
= 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
= max(Sbsel; Sbmin)
= max(156.647; 290.0)
= 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
= min(Sbsel; Sfmax)
= min(290.0; 540.399)
= 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
:: Sbsel ≥ Sgmin-S(Ag / (nb • Abs))
:: 290.0 ≥ 140.0(40714.922/(24 • 1862.725))
:: 290.0 ≥ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
:: Sbsel ≥ (Sgmin-O • Ag + π/4 • Pmax • GI.D.²)/(φG • Abs • nb)
:: 290.0 ≥ (97.0 • 40714.922 + π/4 • 9.515 • 794.0²)/(0.7 • 1862.725 • 24)
:: 290.0 ≥ 276.8 MPa Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
:: Sbsel ≤ Sgmax(Ag / (nb • Abs))
:: 290.0 ≤ 380.0(40714.922/(24 • 1862.725))
:: 290.0 ≤ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
:: Sbsel ≤ Sfmax(thetagmax / thetafmax)
:: 290.0 ≤ 540.399(1.0/0.336)
:: 290.0 ≤ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 290.0 • 0.2 • 1862.725 • 56.0
= 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118	MPa
Bolt Allowable Stress for: SA-193 B7	Sb	171	MPa

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.021•202713•29.0²•149.238•0.3)
= 0.546 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.63951E+09/1.0•1.•0.327/(3.021•199226•29.0²•149.238•0.3)
= 0.480 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.631•202713•29.0²•149.238•0.3)
= 0.454 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.63951E+09/1.0•1.•0.327/(3.631•199226•29.0²•149.238•0.3)
= 0.400 (should be ≤ 1)

Minimum Design Metal Temperature Results:
Thickness Ratio = 0.903, Temperature Reduction per Fig. UCS 66.1 = 5 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B              6 °C
Min Metal Temp. at Required thickness (UCS 66.1)            0 °C
```

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	114.88	207.00	130.78	207.00
Radial Flange	19.66	138.00	22.38	138.00
Tangential Flange	88.60	138.00	100.86	138.00
Maximum Average	101.74	138.00	115.82	138.00
Bolting	155.27	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

Flange Input Data Values **Description: FLANGE-2 :**

Flange-2

Flange Type		Integral Weld Neck	
Design Pressure	P	9.51	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1126.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	g1	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1004.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		108484.00	N
Bending Moment on Flange		170335008.00	N-mm

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
= (P•(D/2+Ca))/(S•E-0.6•P) per UG-27 (c)(1)
= (9.51•(762.002/2+3.0))/(138.0•1.0-0.6•9.51)+Ca
= 30.6151 mm

Small End Hub MAWP:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 29.0)/(384.001 + 0.6 • 29.0)
= 9.970 MPa

Hub Small End Hub MAPnc:
= (Sa•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 32.0)/(381.001 + 0.6 • 32.0)
= 11.034 MPa

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	72.999 mm

```
Gasket Contact Width,       N = (Go - Gi)/2          16.000 mm
Basic Gasket Width,        bo = N/2                   8.000 mm
Effective Gasket Width,     b = Cb*sqrt(bo)           7.127 mm
Gasket Reaction Diameter,   G = Go - 2 * b          811.745 mm
```

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
= π/4 • G² • P
= π/4 • 811.7452² • 9.513
= 4923022.500 N

Contact Load on Gasket Surfaces [Hp or Wo,2]:
= 2 • b • π • G • m • P
= 2 • 7.1274 • π • 811.7452 • 3.0 • 9.51
= 1037420.250 N

Hydrostatic End Load at Flange ID [Hd]:
= π/4 • Bcor² • P
= π/4 • 768.002² • 9.5135
= 4406737.000 N

Pressure Force on Flange Face [Ht]:
= H - Hd
= 4923023 - 4406737
= 516285.469 N

Design Bolt Load for the Operating Condition [Wo]:
= max(Wo,1 + Wo,2 + H'p, 0)
= max(4923023 + 1037420 + 0, 0)
= 5960443.000 N

Design Bolt Load for the Gasket Seating Condition [Wg]:
= y • b • π • G + yPart • bPart • lp
= 68.95•7.1274•π•811.745+0.0•0.0•0.0
= 1253136.250 N

Required Bolt Area [Am]:
= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
= max((5960443 + 108484 + 4•.17034E+09/811.75)/172, 1253136/172)
= 40166.574 mm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
= 2a + 6t/(m + 0.5)
= 2 • 56.0 + 6 • 176.0/(3.0 + 0.5)
= 413.714 mm

Actual Circumferential Bolt Spacing [Bs]:
= C • sin(π / n)
= 1004.0 • sin(π/24)
= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
= max(√(Bs/(2a + t)), 1)
= max(√(131.048/(2 • 56.0 + 176.0)), 1)
= 1.0000

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	40166.574	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
= Ab • Sa/(y • π(Go + Gi))

= 44705.398 • 172.0/(68.95 • π(826.0 + 794.0))
= 21.912 mm[Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2
Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
= Sa • Ab
= 172.0 • 44705.3984
= 7688536.50 N

Gasket Load for the Operating Condition [HG]:
= Wo - H
= 5960443 - 4923023
= 1037420.38 N

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
= (C - G) / 2
= (1004.0 - 811.7452)/2
= 96.1274 mm

Distance to Face Pressure Reaction [ht]:
= 0.5((C - B)/2 + hg)
= 0.5((1004.0 - 768.002)/2 + 96.1274)
= 107.0632 mm

Distance to End Pressure Reaction [hd]:
= (C - B - g1)/2
= (1004.0 - 768.002 - 45.0)/2
= 95.4990 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:
Aa = 179.0, Bb = 176.0, Cc = 60.0 and Ddg = 37.0 mm

Compute the value of [Kab]:
= Aa • Bb^3 • (1/3 - 0.210 • (Bb/Aa) • (1-1/12 • $(Bb/Aa)^4$))
= 179.0 • 176.0^3 • (1/3 - 0.210 • (176.0/179.0) • (1-1/12 • $(176.0/179.0)^4$))
= 139483776.000 mm^4

Compute the value of [Kcd]:
= Cc • Ddg^3 • (1/3 - 0.105 • (Ddg/Cc) • (1-1/192 • $(Ddg/Cc)^4$))
= 60.0•37.0^3•(1/3-0.105•(37.0/60.0)•(1-1/192•$(37.0/60.0)^4$))
= 816420.500 mm^4

Flange Inertia Term [Ip,term1]:
= Kab + Kcd = 8511.831 + 49.821
= 140300192.000 mm^4

Flange Inertia Term [Ip,term2]:
= Ar • t^3(1/3 - 0.21(t/AR)(1 - 1/12$(t/AR)^4$))
= 179.0 • 176.0^3(1/3 - 0.21(176.0/179.0)(1 - 1/12$(176.0/7.047)^4$))
= 139483776.000 mm^4

Flange Inertia [Ip]:
= max(Ip,term1, Ip,term2)
= max(0.14030E+09, 0.13948E+09)
= 140300192.0 mm^4

Flange Moment of Inertia for Integral Types [I]:
= 0.0874 • L • go^2 • ho • B / V
= 0.0874 • 3.631 • 29.0^2 • 149.238 • 768.002/0.327
= 93579992.000 mm^4

Summary of Moments for Internal Pressure: (N-mm)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	4406737.	95.4990	1.0000	421009568.
Face Pressure,	Mt	516285.	107.0632	1.0000	55297600.
Gasket Load,	Mg	1037420.	96.1274	1.0000	99764992.
Gasket Seating,	Matm	7688536.	96.1274	1.0000	739379008.
Moment Load,	Moe	...	...	...	61156136.

Moment Value, eq. 4.16.16, [Moe]:
= 4Me[I/(0.384 • IP + I))•(hD /(C-2•hD)) + FA • hD
= 4•.17034E+09[93579992/(0.384•.14030E+09+93579992))•(95.5/(1004.0-2•95.5))+
108484•95.499
= 61156136.000 N-mm

```
Total Moment for Operation,      Mo                          637228288. N-mm
Total Moment for Gasket seating, Mg                          739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)         149.238 mm
Hub Ratio,           h/h0 = HL / H0                   0.402
Thickness Ratio,    g1/g0 = (g1Cor/goCor)             1.552
```

Flange Factors for Integral Flange:

```
Factor F per 4.16.5                                   0.860
Factor V per 4.16.5                                   0.327
Factor f per 4.16.5                                   1.000
```

Stress factor from Table 4.16.4 [K]:
= A / B
= 1126.0/768.002
= 1.466

Stress factor from Table 4.16.4 [Y]:
= (0.66845 + (5.7169(K^2 • log10(K))) / (K^2 - 1))/(K - 1)
= (0.66845 + (5.7169(1.466^2 • log10(1.466)))/(1.466^2 - 1))/(1.466 - 1)
= 5.245

Stress factor from Table 4.16.4 [T]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K^2) • (K - 1))
= (1.466^2(1 + 8.55246 • log10(1.466)) - 1)/((1.0472 + 1.9448 • 1.466^2) • (1.466 - 1))
= 1.725

Stress factor from Table 4.16.4 [U]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/(1.36136(K^2 - 1) • (K - 1))
= (1.466^2(1 + 8.55246 • log10(1.466)) - 1)/(1.36136(1.466^2 - 1) • (1.466 - 1))
= 5.764

Stress factor from Table 4.16.4 [Z]:
= (K^2 + 1)/(K^2 - 1)
= (1.466^2 + 1)/(1.466^2 - 1)
= 2.740

Stress factor from Table 4.16.4 for integral flanges [d]:
= U • $g0^2$ • h0 / V
= 5.764 • 29.0^2 • 149.238/0.327
= 2213042.750 mm^3

Stress factor from Table 4.16.4 for integral flanges [e]:
= F / h0
= 0.86/149.238
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:
= (t • e + 1) / T + t^3/d
= (176.0 • 0.006 + 1)/1.725 + 176.0^3/2213043
= 3.631

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.0•.63723E+09/768.002)/(3.6309•45.0²)
= 112.81 MPa

Longitudinal Hub Stress, Seating [SHa]:
= (f • Matm / Bcor) / (L • g1²)
= (1.0•0.73938E+09/768.002)/(3.6309•45.0²)
= 130.90 MPa

Radial Flange Stress, Operating [SRo]:
= (β • Mop / Bcor) / (L • t²)
= (2.3524•.63723E+09/768.002)/(3.6309•176.0²)
= 17.35 MPa

Radial Flange Stress, Seating [SRa]:
= (β • Matm/Bcor) / (L • t²)
= (2.3524•.73938E+09/768.002)/(3.6309•176.0²)
= 20.13 MPa

Tangential Flange Stress, Operating [STo]:
= (Y • Mo / (t² • Bcor)) - Z • SRo
= (5.2449•.63723E+09/(176.0²•768.002))-2.7398•17
= 92.91 MPa

Tangential Flange Stress, Seating [STa]:
= (y • Matm / (t² • Bcor)) - Z • SRa
= (5.2449•.73938E+09/(176.0²•768.002))-2.7398•20
= 107.81 MPa

Average Flange Stress, Operating [SAo]:
= (SHo + max(SRo, STo)) / 2
= (113+max(17,93))/2
= 102.86 MPa

Average Flange Stress, Seating [SAa]:
= (SHa + max(SRa, STa)) / 2
= (131+max(20,108))/2
= 119.35 MPa

Bolt Stress, Operating [BSo]:
= (Wm1 + Fa + 4Me/G) / Ab
= 6907939/44705.3984
= 154.53 MPa

Bolt Stress, Seating [BSa]:
= (Wm2 / Ab)
= (1253136/44705.3984)
= 28.03 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	112.81	207.00	130.90	207.00
Radial Flange	17.35	138.00	20.13	138.00
Tangential Flange	92.91	138.00	107.81	138.00
Maximum Average	102.86	138.00	119.35	138.00
Bolting	154.53	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 10 MPa
Estimated Finished Weight of Flange at given Thk. 783.1 kgm
Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Seating Actual	Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	f	*f2*	*f3*
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c\left(\frac{g1}{g0}\right) + e\left(\frac{h}{h0}\right) + g\left(\frac{g1}{g0}\right)^2 + i\left(\frac{h}{h0}\right)^2 + k\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}{\left[1 + b\left(\frac{g1}{g0}\right) + d\left(\frac{h}{h0}\right) + f\left(\frac{g1}{g0}\right)^2 + h\left(\frac{h}{h0}\right)^2 + j\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}$$

Hub Stress Correction Factor [f]:
= max(1; F1num / F1den)
= max(1; 1.3704/1.4405)
= 1.0000

Hub Stress Correction Factor [f2]:
= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
= max(0.25; 0.5259/1.4506) if h/ho < 0.35, else max (0.25; 0.2734/0.7571)
= 0.3611

In this case, the hub ratio h/ho, 0.4020 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress Sbmax 507.9 MPa

Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at Sfmax	Thetafmax	0.336 deg

Gasket Contact Area [Ag]:
= π/4(GO.D.² - GI.D.²)
= π/4(826.0² - 794.0²)
= 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)
= SgT • Ag / (nb • Ab)
= 172.0 • 40714.922/(24 • 1862.725)
= 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
= min(Sbsel; Sbmax)
= min(156.647; 507.9)
= 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
= max(Sbsel; Sbmin)
= max(156.647; 290.0)
= 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
= min(Sbsel; Sfmax)
= min(290.0; 540.399)
= 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
:: Sbsel ≥ Sgmin-S(Ag / (nb • Abs))
:: 290.0 ≥ 140.0(40714.922/(24 • 1862.725))
:: 290.0 ≥ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
:: Sbsel ≥ (Sgmin-O • Ag + π/4 • Pmax • GI.D.²)/(φG • Abs • nb)
:: 290.0 ≥ (97.0 • 40714.922 + π/4 • 9.513 • 794.0²)/(0.7 • 1862.725 • 24)
:: 290.0 ≥ 276.7 MPa Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
:: Sbsel ≤ Sgmax(Ag / (nb • Abs))
:: 290.0 ≤ 380.0(40714.922/(24 • 1862.725))
:: 290.0 ≤ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
:: Sbsel ≤ Sfmax(thetagmax / thetafmax)
:: 290.0 ≤ 540.399(1.0/0.336)
:: 290.0 ≤ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 290.0 • 0.2 • 1862.725 • 56.0
= 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118 MPa
Bolt Allowable Stress for: SA-193 B7	Sb	171 MPa

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.021•202713•29.0²•149.238•0.3)
= 0.546 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.63723E+09/1.0•1.•0.327/(3.021•199226•29.0²•149.238•0.3)
= 0.479 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.631•202713•29.0²•149.238•0.3)
= 0.454 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.63723E+09/1.0•1.•0.327/(3.631•199226•29.0²•149.238•0.3)
= 0.398 (should be ≤ 1)

Minimum Design Metal Temperature Results:
Thickness Ratio = 0.898, Temperature Reduction per Fig. UCS 66.1 = 6 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B           6 °C
Min Metal Temp. at Required thickness (UCS 66.1)         0 °C
```

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	114.47	207.00	130.78	207.00
Radial Flange	19.59	138.00	22.38	138.00
Tangential Flange	88.29	138.00	100.86	138.00
Maximum Average	101.38	138.00	115.82	138.00
Bolting	154.53	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

Flange Input Data Values **Description: FLANGE-3 :**

Flange-3

Flange Type		Integral Weld Neck	
Design Pressure	P	9.47	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1126.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	g1	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1004.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		76109.61	N
Bending Moment on Flange		83714096.00	N-mm

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
= (P•(D/2+Ca))/(S•E-0.6•P) per UG-27 (c)(1)
= (9.47•(762.002/2+3.0))/(138.0•1.0-0.6•9.47)+Ca
= 30.4957 mm

Small End Hub MAWP:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 29.0)/(384.001 + 0.6 • 29.0)
= 9.970 MPa

Hub Small End Hub MAPnc:
= (Sa•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 32.0)/(381.001 + 0.6 • 32.0)
= 11.034 MPa

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	72.999 mm

```
Gasket Contact Width,       N = (Go - Gi)/2            16.000 mm
Basic Gasket Width,        bo = N/2                     8.000 mm
Effective Gasket Width,     b = Cb*sqrt(bo)             7.127 mm
Gasket Reaction Diameter,   G = Go - 2 * b            811.745 mm
```

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
= π/4 • G² • P
= π/4 • 811.7452² • 9.474
= 4902620.000 N

Contact Load on Gasket Surfaces [Hp or Wo,2]:
= 2 • b • π • G • m • P
= 2 • 7.1274 • π • 811.7452 • 3.0 • 9.47
= 1033120.812 N

Hydrostatic End Load at Flange ID [Hd]:
= π/4 • Bcor² • P
= π/4 • 768.002² • 9.4741
= 4388474.000 N

Pressure Force on Flange Face [Ht]:
= H - Hd
= 4902620 - 4388474
= 514145.969 N

Design Bolt Load for the Operating Condition [Wo]:
= max(Wo,1 + Wo,2 + H'p, 0)
= max(4902620 + 1033121 + 0, 0)
= 5935741.000 N

Design Bolt Load for the Gasket Seating Condition [Wg]:
= y • b • π • G + yPart • bPart • lp
= 68.95•7.1274•π•811.745+0.0•0.0•0.0
= 1253136.250 N

Required Bolt Area [Am]:
= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
= max((5935741 + 76110 + 4•83714096/811.75)/172, 1253136/172)
= 37353.832 mm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
= 2a + 6t/(m + 0.5)
= 2 • 56.0 + 6 • 176.0/(3.0 + 0.5)
= 413.714 mm

Actual Circumferential Bolt Spacing [Bs]:
= C • sin(π / n)
= 1004.0 • sin(π/24)
= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
= max($\sqrt{(Bs/(2a + t))}$, 1)
= max($\sqrt{(131.048/(2 • 56.0 + 176.0))}$, 1)
= 1.0000

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	37353.832	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
= Ab • Sa/(y • π(Go + Gi))

= 44705.398 • 172.0/(68.95 • π(826.0 + 794.0))
= 21.912 mm[Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

```
Sheet Gaskets including laminated sheet gaskets with or without a metal core:     16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm
```

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
= Sa • Ab
= 172.0 • 44705.3984
= 7688536.50 N

Gasket Load for the Operating Condition [HG]:
= Wo - H
= 5935741 - 4902620
= 1033120.81 N

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
= (C - G) / 2
= (1004.0 - 811.7452)/2
= 96.1274 mm

Distance to Face Pressure Reaction [ht]:
= 0.5((C - B)/2 + hg)
= 0.5((1004.0 - 768.002)/2 + 96.1274)
= 107.0632 mm

Distance to End Pressure Reaction [hd]:
= (C - B - g1)/2
= (1004.0 - 768.002 - 45.0)/2
= 95.4990 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

```
Aa = 179.0, Bb = 176.0, Cc = 60.0 and Ddg = 37.0 mm
```

Compute the value of [Kab]:
= Aa • Bb^3 • (1/3 - 0.210 • (Bb/Aa) • (1-1/12 • $(Bb/Aa)^4$))
= 179.0 • 176.0^3 • (1/3 - 0.210 • (176.0/179.0) • (1-1/12 • $(176.0/179.0)^4$))
= 139483776.000 mm^4

Compute the value of [Kcd]:
= Cc • Ddg^3 • (1/3 - 0.105 • (Ddg/Cc) • (1-1/192 • $(Ddg/Cc)^4$))
= 60.0•37.0^3•(1/3-0.105•(37.0/60.0)•(1-1/192•$(37.0/60.0)^4$))
= 816420.500 mm^4

Flange Inertia Term [Ip,term1]:
= Kab + Kcd = 8511.831 + 49.821
= 140300192.000 mm^4

Flange Inertia Term [Ip,term2]:
= Ar • t^3(1/3 - 0.21(t/AR)(1 - 1/12$(t/AR)^4$))
= 179.0 • 176.0^3(1/3 - 0.21(176.0/179.0)(1 - 1/12$(176.0/7.047)^4$))
= 139483776.000 mm^4

Flange Inertia [Ip]:
= max(Ip,term1, Ip,term2)
= max(0.14030E+09, 0.13948E+09)
= 140300192.0 mm^4

Flange Moment of Inertia for Integral Types [I]:
= 0.0874 • L • go^2 • ho • B / V
= 0.0874 • 3.631 • 29.0^2 • 149.238 • 768.002/0.327
= 93579992.000 mm^4

Summary of Moments for Internal Pressure: (N-mm)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	4388474.	95.4990	1.0000	419264768.
Face Pressure,	Mt	514146.	107.0632	1.0000	55068440.
Gasket Load,	Mg	1033121.	96.1274	1.0000	99351536.
Gasket Seating,	Matm	7688536.	96.1274	1.0000	739379008.
Moment Load,	Moe	...	...	...	32233858.

Moment Value, eq. 4.16.16, [Moe]:
= 4Me[I/(0.384 • IP + I))•(hD /(C-2•hD)) + FA • hD
= 4•83714096[93579992/(0.384•.14030E+09+93579992))•(95.5/(1004.0-2•95.5))+
76110•95.499
= 32233858.000 N-mm

```
Total Moment for Operation,      Mo                        605918592. N-mm
Total Moment for Gasket seating, Mg                        739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)          149.238 mm
Hub Ratio,           h/h0 = HL / H0                    0.402
Thickness Ratio,    g1/g0 = (g1Cor/goCor)              1.552
```

Flange Factors for Integral Flange:

```
Factor F per 4.16.5                                    0.860
Factor V per 4.16.5                                    0.327
Factor f per 4.16.5                                    1.000
```

Stress factor from Table 4.16.4 [K]:
= A / B
= 1126.0/768.002
= 1.466

Stress factor from Table 4.16.4 [Y]:
= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.466² • log10(1.466)))/(1.466² - 1))/(1.466 - 1)
= 5.245

Stress factor from Table 4.16.4 [T]:
= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.466²(1 + 8.55246 • log10(1.466)) - 1)/((1.0472 + 1.9448 • 1.466²) • (1.466 - 1))
= 1.725

Stress factor from Table 4.16.4 [U]:
= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.466²(1 + 8.55246 • log10(1.466)) - 1)/(1.36136(1.466² - 1) • (1.466 - 1))
= 5.764

Stress factor from Table 4.16.4 [Z]:
= (K² + 1)/(K² - 1)
= (1.466² + 1)/(1.466² - 1)
= 2.740

Stress factor from Table 4.16.4 for integral flanges [d]:
= U • g0² • h0 / V
= 5.764 • 29.0² • 149.238/0.327
= 2213042.750 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:
= F / h0
= 0.86/149.238
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:
= (t • e + 1) / T + t³/d
= (176.0 • 0.006 + 1)/1.725 + 176.0³/2213043
= 3.631

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.0•.60592E+09/768.002)/(3.6309•45.0²)
= 107.27 MPa

Longitudinal Hub Stress, Seating [SHa]:
= (f • Matm / Bcor) / (L • g1²)
= (1.0•0.73938E+09/768.002)/(3.6309•45.0²)
= 130.90 MPa

Radial Flange Stress, Operating [SRo]:
= (β • Mop / Bcor) / (L • t²)
= (2.3524•.60592E+09/768.002)/(3.6309•176.0²)
= 16.50 MPa

Radial Flange Stress, Seating [SRa]:
= (β • Matm/Bcor) / (L • t²)
= (2.3524•.73938E+09/768.002)/(3.6309•176.0²)
= 20.13 MPa

Tangential Flange Stress, Operating [STo]:
= (Y • Mo / (t² • Bcor)) - Z • SRo
= (5.2449•.60592E+09/(176.0²•768.002))-2.7398•16
= 88.35 MPa

Tangential Flange Stress, Seating [STa]:
= (y • Matm / (t² • Bcor)) - Z • SRa
= (5.2449•.73938E+09/(176.0²•768.002))-2.7398•20
= 107.81 MPa

Average Flange Stress, Operating [SAo]:
= (SHo + max(SRo, STo)) / 2
= (107+max(16,88))/2
= 97.81 MPa

Average Flange Stress, Seating [SAa]:
= (SHa + max(SRa, STa)) / 2
= (131+max(20,108))/2
= 119.35 MPa

Bolt Stress, Operating [BSo]:
= (Wm1 + Fa + 4Me/G) / Ab
= 6424198/44705.3984
= 143.71 MPa

Bolt Stress, Seating [BSa]:
= (Wm2 / Ab)
= (1253136/44705.3984)
= 28.03 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	107.27	207.00	130.90	207.00
Radial Flange	16.50	138.00	20.13	138.00
Tangential Flange	88.35	138.00	107.81	138.00
Maximum Average	97.81	138.00	119.35	138.00
Bolting	143.71	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 10 MPa
Estimated Finished Weight of Flange at given Thk. 783.1 kgm
Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating Actual	Operating Allowed	stress ratio	Gasket Seating Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	f	*f2*	*f3*
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c\left(\frac{g1}{g0}\right) + e\left(\frac{h}{h0}\right) + g\left(\frac{g1}{g0}\right)^2 + i\left(\frac{h}{h0}\right)^2 + k\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}{\left[1 + b\left(\frac{g1}{g0}\right) + d\left(\frac{h}{h0}\right) + f\left(\frac{g1}{g0}\right)^2 + h\left(\frac{h}{h0}\right)^2 + j\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}$$

Hub Stress Correction Factor [f]:
= max(1; F1num / F1den)
= max(1; 1.3704/1.4405)
= 1.0000

Hub Stress Correction Factor [f2]:
= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
= max(0.25; 0.5259/1.4506) if h/ho < 0.35, else max (0.25; 0.2734/0.7571)
= 0.3611

In this case, the hub ratio h/ho, 0.4020 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress Sbmax 507.5 MPa

Minimum permissible bolt stress	Sbmin	290.0	MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4	MPa
Target assembly gasket stress	SgT	172.0	MPa
Maximum permissible gasket stress	Sgmax	380.0	MPa
Minimum gasket operating stress	Sgmin-O	97.0	MPa
Minimum gasket seating stress	Sgmin-S	140.0	MPa
Nut Factor	K	0.200	
Fraction of Gasket Load remaining after relaxation	Phig	0.700	
Root Area of a Single Bolt	Abs	1862.7250	mm²
Max. permissible single flange rotation for gasket	Thetagmax	1.000	deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0	MPa
Flange Material Operating Yield Stress	Syo	230.1	MPa
Single flange rotation at Sfmax	Thetafmax	0.336	deg

Gasket Contact Area [Ag]:
= π/4(GO.D.² - GI.D.²)
= π/4(826.0² - 794.0²)
= 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)
= SgT • Ag / (nb • Ab)
= 172.0 • 40714.922/(24 • 1862.725)
= 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
= min(Sbsel; Sbmax)
= min(156.647; 507.5)
= 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
= max(Sbsel; Sbmin)
= max(156.647; 290.0)
= 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
= min(Sbsel; Sfmax)
= min(290.0; 540.399)
= 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
:: Sbsel ≥ Sgmin-S(Ag / (nb • Abs))
:: 290.0 ≥ 140.0(40714.922/(24 • 1862.725))
:: 290.0 ≥ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
:: Sbsel ≥ (Sgmin-O • Ag + π/4 • Pmax • GI.D.²)/(φG • Abs • nb)
:: 290.0 ≥ (97.0 • 40714.922 + π/4 • 9.474 • 794.0²)/(0.7 • 1862.725 • 24)
:: 290.0 ≥ 276.1 MPa Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
:: Sbsel ≤ Sgmax(Ag / (nb • Abs))
:: 290.0 ≤ 380.0(40714.922/(24 • 1862.725))
:: 290.0 ≤ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
:: Sbsel ≤ Sfmax(thetagmax / thetafmax)
:: 290.0 ≤ 540.399(1.0/0.336)
:: 290.0 ≤ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 290.0 • 0.2 • 1862.725 • 56.0
= 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118	MPa
Bolt Allowable Stress for: SA-193 B7	Sb	171	MPa

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.021•202713•29.0²•149.238•0.3)
= 0.546 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.60592E+09/1.0•1.•0.327/(3.021•199226•29.0²•149.238•0.3)
= 0.455 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.631•202713•29.0²•149.238•0.3)
= 0.454 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.60592E+09/1.0•1.•0.327/(3.631•199226•29.0²•149.238•0.3)
= 0.379 (should be ≤ 1)

Minimum Design Metal Temperature Results:
Thickness Ratio = 0.872, Temperature Reduction per Fig. UCS 66.1 = 7 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B            6 °C
Min Metal Temp. at Required thickness (UCS 66.1)         -1 °C
```

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	108.77	207.00	130.78	207.00
Radial Flange	18.62	138.00	22.38	138.00
Tangential Flange	83.89	138.00	100.86	138.00
Maximum Average	96.33	138.00	115.82	138.00
Bolting	143.71	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

Flange Input Data Values **Description: FLANGE-4 :**

Flange-4

```
Flange Type                                  Integral Weld Neck

Design Pressure                          P             9.47  MPa
Design Temperature                                       82  °C
Internal Corrosion Allowance            ci           3.0000  mm
External Corrosion Allowance            ce           0.0000  mm
Use Corrosion Allowance in Thickness Calcs.              No

Flange Inside Diameter                   B          762.002  mm
Flange Outside Diameter                  A         1126.000  mm
Flange Thickness                         t         176.0000  mm
Thickness of Hub at Small End           go          32.0000  mm
Thickness of Hub at Large End           g1          48.0000  mm
Length of Hub                            h          60.0000  mm

Flange Material                                    SA-266 2
Flange Material UNS number                           K03506
Flange Allowable Stress At Temperature Sfo          138.00  MPa
Flange Allowable Stress At Ambient     Sfa          138.00  MPa

Bolt Material                                     SA-193 B7
Bolt Allowable Stress At Temperature    Sb          172.00  MPa
Bolt Allowable Stress At Ambient        Sa          172.00  MPa

Diameter of Bolt Circle                  C         1004.000  mm
Nominal Bolt Diameter                    a          56.0000  mm
Type of Threads                                 TEMA Metric
Number of Bolts                                          24

Flange Face Outside Diameter           Fod          936.000  mm
Flange Face Inside Diameter            Fid          762.002  mm
Flange Facing Sketch                              Sketch 1a

Gasket Outside Diameter                 Go          826.000  mm
Gasket Inside Diameter                  Gi          794.000  mm
Gasket Factor                            m           3.0000
Gasket Design Seating Stress             y            68.95  MPa

Gasket Thickness                        tg           4.5000  mm
Raised face depth                       Fd           0.0000  mm

Axial Force on Flange                              68913.40  N
Bending Moment on Flange                        79887400.00  N-mm
```

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
= (P•(D/2+Ca))/(S•E-0.6•P) per UG-27 (c)(1)
= (9.47•(762.002/2+3.0))/(138.0•1.0-0.6•9.47)+Ca
= 30.4898 mm

Small End Hub MAWP:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 29.0)/(384.001 + 0.6 • 29.0)
= 9.970 MPa

Hub Small End Hub MAPnc:
= (Sa•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 32.0)/(381.001 + 0.6 • 32.0)
= 11.034 MPa

```
Corroded Flange ID,     Bcor = B+2*ci             768.002 mm
Corroded Large Hub,    g1Cor = g1-ci               45.000 mm
Corroded Small Hub,    g0Cor = go-ci               29.000 mm
Code R Dimension,          R = ((C-Bcor)/2)-g1cor  72.999 mm
```

```
Gasket Contact Width,       N = (Go - Gi)/2            16.000 mm
Basic Gasket Width,        bo = N/2                     8.000 mm
Effective Gasket Width,     b = Cb*sqrt(bo)             7.127 mm
Gasket Reaction Diameter,   G = Go - 2 * b            811.745 mm
```

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
= $\pi/4 \cdot G^2 \cdot P$
= $\pi/4 \cdot 811.7452^2 \cdot 9.472$
= 4901612.000 N

Contact Load on Gasket Surfaces [Hp or Wo,2]:
= $2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
= $2 \cdot 7.1274 \cdot \pi \cdot 811.7452 \cdot 3.0 \cdot 9.47$
= 1032908.375 N

Hydrostatic End Load at Flange ID [Hd]:
= $\pi/4 \cdot Bcor^2 \cdot P$
= $\pi/4 \cdot 768.002^2 \cdot 9.4721$
= 4387571.500 N

Pressure Force on Flange Face [Ht]:
= H - Hd
= 4901612 - 4387572
= 514040.344 N

Design Bolt Load for the Operating Condition [Wo]:
= max(Wo,1 + Wo,2 + H'p, 0)
= max(4901612 + 1032908 + 0, 0)
= 5934520.500 N

Design Bolt Load for the Gasket Seating Condition [Wg]:
= $y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
= $68.95 \cdot 7.1274 \cdot \pi \cdot 811.745 + 0.0 \cdot 0.0 \cdot 0.0$
= 1253136.250 N

Required Bolt Area [Am]:
= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
= max((5934521 + 68913 + 4•79887400/811.75)/172, 1253136/172)
= 37195.297 mm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
= 2a + 6t/(m + 0.5)
= $2 \cdot 56.0 + 6 \cdot 176.0/(3.0 + 0.5)$
= 413.714 mm

Actual Circumferential Bolt Spacing [Bs]:
= $C \cdot \sin(\pi / n)$
= $1004.0 \cdot \sin(\pi/24)$
= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
= max($\sqrt{Bs/(2a + t)}$, 1)
= max($\sqrt{131.048/(2 \cdot 56.0 + 176.0)}$, 1)
= 1.0000

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	37195.297	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
= $Ab \cdot Sa/(y \cdot \pi(Go + Gi))$

= 44705.398 • 172.0/(68.95 • π(826.0 + 794.0))
= 21.912 mm[Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
= Sa • Ab
= 172.0 • 44705.3984
= 7688536.50 N

Gasket Load for the Operating Condition [HG]:
= Wo - H
= 5934521 - 4901612
= 1032908.44 N

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
= (C - G) / 2
= (1004.0 - 811.7452)/2
= 96.1274 mm

Distance to Face Pressure Reaction [ht]:
= 0.5((C - B)/2 + hg)
= 0.5((1004.0 - 768.002)/2 + 96.1274)
= 107.0632 mm

Distance to End Pressure Reaction [hd]:
= (C - B - g1)/2
= (1004.0 - 768.002 - 45.0)/2
= 95.4990 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:
Aa = 179.0, Bb = 176.0, Cc = 60.0 and Ddg = 37.0 mm

Compute the value of [Kab]:
= Aa • Bb^3 • (1/3 - 0.210 • (Bb/Aa) • (1-1/12 • $(Bb/Aa)^4$))
= 179.0 • 176.0^3 • (1/3 - 0.210 • (176.0/179.0) • (1-1/12 • $(176.0/179.0)^4$))
= 139483776.000 mm^4

Compute the value of [Kcd]:
= Cc • Ddg^3 • (1/3 - 0.105 • (Ddg/Cc) • (1-1/192 • $(Ddg/Cc)^4$))
= 60.0•37.0^3•(1/3-0.105•(37.0/60.0)•(1-1/192•$(37.0/60.0)^4$))
= 816420.500 mm^4

Flange Inertia Term [Ip,term1]:
= Kab + Kcd = 8511.831 + 49.821
= 140300192.000 mm^4

Flange Inertia Term [Ip,term2]:
= Ar • t^3(1/3 - 0.21(t/AR)(1 - 1/12(t/AR $)^4$))
= 179.0 • 176.0^3(1/3 - 0.21(176.0/179.0)(1 - 1/12(176.0/7.047 $)^4$))
= 139483776.000 mm^4

Flange Inertia [Ip]:
= max(Ip,term1, Ip,term2)
= max(0.14030E+09, 0.13948E+09)
= 140300192.0 mm^4

Flange Moment of Inertia for Integral Types [I]:
= 0.0874 • L • go^2 • ho • B / V
= 0.0874 • 3.631 • 29.0^2 • 149.238 • 768.002/0.327
= 93579992.000 mm^4

Summary of Moments for Internal Pressure: (N-mm)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	4387572.	95.4990	1.0000	419178560.
Face Pressure,	Mt	514040.	107.0632	1.0000	55057132.
Gasket Load,	Mg	1032908.	96.1274	1.0000	99331104.
Gasket Seating,	Matm	7688536.	96.1274	1.0000	739379008.
Moment Load,	Moe	...	...	...	30405278.

Moment Value, eq. 4.16.16, [Moe]:
= 4Me[I/(0.384 • IP + I))•(hD /(C-2•hD)) + FA • hD
= 4•79887400[93579992/(0.384•.14030E+09+93579992))•(95.5/(1004.0-2•95.5))+
68913•95.499
= 30405278.000 N-mm

```
Total Moment for Operation,      Mo                       603972096. N-mm
Total Moment for Gasket seating, Mg                       739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)         149.238 mm
Hub Ratio,            h/h0 = HL / H0                  0.402
Thickness Ratio,     g1/g0 = (g1Cor/goCor)            1.552
```

Flange Factors for Integral Flange:

```
Factor F per 4.16.5                                   0.860
Factor V per 4.16.5                                   0.327
Factor f per 4.16.5                                   1.000
```

Stress factor from Table 4.16.4 [K]:
= A / B
= 1126.0/768.002
= 1.466

Stress factor from Table 4.16.4 [Y]:
= $(0.66845 + (5.7169(K^2 \cdot \log10(K))) / (K^2 - 1))/(K - 1)$
= $(0.66845 + (5.7169(1.466^2 \cdot \log10(1.466)))/(1.466^2 - 1))/(1.466 - 1)$
= 5.245

Stress factor from Table 4.16.4 [T]:
= $(K^2(1 + 8.55246 \cdot \log10(K)) - 1)/((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
= $(1.466^2(1 + 8.55246 \cdot \log10(1.466)) - 1)/((1.0472 + 1.9448 \cdot 1.466^2) \cdot (1.466 - 1))$
= 1.725

Stress factor from Table 4.16.4 [U]:
= $(K^2(1 + 8.55246 \cdot \log10(K)) - 1)/(1.36136(K^2 - 1) \cdot (K - 1))$
= $(1.466^2(1 + 8.55246 \cdot \log10(1.466)) - 1)/(1.36136(1.466^2 - 1) \cdot (1.466 - 1))$
= 5.764

Stress factor from Table 4.16.4 [Z]:
= $(K^2 + 1)/(K^2 - 1)$
= $(1.466^2 + 1)/(1.466^2 - 1)$
= 2.740

Stress factor from Table 4.16.4 for integral flanges [d]:
= $U \cdot g0^2 \cdot h0 / V$
= $5.764 \cdot 29.0^2 \cdot 149.238/0.327$
= 2213042.750 mm^3

Stress factor from Table 4.16.4 for integral flanges [e]:
= F / h0
= 0.86/149.238
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:
= $(t \cdot e + 1) / T + t^3/d$
= $(176.0 \cdot 0.006 + 1)/1.725 + 176.0^3/2213043$
= 3.631

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.0•.60397E+09/768.002)/(3.6309•45.0²)
= 106.92 MPa

Longitudinal Hub Stress, Seating [SHa]:
= (f • Matm / Bcor) / (L • g1²)
= (1.0•0.73938E+09/768.002)/(3.6309•45.0²)
= 130.90 MPa

Radial Flange Stress, Operating [SRo]:
= (β • Mop / Bcor) / (L • t²)
= (2.3524•.60397E+09/768.002)/(3.6309•176.0²)
= 16.44 MPa

Radial Flange Stress, Seating [SRa]:
= (β • Matm/Bcor) / (L • t²)
= (2.3524•.73938E+09/768.002)/(3.6309•176.0²)
= 20.13 MPa

Tangential Flange Stress, Operating [STo]:
= (Y • Mo / (t² • Bcor)) - Z • SRo
= (5.2449•.60397E+09/(176.0²•768.002))-2.7398•16
= 88.06 MPa

Tangential Flange Stress, Seating [STa]:
= (y • Matm / (t² • Bcor)) - Z • SRa
= (5.2449•.73938E+09/(176.0²•768.002))-2.7398•20
= 107.81 MPa

Average Flange Stress, Operating [SAo]:
= (SHo + max(SRo, STo)) / 2
= (107+max(16,88))/2
= 97.49 MPa

Average Flange Stress, Seating [SAa]:
= (SHa + max(SRa, STa)) / 2
= (131+max(20,108))/2
= 119.35 MPa

Bolt Stress, Operating [BSo]:
= (Wm1 + Fa + 4Me/G) / Ab
= 6396932/44705.3984
= 143.10 MPa

Bolt Stress, Seating [BSa]:
= (Wm2 / Ab)
= (1253136/44705.3984)
= 28.03 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	106.92	207.00	130.90	207.00
Radial Flange	16.44	138.00	20.13	138.00
Tangential Flange	88.06	138.00	107.81	138.00
Maximum Average	97.49	138.00	119.35	138.00
Bolting	143.10	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 10 MPa
Estimated Finished Weight of Flange at given Thk. 783.1 kgm
Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating Actual	Operating Allowed	stress ratio	Gasket Seating Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	f	*f2*	*f3*
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c\left(\frac{g1}{g0}\right) + e\left(\frac{h}{h0}\right) + g\left(\frac{g1}{g0}\right)^2 + i\left(\frac{h}{h0}\right)^2 + k\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}{\left[1 + b\left(\frac{g1}{g0}\right) + d\left(\frac{h}{h0}\right) + f\left(\frac{g1}{g0}\right)^2 + h\left(\frac{h}{h0}\right)^2 + j\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}$$

Hub Stress Correction Factor [f]:
= max(1; F1num / F1den)
= max(1; 1.3704/1.4405)
= 1.0000

Hub Stress Correction Factor [f2]:
= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
= max(0.25; 0.5259/1.4506) if h/ho < 0.35, else max (0.25; 0.2734/0.7571)
= 0.3611

In this case, the hub ratio h/ho, 0.4020 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress   Sbmax   507.5 MPa

Minimum permissible bolt stress	Sbmin	290.0	MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4	MPa
Target assembly gasket stress	SgT	172.0	MPa
Maximum permissible gasket stress	Sgmax	380.0	MPa
Minimum gasket operating stress	Sgmin-O	97.0	MPa
Minimum gasket seating stress	Sgmin-S	140.0	MPa
Nut Factor	K	0.200	
Fraction of Gasket Load remaining after relaxation	Phig	0.700	
Root Area of a Single Bolt	Abs	1862.7250	mm²
Max. permissible single flange rotation for gasket	Thetagmax	1.000	deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0	MPa
Flange Material Operating Yield Stress	Syo	230.1	MPa
Single flange rotation at Sfmax	Thetafmax	0.336	deg

Gasket Contact Area [Ag]:
= $\pi/4(GO.D.^2 - GI.D.^2)$
= $\pi/4(826.0^2 - 794.0^2)$
= $40714.9\ mm^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
= SgT • Ag / (nb • Ab)
= 172.0 • 40714.922/(24 • 1862.725)
= 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
= min(Sbsel; Sbmax)
= min(156.647; 507.5)
= 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
= max(Sbsel; Sbmin)
= max(156.647; 290.0)
= 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
= min(Sbsel; Sfmax)
= min(290.0; 540.399)
= 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
:: Sbsel ≥ Sgmin-S(Ag / (nb • Abs))
:: 290.0 ≥ 140.0(40714.922/(24 • 1862.725))
:: 290.0 ≥ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
:: Sbsel ≥ $(Sgmin\text{-}O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2)/(\varphi G \cdot Abs \cdot nb)$
:: 290.0 ≥ $(97.0 \cdot 40714.922 + \pi/4 \cdot 9.472 \cdot 794.0^2)/(0.7 \cdot 1862.725 \cdot 24)$
:: 290.0 ≥ 276.1 MPa Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
:: Sbsel ≤ Sgmax(Ag / (nb • Abs))
:: 290.0 ≤ 380.0(40714.922/(24 • 1862.725))
:: 290.0 ≤ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
:: Sbsel ≤ Sfmax(thetagmax / thetafmax)
:: 290.0 ≤ 540.399(1.0/0.336)
:: 290.0 ≤ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 290.0 • 0.2 • 1862.725 • 56.0
= 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118	MPa
Bolt Allowable Stress for: SA-193 B7	Sb	171	MPa

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.021•202713•29.0²•149.238•0.3)
= 0.546 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.60397E+09/1.0•1.•0.327/(3.021•199226•29.0²•149.238•0.3)
= 0.454 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.631•202713•29.0²•149.238•0.3)
= 0.454 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.60397E+09/1.0•1.•0.327/(3.631•199226•29.0²•149.238•0.3)
= 0.377 (should be ≤ 1)

Minimum Design Metal Temperature Results:
Thickness Ratio = 0.872, Temperature Reduction per Fig. UCS 66.1 = 7 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B             6 °C
Min Metal Temp. at Required thickness (UCS 66.1)          -1 °C
```

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	108.42	207.00	130.78	207.00
Radial Flange	18.56	138.00	22.38	138.00
Tangential Flange	83.62	138.00	100.86	138.00
Maximum Average	96.02	138.00	115.82	138.00
Bolting	143.10	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

Flange Input Data Values **Description: FLANGE-5 :**

Flange-5

```
Flange Type                                   Integral Weld Neck

Design Pressure                         P               9.43  MPa
Design Temperature                                        82  °C
Internal Corrosion Allowance           ci             3.0000  mm
External Corrosion Allowance           ce             0.0000  mm
Use Corrosion Allowance in Thickness Calcs.               No

Flange Inside Diameter                  B            762.002  mm
Flange Outside Diameter                 A           1126.000  mm
Flange Thickness                        t           176.0000  mm
Thickness of Hub at Small End          go            32.0000  mm
Thickness of Hub at Large End          g1            48.0000  mm
Length of Hub                           h            60.0000  mm

Flange Material                                     SA-266 2
Flange Material UNS number                            K03506
Flange Allowable Stress At Temperature Sfo            138.00  MPa
Flange Allowable Stress At Ambient     Sfa            138.00  MPa

Bolt Material                                      SA-193 B7
Bolt Allowable Stress At Temperature   Sb             172.00  MPa
Bolt Allowable Stress At Ambient       Sa             172.00  MPa

Diameter of Bolt Circle                 C           1004.000  mm
Nominal Bolt Diameter                   a            56.0000  mm
Type of Threads                                  TEMA Metric
Number of Bolts                                           24

Flange Face Outside Diameter          Fod            936.000  mm
Flange Face Inside Diameter           Fid            762.002  mm
Flange Facing Sketch                                Sketch 1a

Gasket Outside Diameter                Go            826.000  mm
Gasket Inside Diameter                 Gi            794.000  mm
Gasket Factor                           m             3.0000
Gasket Design Seating Stress            y              68.95  MPa

Gasket Thickness                       tg             4.5000  mm
Raised face depth                      Fd             0.0000  mm

Axial Force on Flange                               37417.30  N
Bending Moment on Flange                         20375300.00  N-mm
```

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
= (P•(D/2+Ca))/(S•E-0.6•P) per UG-27 (c)(1)
= (9.43•(762.002/2+3.0))/(138.0•1.0-0.6•9.43)+Ca
= 30.3713 mm

Small End Hub MAWP:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 29.0)/(384.001 + 0.6 • 29.0)
= 9.970 MPa

Hub Small End Hub MAPnc:
= (Sa•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 32.0)/(381.001 + 0.6 • 32.0)
= 11.034 MPa

```
Corroded Flange ID,     Bcor = B+2*ci              768.002 mm
Corroded Large Hub,     g1Cor = g1-ci               45.000 mm
Corroded Small Hub,     g0Cor = go-ci               29.000 mm
Code R Dimension,          R = ((C-Bcor)/2)-g1cor   72.999 mm
```

Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
= π/4 • G² • P
= π/4 • 811.7452² • 9.433
= 4881340.000 N

Contact Load on Gasket Surfaces [Hp or Wo,2]:
= 2 • b • π • G • m • P
= 2 • 7.1274 • π • 811.7452 • 3.0 • 9.43
= 1028636.562 N

Hydrostatic End Load at Flange ID [Hd]:
= π/4 • Bcor² • P
= π/4 • 768.002² • 9.4329
= 4369426.000 N

Pressure Force on Flange Face [Ht]:
= H - Hd
= 4881340 - 4369426
= 511914.469 N

Design Bolt Load for the Operating Condition [Wo]:
= max(Wo,1 + Wo,2 + H'p, 0)
= max(4881340 + 1028637 + 0, 0)
= 5909977.000 N

Design Bolt Load for the Gasket Seating Condition [Wg]:
= y • b • π • G + yPart • bPart • lp
= 68.95•7.1274•π•811.745+0.0•0.0•0.0
= 1253136.250 N

Required Bolt Area [Am]:
= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
= max((5909977 + 37417 + 4•20375300/811.75)/172, 1253136/172)
= 35164.996 mm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
= 2a + 6t/(m + 0.5)
= 2 • 56.0 + 6 • 176.0/(3.0 + 0.5)
= 413.714 mm

Actual Circumferential Bolt Spacing [Bs]:
= C • sin(π / n)
= 1004.0 • sin(π/24)
= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
= max($\sqrt{(\text{Bs}/(2a + t))}$, 1)
= max($\sqrt{(131.048/(2 \cdot 56.0 + 176.0))}$, 1)
= 1.0000

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	35164.996	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
= Ab • Sa/(y • π(Go + Gi))

= 44705.398 • 172.0/(68.95 • π(826.0 + 794.0))
= 21.912 mm[Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
= Sa • Ab
= 172.0 • 44705.3984
= 7688536.50 N

Gasket Load for the Operating Condition [HG]:
= Wo - H
= 5909977 - 4881340
= 1028636.69 N

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
= (C - G) / 2
= (1004.0 - 811.7452)/2
= 96.1274 mm

Distance to Face Pressure Reaction [ht]:
= 0.5((C - B)/2 + hg)
= 0.5((1004.0 - 768.002)/2 + 96.1274)
= 107.0632 mm

Distance to End Pressure Reaction [hd]:
= (C - B - g1)/2
= (1004.0 - 768.002 - 45.0)/2
= 95.4990 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:
Aa = 179.0, Bb = 176.0, Cc = 60.0 and Ddg = 37.0 mm

Compute the value of [Kab]:
= Aa • Bb^3 • (1/3 - 0.210 • (Bb/Aa) • (1-1/12 • $(Bb/Aa)^4$))
= 179.0 • 176.0^3 • (1/3 - 0.210 • (176.0/179.0) • (1-1/12 • $(176.0/179.0)^4$))
= 139483776.000 mm^4

Compute the value of [Kcd]:
= Cc • Ddg^3 • (1/3 - 0.105 • (Ddg/Cc) • (1-1/192 • $(Ddg/Cc)^4$))
= 60.0•37.0^3•(1/3-0.105•(37.0/60.0)•(1-1/192•$(37.0/60.0)^4$))
= 816420.500 mm^4

Flange Inertia Term [Ip,term1]:
= Kab + Kcd = 8511.831 + 49.821
= 140300192.000 mm^4

Flange Inertia Term [Ip,term2]:
= Ar • t^3(1/3 - 0.21(t/AR)(1 - 1/12$(t/AR)^4$))
= 179.0 • 176.0^3(1/3 - 0.21(176.0/179.0)(1 - 1/12$(176.0/7.047)^4$))
= 139483776.000 mm^4

Flange Inertia [Ip]:
= max(Ip,term1, Ip,term2)
= max(0.14030E+09, 0.13948E+09)
= 140300192.0 mm^4

Flange Moment of Inertia for Integral Types [I]:
= 0.0874 • L • go^2 • ho • B / V
= 0.0874 • 3.631 • 29.0^2 • 149.238 • 768.002/0.327
= 93579992.000 mm^4

Summary of Moments for Internal Pressure: (N-mm)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	4369426.	95.4990	1.0000	417444960.
Face Pressure,	Mt	511914.	107.0632	1.0000	54829432.
Gasket Load,	Mg	1028637.	96.1274	1.0000	98920304.
Gasket Seating,	Matm	7688536.	96.1274	1.0000	739379008.
Moment Load,	Moe	...	...	...	9650429.

Moment Value, eq. 4.16.16, [Moe]:
= 4Me[I/(0.384 • IP + I))•(hD /(C-2•hD)) + FA • hD
= 4•20375300[93579992/(0.384•.14030E+09+93579992))•(95.5/(1004.0-2•95.5))+ 37417•95.499
= 9650429.000 N-mm

Total Moment for Operation, Mo 580845120. N-mm
Total Moment for Gasket seating, Mg 739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
Hub Ratio, h/h0 = HL / H0 0.402
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:
Factor F per 4.16.5 0.860
Factor V per 4.16.5 0.327
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:
= A / B
= 1126.0/768.002
= 1.466

Stress factor from Table 4.16.4 [Y]:
= (0.66845 + (5.7169(K^2 • log10(K))) / (K^2 - 1))/(K - 1)
= (0.66845 + (5.7169(1.466^2 • log10(1.466)))/(1.466^2 - 1))/(1.466 - 1)
= 5.245

Stress factor from Table 4.16.4 [T]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K^2) • (K - 1))
= (1.466^2(1 + 8.55246 • log10(1.466)) - 1)/((1.0472 + 1.9448 • 1.466^2) • (1.466 - 1))
= 1.725

Stress factor from Table 4.16.4 [U]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/(1.36136(K^2 - 1) • (K - 1))
= (1.466^2(1 + 8.55246 • log10(1.466)) - 1)/(1.36136(1.466^2 - 1) • (1.466 - 1))
= 5.764

Stress factor from Table 4.16.4 [Z]:
= (K^2 + 1)/(K^2 - 1)
= (1.466^2 + 1)/(1.466^2 - 1)
= 2.740

Stress factor from Table 4.16.4 for integral flanges [d]:
= U • $g0^2$ • h0 / V
= 5.764 • 29.0^2 • 149.238/0.327
= 2213042.750 mm^3

Stress factor from Table 4.16.4 for integral flanges [e]:
= F / h0
= 0.86/149.238
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:
= (t • e + 1) / T + t^3/d
= (176.0 • 0.006 + 1)/1.725 + 176.0^3/2213043
= 3.631

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.0•.58085E+09/768.002)/(3.6309•45.0²)
= 102.83 MPa

Longitudinal Hub Stress, Seating [SHa]:
= (f • Matm / Bcor) / (L • g1²)
= (1.0•0.73938E+09/768.002)/(3.6309•45.0²)
= 130.90 MPa

Radial Flange Stress, Operating [SRo]:
= (β • Mop / Bcor) / (L • t²)
= (2.3524•.58085E+09/768.002)/(3.6309•176.0²)
= 15.81 MPa

Radial Flange Stress, Seating [SRa]:
= (β • Matm/Bcor) / (L • t²)
= (2.3524•.73938E+09/768.002)/(3.6309•176.0²)
= 20.13 MPa

Tangential Flange Stress, Operating [STo]:
= (Y • Mo / (t² • Bcor)) - Z • SRo
= (5.2449•.58085E+09/(176.0²•768.002))-2.7398•16
= 84.69 MPa

Tangential Flange Stress, Seating [STa]:
= (y • Matm / (t² • Bcor)) - Z • SRa
= (5.2449•.73938E+09/(176.0²•768.002))-2.7398•20
= 107.81 MPa

Average Flange Stress, Operating [SAo]:
= (SHo + max(SRo, STo)) / 2
= (103+max(16,85))/2
= 93.76 MPa

Average Flange Stress, Seating [SAa]:
= (SHa + max(SRa, STa)) / 2
= (131+max(20,108))/2
= 119.35 MPa

Bolt Stress, Operating [BSo]:
= (Wm1 + Fa + 4Me/G) / Ab
= 6047756/44705.3984
= 135.29 MPa

Bolt Stress, Seating [BSa]:
= (Wm2 / Ab)
= (1253136/44705.3984)
= 28.03 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	102.83	207.00	130.90	207.00
Radial Flange	15.81	138.00	20.13	138.00
Tangential Flange	84.69	138.00	107.81	138.00
Maximum Average	93.76	138.00	119.35	138.00
Bolting	135.29	172.00	28.03	172.00

```
Minimum Required Flange Thickness                      161.468 mm

Estimated M.A.W.P. ( Operating )                            10 MPa
Estimated Finished Weight of Flange at given Thk.        783.1 kgm
Estimated Unfinished Weight of Forging at given Thk      987.3 kgm
```

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating		stress	Gasket Seating		stress
		Actual	Allowed	ratio	Actual	Allowed	ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	f	*f2*	*f3*
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c\left(\frac{g1}{g0}\right) + e\left(\frac{h}{h0}\right) + g\left(\frac{g1}{g0}\right)^2 + i\left(\frac{h}{h0}\right)^2 + k\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}{\left[1 + b\left(\frac{g1}{g0}\right) + d\left(\frac{h}{h0}\right) + f\left(\frac{g1}{g0}\right)^2 + h\left(\frac{h}{h0}\right)^2 + j\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}$$

Hub Stress Correction Factor [f]:
= max(1; F1num / F1den)
= max(1; 1.3704/1.4405)
= 1.0000

Hub Stress Correction Factor [f2]:
= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
= max(0.25; 0.5259/1.4506) if h/ho < 0.35, else max (0.25; 0.2734/0.7571)
= 0.3611

In this case, the hub ratio h/ho, 0.4020 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress Sbmax 507.5 MPa

Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm^2
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at Sfmax	Thetafmax	0.336 deg

Gasket Contact Area [Ag]:
= π/4(GO.D.² - GI.D.²)
= π/4(826.0² - 794.0²)
= 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)
= SgT • Ag / (nb • Ab)
= 172.0 • 40714.922/(24 • 1862.725)
= 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
= min(Sbsel; Sbmax)
= min(156.647; 507.5)
= 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
= max(Sbsel; Sbmin)
= max(156.647; 290.0)
= 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
= min(Sbsel; Sfmax)
= min(290.0; 540.399)
= 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
:: Sbsel ≥ Sgmin-S(Ag / (nb • Abs))
:: 290.0 ≥ 140.0(40714.922/(24 • 1862.725))
:: 290.0 ≥ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
:: Sbsel ≥ (Sgmin-O • Ag + π/4 • Pmax • GI.D.²)/(φG • Abs • nb)
:: 290.0 ≥ (97.0 • 40714.922 + π/4 • 9.433 • 794.0²)/(0.7 • 1862.725 • 24)
:: 290.0 ≥ 275.5 MPa Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
:: Sbsel ≤ Sgmax(Ag / (nb • Abs))
:: 290.0 ≤ 380.0(40714.922/(24 • 1862.725))
:: 290.0 ≤ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
:: Sbsel ≤ Sfmax(thetagmax / thetafmax)
:: 290.0 ≤ 540.399(1.0/0.336)
:: 290.0 ≤ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 290.0 • 0.2 • 1862.725 • 56.0
= 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118 MPa
Bolt Allowable Stress for: SA-193 B7	Sb	171 MPa

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.021•202713•29.0²•149.238•0.3)
= 0.546 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.58085E+09/1.0•1.•0.327/(3.021•199226•29.0²•149.238•0.3)
= 0.436 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.631•202713•29.0²•149.238•0.3)
= 0.454 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.58085E+09/1.0•1.•0.327/(3.631•199226•29.0²•149.238•0.3)
= 0.363 (should be ≤ 1)

Minimum Design Metal Temperature Results:
Thickness Ratio = 0.869, Temperature Reduction per Fig. UCS 66.1 = 7 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B             6 °C
Min Metal Temp. at Required thickness (UCS 66.1)          -2 °C
```

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	104.21	207.00	130.78	207.00
Radial Flange	17.84	138.00	22.38	138.00
Tangential Flange	80.38	138.00	100.86	138.00
Maximum Average	92.30	138.00	115.82	138.00
Bolting	135.29	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

Flange Input Data Values **Description: FLANGE-6 :**

Flange-6

```
Flange Type                                        Integral Weld Neck

Design Pressure                           P                  9.43  MPa
Design Temperature                                             82  °C
Internal Corrosion Allowance              ci               3.0000  mm
External Corrosion Allowance              ce               0.0000  mm
Use Corrosion Allowance in Thickness Calcs.                    No

Flange Inside Diameter                    B               762.002  mm
Flange Outside Diameter                   A              1126.000  mm
Flange Thickness                          t              176.0000  mm
Thickness of Hub at Small End             go              32.0000  mm
Thickness of Hub at Large End             g1              48.0000  mm
Length of Hub                             h               60.0000  mm

Flange Material                                          SA-266 2
Flange Material UNS number                                 K03506
Flange Allowable Stress At Temperature   Sfo               138.00  MPa
Flange Allowable Stress At Ambient       Sfa               138.00  MPa

Bolt Material                                           SA-193 B7
Bolt Allowable Stress At Temperature      Sb               172.00  MPa
Bolt Allowable Stress At Ambient          Sa               172.00  MPa

Diameter of Bolt Circle                   C              1004.000  mm
Nominal Bolt Diameter                     a               56.0000  mm
Type of Threads                                       TEMA Metric
Number of Bolts                                                24

Flange Face Outside Diameter             Fod              936.000  mm
Flange Face Inside Diameter              Fid              762.002  mm
Flange Facing Sketch                                    Sketch 1a

Gasket Outside Diameter                   Go              826.000  mm
Gasket Inside Diameter                    Gi              794.000  mm
Gasket Factor                             m                3.0000
Gasket Design Seating Stress              y                 68.95  MPa

Gasket Thickness                          tg               4.5000  mm
Raised face depth                         Fd               0.0000  mm

Axial Force on Flange                                    30965.80  N
Bending Moment on Flange                              18257800.00  N-mm
```

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
= (P•(D/2+Ca))/(S•E-0.6•P) per UG-27 (c)(1)
= (9.43•(762.002/2+3.0))/(138.0•1.0-0.6•9.43)+Ca
= 30.3656 mm

Small End Hub MAWP:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 29.0)/(384.001 + 0.6 • 29.0)
= 9.970 MPa

Hub Small End Hub MAPnc:
= (Sa•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 32.0)/(381.001 + 0.6 • 32.0)
= 11.034 MPa

```
Corroded Flange ID,      Bcor = B+2*ci                 768.002 mm
Corroded Large Hub,      g1Cor = g1-ci                  45.000 mm
Corroded Small Hub,      g0Cor = go-ci                  29.000 mm
Code R Dimension,           R = ((C-Bcor)/2)-g1cor      72.999 mm
```

```
Gasket Contact Width,       N = (Go - Gi)/2              16.000 mm
Basic Gasket Width,        bo = N/2                       8.000 mm
Effective Gasket Width,     b = Cb*sqrt(bo)               7.127 mm
Gasket Reaction Diameter,   G = Go - 2 * b              811.745 mm
```

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
= π/4 • G² • P
= π/4 • 811.7452² • 9.431
= 4880369.000 N

Contact Load on Gasket Surfaces [Hp or Wo,2]:
= 2 • b • π • G • m • P
= 2 • 7.1274 • π • 811.7452 • 3.0 • 9.43
= 1028431.938 N

Hydrostatic End Load at Flange ID [Hd]:
= π/4 • Bcor² • P
= π/4 • 768.002² • 9.4311
= 4368556.500 N

Pressure Force on Flange Face [Ht]:
= H - Hd
= 4880369 - 4368557
= 511812.188 N

Design Bolt Load for the Operating Condition [Wo]:
= max(Wo,1 + Wo,2 + H'p, 0)
= max(4880369 + 1028432 + 0, 0)
= 5908801.000 N

Design Bolt Load for the Gasket Seating Condition [Wg]:
= y • b • π • G + yPart • bPart • lp
= 68.95•7.1274•π•811.745+0.0•0.0•0.0
= 1253136.250 N

Required Bolt Area [Am]:
= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
= max((5908801 + 30966 + 4•18257800/811.75)/172, 1253136/172)
= 35060.000 mm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
= 2a + 6t/(m + 0.5)
= 2 • 56.0 + 6 • 176.0/(3.0 + 0.5)
= 413.714 mm

Actual Circumferential Bolt Spacing [Bs]:
= C • sin(π / n)
= 1004.0 • sin(π/24)
= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
= max($\sqrt{\text{(Bs/(2a + t)}}$), 1)
= max($\sqrt{\text{(131.048/(2 • 56.0 + 176.0)}}$), 1)
= 1.0000

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	35060.000	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
= Ab • Sa/(y • π(Go + Gi))

= 44705.398 • 172.0/(68.95 • π(826.0 + 794.0))
= 21.912 mm[Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2
Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
= Sa • Ab
= 172.0 • 44705.3984
= 7688536.50 N

Gasket Load for the Operating Condition [HG]:
= Wo - H
= 5908801 - 4880369
= 1028432.06 N

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
= (C - G) / 2
= (1004.0 - 811.7452)/2
= 96.1274 mm

Distance to Face Pressure Reaction [ht]:
= 0.5((C - B)/2 + hg)
= 0.5((1004.0 - 768.002)/2 + 96.1274)
= 107.0632 mm

Distance to End Pressure Reaction [hd]:
= (C - B - g1)/2
= (1004.0 - 768.002 - 45.0)/2
= 95.4990 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:
Aa = 179.0, Bb = 176.0, Cc = 60.0 and Ddg = 37.0 mm

Compute the value of [Kab]:
= Aa • Bb^3 • (1/3 - 0.210 • (Bb/Aa) • (1-1/12 • (Bb/Aa)^4))
= 179.0 • 176.0^3 • (1/3 - 0.210 • (176.0/179.0) • (1-1/12 • (176.0/179.0)^4))
= 139483776.000 mm^4

Compute the value of [Kcd]:
= Cc • Ddg^3 • (1/3 - 0.105 • (Ddg/Cc) • (1-1/192 • (Ddg/Cc)^4))
= 60.0•37.0^3•(1/3-0.105•(37.0/60.0)•(1-1/192•(37.0/60.0)^4))
= 816420.500 mm^4

Flange Inertia Term [Ip,term1]:
= Kab + Kcd = 8511.831 + 49.821
= 140300192.000 mm^4

Flange Inertia Term [Ip,term2]:
= Ar • t^3(1/3 - 0.21(t/AR)(1 - 1/12(t/AR)^4))
= 179.0 • 176.0^3(1/3 - 0.21(176.0/179.0)(1 - 1/12(176.0/7.047)^4))
= 139483776.000 mm^4

Flange Inertia [Ip]:
= max(Ip,term1, Ip,term2)
= max(0.14030E+09, 0.13948E+09)
= 140300192.0 mm^4

Flange Moment of Inertia for Integral Types [I]:
= 0.0874 • L • go^2 • ho • B / V
= 0.0874 • 3.631 • 29.0^2 • 149.238 • 768.002/0.327
= 93579992.000 mm^4

Summary of Moments for Internal Pressure: (N-mm)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	4368556.	95.4990	1.0000	417361920.
Face Pressure,	Mt	511812.	107.0632	1.0000	54818476.
Gasket Load,	Mg	1028432.	96.1274	1.0000	98900624.
Gasket Seating,	Matm	7688536.	96.1274	1.0000	739379008.
Moment Load,	Moe	...	...	...	8402655.

Moment Value, eq. 4.16.16, [Moe]:
= 4Me[I/(0.384 • IP + I))•(hD /(C-2•hD)) + FA • hD
= 4•18257800[93579992/(0.384•.14030E+09+93579992))•(95.5/(1004.0-2•95.5))+
30966•95.499
= 8402655.000 N-mm

Total Moment for Operation,	Mo	579483712. N-mm
Total Moment for Gasket seating,	Mg	739379008. N-mm

Effective Hub Length,	ho = sqrt(Bcor*goCor)	149.238 mm
Hub Ratio,	h/h0 = HL / H0	0.402
Thickness Ratio,	g1/g0 = (g1Cor/goCor)	1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.860
Factor V per 4.16.5	0.327
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:
= A / B
= 1126.0/768.002
= 1.466

Stress factor from Table 4.16.4 [Y]:
= (0.66845 + (5.7169(K^2 • log10(K))) / (K^2 - 1))/(K - 1)
= (0.66845 + (5.7169(1.466^2 • log10(1.466)))/(1.466^2 - 1))/(1.466 - 1)
= 5.245

Stress factor from Table 4.16.4 [T]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K^2) • (K - 1))
= (1.466^2(1 + 8.55246 • log10(1.466)) - 1)/((1.0472 + 1.9448 • 1.466^2) • (1.466 - 1))
= 1.725

Stress factor from Table 4.16.4 [U]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/(1.36136(K^2 - 1) • (K - 1))
= (1.466^2(1 + 8.55246 • log10(1.466)) - 1)/(1.36136(1.466^2 - 1) • (1.466 - 1))
= 5.764

Stress factor from Table 4.16.4 [Z]:
= (K^2 + 1)/(K^2 - 1)
= (1.466^2 + 1)/(1.466^2 - 1)
= 2.740

Stress factor from Table 4.16.4 for integral flanges [d]:
= U • $g0^2$ • h0 / V
= 5.764 • 29.0^2 • 149.238/0.327
= 2213042.750 mm^3

Stress factor from Table 4.16.4 for integral flanges [e]:
= F / h0
= 0.86/149.238
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:
= (t • e + 1) / T + t^3/d
= (176.0 • 0.006 + 1)/1.725 + 176.0^3/2213043
= 3.631

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • $g1^2$)
= (1.0•.57948E+09/768.002)/(3.6309•45.0^2)
= 102.59 MPa

Longitudinal Hub Stress, Seating [SHa]:
= (f • Matm / Bcor) / (L • $g1^2$)
= (1.0•0.73938E+09/768.002)/(3.6309•45.0^2)
= 130.90 MPa

Radial Flange Stress, Operating [SRo]:
= (β • Mop / Bcor) / (L • t^2)
= (2.3524•.57948E+09/768.002)/(3.6309•176.0^2)
= 15.78 MPa

Radial Flange Stress, Seating [SRa]:
= (β • Matm/Bcor) / (L • t^2)
= (2.3524•.73938E+09/768.002)/(3.6309•176.0^2)
= 20.13 MPa

Tangential Flange Stress, Operating [STo]:
= (Y • Mo / (t^2 • Bcor)) - Z • SRo
= (5.2449•.57948E+09/(176.0^2•768.002))-2.7398•16
= 84.49 MPa

Tangential Flange Stress, Seating [STa]:
= (y • Matm / (t^2 • Bcor)) - Z • SRa
= (5.2449•.73938E+09/(176.0^2•768.002))-2.7398•20
= 107.81 MPa

Average Flange Stress, Operating [SAo]:
= (SHo + max(SRo, STo)) / 2
= (103+max(16,84))/2
= 93.54 MPa

Average Flange Stress, Seating [SAa]:
= (SHa + max(SRa, STa)) / 2
= (131+max(20,108))/2
= 119.35 MPa

Bolt Stress, Operating [BSo]:
= (Wm1 + Fa + 4Me/G) / Ab
= 6029699/44705.3984
= 134.89 MPa

Bolt Stress, Seating [BSa]:
= (Wm2 / Ab)
= (1253136/44705.3984)
= 28.03 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	102.59	207.00	130.90	207.00
Radial Flange	15.78	138.00	20.13	138.00
Tangential Flange	84.49	138.00	107.81	138.00
Maximum Average	93.54	138.00	119.35	138.00
Bolting	134.89	172.00	28.03	172.00

Minimum Required Flange Thickness	161.468	mm
Estimated M.A.W.P. (Operating)	10	MPa
Estimated Finished Weight of Flange at given Thk.	783.1	kgm
Estimated Unfinished Weight of Forging at given Thk	987.3	kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating Actual	Operating Allowed	stress ratio	Gasket Seating Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	f	***f2***	***f3***
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c\left(\frac{g1}{g0}\right) + e\left(\frac{h}{h0}\right) + g\left(\frac{g1}{g0}\right)^2 + i\left(\frac{h}{h0}\right)^2 + k\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}{\left[1 + b\left(\frac{g1}{g0}\right) + d\left(\frac{h}{h0}\right) + f\left(\frac{g1}{g0}\right)^2 + h\left(\frac{h}{h0}\right)^2 + j\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}$$

Hub Stress Correction Factor [f]:
= max(1; F1num / F1den)
= max(1; 1.3704/1.4405)
= 1.0000

Hub Stress Correction Factor [f2]:
= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
= max(0.25; 0.5259/1.4506) if h/ho < 0.35, else max (0.25; 0.2734/0.7571)
= 0.3611

In this case, the hub ratio h/ho, 0.4020 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress Sbmax 507.5 MPa

Minimum permissible bolt stress	Sbmin	290.0	MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4	MPa
Target assembly gasket stress	SgT	172.0	MPa
Maximum permissible gasket stress	Sgmax	380.0	MPa
Minimum gasket operating stress	Sgmin-O	97.0	MPa
Minimum gasket seating stress	Sgmin-S	140.0	MPa
Nut Factor	K	0.200	
Fraction of Gasket Load remaining after relaxation	Phig	0.700	
Root Area of a Single Bolt	Abs	1862.7250	mm^2
Max. permissible single flange rotation for gasket	Thetagmax	1.000	deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0	MPa
Flange Material Operating Yield Stress	Syo	230.1	MPa
Single flange rotation at Sfmax	Thetafmax	0.336	deg

Gasket Contact Area [Ag]:
= π/4(GO.D.² - GI.D.²)
= π/4(826.0² - 794.0²)
= 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)
= SgT • Ag / (nb • Ab)
= 172.0 • 40714.922/(24 • 1862.725)
= 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
= min(Sbsel; Sbmax)
= min(156.647; 507.5)
= 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
= max(Sbsel; Sbmin)
= max(156.647; 290.0)
= 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
= min(Sbsel; Sfmax)
= min(290.0; 540.399)
= 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
:: Sbsel ≥ Sgmin-S(Ag / (nb • Abs))
:: 290.0 ≥ 140.0(40714.922/(24 • 1862.725))
:: 290.0 ≥ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
:: Sbsel ≥ (Sgmin-O • Ag + π/4 • Pmax • GI.D.²)/(φG • Abs • nb)
:: 290.0 ≥ (97.0 • 40714.922 + π/4 • 9.431 • 794.0²)/(0.7 • 1862.725 • 24)
:: 290.0 ≥ 275.4 MPa Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
:: Sbsel ≤ Sgmax(Ag / (nb • Abs))
:: 290.0 ≤ 380.0(40714.922/(24 • 1862.725))
:: 290.0 ≤ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
:: Sbsel ≤ Sfmax(thetagmax / thetafmax)
:: 290.0 ≤ 540.399(1.0/0.336)
:: 290.0 ≤ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 290.0 • 0.2 • 1862.725 • 56.0
= 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118	MPa
Bolt Allowable Stress for: SA-193 B7	Sb	171	MPa

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.021•202713•29.0²•149.238•0.3)
= 0.546 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.57948E+09/1.0•1.•0.327/(3.021•199226•29.0²•149.238•0.3)
= 0.435 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.73938E+09/1.0•1.•0.327/(3.631•202713•29.0²•149.238•0.3)
= 0.454 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.57948E+09/1.0•1.•0.327/(3.631•199226•29.0²•149.238•0.3)
= 0.362 (should be ≤ 1)

Minimum Design Metal Temperature Results:
Thickness Ratio = 0.868, Temperature Reduction per Fig. UCS 66.1 = 7 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B             6 °C
Min Metal Temp. at Required thickness (UCS 66.1)          -2 °C
```

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	103.97	207.00	130.78	207.00
Radial Flange	17.80	138.00	22.38	138.00
Tangential Flange	80.19	138.00	100.86	138.00
Maximum Average	92.08	138.00	115.82	138.00
Bolting	134.89	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

Flange Input Data Values **Description: FLANGE-7 :**

Flange-7

Flange Type		Integral Weld Neck	
Design Pressure	P	9.41	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1128.000	mm
Flange Thickness	t	178.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	g1	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1014.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		9228.11	N
Bending Moment on Flange		1188450.00	N-mm

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
= (P•(D/2+Ca))/(S•E-0.6•P) per UG-27 (c)(1)
= (9.41•(762.002/2+3.0))/(138.0•1.0-0.6•9.41)+Ca
= 30.2945 mm

Small End Hub MAWP:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 29.0)/(384.001 + 0.6 • 29.0)
= 9.970 MPa

Hub Small End Hub MAPnc:
= (Sa•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 32.0)/(381.001 + 0.6 • 32.0)
= 11.034 MPa

Corroded Flange ID,	Bcor = B+2*ci	768.002	mm
Corroded Large Hub,	g1Cor = g1-ci	45.000	mm
Corroded Small Hub,	g0Cor = go-ci	29.000	mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	77.999	mm

```
Gasket Contact Width,       N = (Go - Gi)/2               16.000 mm
Basic Gasket Width,        bo = N/2                        8.000 mm
Effective Gasket Width,     b = Cb*sqrt(bo)                7.127 mm
Gasket Reaction Diameter,   G = Go - 2 * b               811.745 mm
```

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
= π/4 • G² • P
= π/4 • 811.7452² • 9.408
= 4868198.500 N

Contact Load on Gasket Surfaces [Hp or Wo,2]:
= 2 • b • π • G • m • P
= 2 • 7.1274 • π • 811.7452 • 3.0 • 9.41
= 1025867.250 N

Hydrostatic End Load at Flange ID [Hd]:
= π/4 • Bcor² • P
= π/4 • 768.002² • 9.4075
= 4357662.500 N

Pressure Force on Flange Face [Ht]:
= H - Hd
= 4868199 - 4357663
= 510536.156 N

Design Bolt Load for the Operating Condition [Wo]:
= max(Wo,1 + Wo,2 + H'p, 0)
= max(4868199 + 1025867 + 0, 0)
= 5894066.000 N

Design Bolt Load for the Gasket Seating Condition [Wg]:
= y • b • π • G + yPart • bPart • lp
= 68.95•7.1274•π•811.745+0.0•0.0•0.0
= 1253136.250 N

Required Bolt Area [Am]:
= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
= max((5894066 + 9228 + 4•1188450/811.75)/172, 1253136/172)
= 34359.051 mm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
= 2a + 6t/(m + 0.5)
= 2 • 56.0 + 6 • 178.0/(3.0 + 0.5)
= 417.143 mm

Actual Circumferential Bolt Spacing [Bs]:
= C • sin(π / n)
= 1014.0 • sin(π/24)
= 132.354 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
= max($\sqrt{(\text{ Bs/(2a + t)}}$), 1)
= max($\sqrt{(\text{ 132.354/(2 • 56.0 + 178.0)}}$), 1)
= 1.0000

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	34359.051	44705.398	
Radial Distance between Hub and Bolts:	71.500	77.999	
Radial Distance between Bolts and Edge:	56.500	57.000	
Circ. Spacing between the Bolts:	124.700	132.354	417.143

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
= Ab • Sa/(y • π(Go + Gi))

= 44705.398 • 172.0/(68.95 • π(826.0 + 794.0))
= 21.912 mm[Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
= Sa • Ab
= 172.0 • 44705.3984
= 7688536.50 N

Gasket Load for the Operating Condition [HG]:
= Wo - H
= 5894066 - 4868199
= 1025867.25 N

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
= (C - G) / 2
= (1014.0 - 811.7452)/2
= 101.1274 mm

Distance to Face Pressure Reaction [ht]:
= 0.5((C - B)/2 + hg)
= 0.5((1014.0 - 768.002)/2 + 101.1274)
= 112.0632 mm

Distance to End Pressure Reaction [hd]:
= (C - B - g1)/2
= (1014.0 - 768.002 - 45.0)/2
= 100.4990 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:
Aa = 180.0, Bb = 178.0, Cc = 60.0 and Ddg = 37.0 mm

Compute the value of [Kab]:
= Aa • Bb^3 • (1/3 - 0.210 • (Bb/Aa) • (1-1/12 • $(Bb/Aa)^4$))
= 180.0 • 178.0^3 • (1/3 - 0.210 • (178.0/180.0) • (1-1/12 • $(178.0/180.0)^4$))
= 144369504.000 mm^4

Compute the value of [Kcd]:
= Cc • Ddg^3 • (1/3 - 0.105 • (Ddg/Cc) • (1-1/192 • $(Ddg/Cc)^4$))
= 60.0•37.0^3•(1/3-0.105•(37.0/60.0)•(1-1/192•$(37.0/60.0)^4$))
= 816420.500 mm^4

Flange Inertia Term [Ip,term1]:
= Kab + Kcd = 8809.977 + 49.821
= 145185920.000 mm^4

Flange Inertia Term [Ip,term2]:
= Ar • t^3(1/3 - 0.21(t/AR)(1 - 1/12$(t/AR)^4$))
= 180.0 • 178.0^3(1/3 - 0.21(178.0/180.0)(1 - 1/12$(178.0/7.087)^4$))
= 144369504.000 mm^4

Flange Inertia [Ip]:
= max(Ip,term1, Ip,term2)
= max(0.14519E+09, 0.14437E+09)
= 145185920.0 mm^4

Flange Moment of Inertia for Integral Types [I]:
= 0.0874 • L • go^2 • ho • B / V
= 0.0874 • 3.735 • 29.0^2 • 149.238 • 768.002/0.327
= 96253848.000 mm^4

Summary of Moments for Internal Pressure: (N-mm)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	4357662.	100.4990	1.0000	438118176.
Face Pressure,	Mt	510536.	112.0632	1.0000	57235512.
Gasket Load,	Mg	1025867.	101.1274	1.0000	103785384.
Gasket Seating,	Matm	7688536.	101.1274	1.0000	777837120.
Moment Load,	Moe	...	...	...	1299901.

Moment Value, eq. 4.16.16, [Moe]:
= 4Me[I/(0.384 • IP + I))•(hD /(C-2•hD)) + FA • hD
= 4•1188450[96253848/(0.384•.14519E+09+96253848))•(100.5/(1014.0-2•100.5))+
9228•100.499
= 1299900.875 N-mm

Total Moment for Operation, Mo 600438976. N-mm
Total Moment for Gasket seating, Mg 777837120. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
Hub Ratio, h/h0 = HL / H0 0.402
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:
Factor F per 4.16.5 0.860
Factor V per 4.16.5 0.327
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:
= A / B
= 1128.0/768.002
= 1.469

Stress factor from Table 4.16.4 [Y]:
= (0.66845 + (5.7169(K^2 • log10(K))) / (K^2 - 1))/(K - 1)
= (0.66845 + (5.7169(1.469^2 • log10(1.469)))/(1.469^2 - 1))/(1.469 - 1)
= 5.222

Stress factor from Table 4.16.4 [T]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K^2) • (K - 1))
= (1.469^2(1 + 8.55246 • log10(1.469)) - 1)/((1.0472 + 1.9448 • 1.469^2) • (1.469 - 1))
= 1.724

Stress factor from Table 4.16.4 [U]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/(1.36136(K^2 - 1) • (K - 1))
= (1.469^2(1 + 8.55246 • log10(1.469)) - 1)/(1.36136(1.469^2 - 1) • (1.469 - 1))
= 5.738

Stress factor from Table 4.16.4 [Z]:
= (K^2 + 1)/(K^2 - 1)
= (1.469^2 + 1)/(1.469^2 - 1)
= 2.728

Stress factor from Table 4.16.4 for integral flanges [d]:
= U • $g0^2$ • h0 / V
= 5.738 • 29.0^2 • 149.238/0.327
= 2203228.000 mm^3

Stress factor from Table 4.16.4 for integral flanges [e]:
= F / h0
= 0.86/149.238
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:
= (t • e + 1) / T + t^3/d
= (178.0 • 0.006 + 1)/1.724 + 178.0^3/2203228
= 3.735

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.0•.60044E+09/768.002)/(3.7346•45.0²)
= 103.35 MPa

Longitudinal Hub Stress, Seating [SHa]:
= (f • Matm / Bcor) / (L • g1²)
= (1.0•0.77784E+09/768.002)/(3.7346•45.0²)
= 133.88 MPa

Radial Flange Stress, Operating [SRo]:
= (β • Mop / Bcor) / (L • t²)
= (2.3678•.60044E+09/768.002)/(3.7346•178.0²)
= 15.64 MPa

Radial Flange Stress, Seating [SRa]:
= (β • Matm/Bcor) / (L • t²)
= (2.3678•.77784E+09/768.002)/(3.7346•178.0²)
= 20.26 MPa

Tangential Flange Stress, Operating [STo]:
= (Y • Mo / (t² • Bcor)) - Z • SRo
= (5.2216•.60044E+09/(178.0²•768.002))-2.7283•16
= 86.14 MPa

Tangential Flange Stress, Seating [STa]:
= (y • Matm / (t² • Bcor)) - Z • SRa
= (5.2216•.77784E+09/(178.0²•768.002))-2.7283•20
= 111.59 MPa

Average Flange Stress, Operating [SAo]:
= (SHo + max(SRo, STo)) / 2
= (103+max(16,86))/2
= 94.74 MPa

Average Flange Stress, Seating [SAa]:
= (SHa + max(SRa, STa)) / 2
= (134+max(20,112))/2
= 122.73 MPa

Bolt Stress, Operating [BSo]:
= (Wm1 + Fa + 4Me/G) / Ab
= 5909148/44705.3984
= 132.19 MPa

Bolt Stress, Seating [BSa]:
= (Wm2 / Ab)
= (1253136/44705.3984)
= 28.03 MPa

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	103.35	207.00	133.88	207.00
Radial Flange	15.64	138.00	20.26	138.00
Tangential Flange	86.14	138.00	111.59	138.00
Maximum Average	94.74	138.00	122.73	138.00
Bolting	132.19	172.00	28.03	172.00

Minimum Required Flange Thickness 166.218 mm

Estimated M.A.W.P. (Operating) 10 MPa
Estimated Finished Weight of Flange at given Thk. 796.4 kgm
Estimated Unfinished Weight of Forging at given Thk 1002.2 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Seating Actual	Allowed	stress ratio
Longitudinal Hub	Sh	409.05	499.99	0.818	409.05	499.99	0.818
Tangential Flange	St	340.94	374.99	0.909	340.94	374.99	0.909
Tangential Flange	Sto	147.70	250.00	0.591	147.70	250.00	0.591
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	402.84	749.99	0.537	402.84	749.99	0.537
Combined	Sh + Sto	556.75	749.99	0.742	556.75	749.99	0.742
Radial Flange	Sr	61.90	499.99	0.124	61.90	499.99	0.124
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	f	***f2***	***f3***
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c\left(\frac{g1}{g0}\right) + e\left(\frac{h}{h0}\right) + g\left(\frac{g1}{g0}\right)^2 + i\left(\frac{h}{h0}\right)^2 + k\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}{\left[1 + b\left(\frac{g1}{g0}\right) + d\left(\frac{h}{h0}\right) + f\left(\frac{g1}{g0}\right)^2 + h\left(\frac{h}{h0}\right)^2 + j\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}$$

Hub Stress Correction Factor [f]:
= max(1; F1num / F1den)
= max(1; 1.3704/1.4405)
= 1.0000

Hub Stress Correction Factor [f2]:
= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
= max(0.25; 0.5259/1.4506) if h/ho < 0.35, else max (0.25; 0.2734/0.7571)
= 0.3611

In this case, the hub ratio h/ho, 0.4020 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.4 MPa

Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	525.5 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at Sfmax	Thetafmax	0.334 deg

Gasket Contact Area [Ag]:
= π/4(GO.D.² - GI.D.²)
= π/4(826.0² - 794.0²)
= 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)
= SgT • Ag / (nb • Ab)
= 172.0 • 40714.922/(24 • 1862.725)
= 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
= min(Sbsel; Sbmax)
= min(156.647; 507.4)
= 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
= max(Sbsel; Sbmin)
= max(156.647; 290.0)
= 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
= min(Sbsel; Sfmax)
= min(290.0; 525.514)
= 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
:: Sbsel ≥ Sgmin-S(Ag / (nb • Abs))
:: 290.0 ≥ 140.0(40714.922/(24 • 1862.725))
:: 290.0 ≥ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
:: Sbsel ≥ (Sgmin-O • Ag + π/4 • Pmax • GI.D.²)/(φG • Abs • nb)
:: 290.0 ≥ (97.0 • 40714.922 + π/4 • 9.408 • 794.0²)/(0.7 • 1862.725 • 24)
:: 290.0 ≥ 275.1 MPa Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
:: Sbsel ≤ Sgmax(Ag / (nb • Abs))
:: 290.0 ≤ 380.0(40714.922/(24 • 1862.725))
:: 290.0 ≤ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
:: Sbsel ≤ Sfmax(thetagmax / thetafmax)
:: 290.0 ≤ 525.514(1.0/0.334)
:: 290.0 ≤ 1571.1 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 290.0 • 0.2 • 1862.725 • 56.0
= 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118 MPa
Bolt Allowable Stress for: SA-193 B7	Sb	171 MPa

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.77784E+09/1.0•1.•0.327/(3.219•202713•29.0²•149.238•0.3)
= 0.539 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.60044E+09/1.0•1.•0.327/(3.219•199226•29.0²•149.238•0.3)
= 0.423 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.77784E+09/1.0•1.•0.327/(3.735•202713•29.0²•149.238•0.3)
= 0.464 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.60044E+09/1.0•1.•0.327/(3.735•199226•29.0²•149.238•0.3)
= 0.365 (should be ≤ 1)

Minimum Design Metal Temperature Results:
Thickness Ratio = 0.866, Temperature Reduction per Fig. UCS 66.1 = 7 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B            6 °C
Min Metal Temp. at Required thickness (UCS 66.1)         -2 °C
```

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	104.77	207.00	133.92	207.00
Radial Flange	17.65	138.00	22.55	138.00
Tangential Flange	81.78	138.00	104.53	138.00
Maximum Average	93.28	138.00	119.22	138.00
Bolting	132.19	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 162.636 mm

Flange Input Data Values **Description: FLANGE-8 :**

Flange-8

Flange Type		Integral Weld Neck	
Design Pressure	P	9.41	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1128.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	36.0000	mm
Thickness of Hub at Large End	g1	54.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1014.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		2428.47	N
Bending Moment on Flange		555057.00	N-mm

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
= (P•(D/2+Ca))/(S•E-0.6•P) per UG-27 (c)(1)
= (9.41•(762.002/2+3.0))/(138.0•1.0-0.6•9.41)+Ca
= 30.2888 mm

Small End Hub MAWP:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 33.0)/(384.001 + 0.6 • 33.0)
= 11.278 MPa

Hub Small End Hub MAPnc:
= (Sa•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 36.0)/(381.001 + 0.6 • 36.0)
= 12.340 MPa

Corroded Flange ID,	Bcor = B+2*ci	768.002	mm
Corroded Large Hub,	g1Cor = g1-ci	51.000	mm
Corroded Small Hub,	g0Cor = go-ci	33.000	mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	71.999	mm

```
Gasket Contact Width,       N = (Go - Gi)/2            16.000 mm
Basic Gasket Width,        bo = N/2                      8.000 mm
Effective Gasket Width,     b = Cb*sqrt(bo)              7.127 mm
Gasket Reaction Diameter,   G = Go - 2 * b             811.745 mm
```

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
= π/4 • G² • P
= π/4 • 811.7452² • 9.406
= 4867228.000 N

Contact Load on Gasket Surfaces [Hp or Wo,2]:
= 2 • b • π • G • m • P
= 2 • 7.1274 • π • 811.7452 • 3.0 • 9.41
= 1025662.688 N

Hydrostatic End Load at Flange ID [Hd]:
= π/4 • Bcor² • P
= π/4 • 768.002² • 9.4057
= 4356793.500 N

Pressure Force on Flange Face [Ht]:
= H - Hd
= 4867228 - 4356794
= 510434.406 N

Design Bolt Load for the Operating Condition [Wo]:
= max(Wo,1 + Wo,2 + H'p, 0)
= max(4867228 + 1025663 + 0, 0)
= 5892890.500 N

Design Bolt Load for the Gasket Seating Condition [Wg]:
= y • b • π • G + yPart • bPart • lp
= 68.95•7.1274•π•811.745+0.0•0.0•0.0
= 1253136.250 N

Required Bolt Area [Am]:
= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
= max((5892891 + 2428 + 4•555057/811.75)/172, 1253136/172)
= 34294.539 mm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
= 2a + 6t/(m + 0.5)
= 2 • 56.0 + 6 • 176.0/(3.0 + 0.5)
= 413.714 mm

Actual Circumferential Bolt Spacing [Bs]:
= C • sin(π / n)
= 1014.0 • sin(π/24)
= 132.354 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
= max($\sqrt{\text{(Bs/(2a + t))}}$, 1)
= max($\sqrt{\text{(132.354/(2 • 56.0 + 176.0))}}$, 1)
= 1.0000

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	34294.539	44705.398	
Radial Distance between Hub and Bolts:	71.500	71.999	
Radial Distance between Bolts and Edge:	56.500	57.000	
Circ. Spacing between the Bolts:	124.700	132.354	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
= Ab • Sa/(y • π(Go + Gi))

= 44705.398 • 172.0/(68.95 • π(826.0 + 794.0))
= 21.912 mm[Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

```
Sheet Gaskets including laminated sheet gaskets with or without a metal core:     16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm
```

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
= Sa • Ab
= 172.0 • 44705.3984
= 7688536.50 N

Gasket Load for the Operating Condition [HG]:
= Wo - H
= 5892891 - 4867228
= 1025662.62 N

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
= (C - G) / 2
= (1014.0 - 811.7452)/2
= 101.1274 mm

Distance to Face Pressure Reaction [ht]:
= 0.5((C - B)/2 + hg)
= 0.5((1014.0 - 768.002)/2 + 101.1274)
= 112.0632 mm

Distance to End Pressure Reaction [hd]:
= (C - B - g1)/2
= (1014.0 - 768.002 - 51.0)/2
= 97.4990 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

```
Aa = 180.0, Bb = 176.0, Cc = 60.0 and Ddg = 42.0 mm
```

Compute the value of [Kab]:
= Aa • Bb^3 • (1/3 - 0.210 • (Bb/Aa) • (1-1/12 • (Bb/Aa)^4))
= 180.0 • 176.0^3 • (1/3 - 0.210 • (176.0/180.0) • (1-1/12 • (176.0/180.0)^4))
= 140955216.000 mm^4

Compute the value of [Kcd]:
= Cc • Ddg^3 • (1/3 - 0.105 • (Ddg/Cc) • (1-1/192 • (Ddg/Cc)^4))
= 60.0•42.0^3•(1/3-0.105•(42.0/60.0)•(1-1/192•(42.0/60.0)^4))
= 1155439.500 mm^4

Flange Inertia Term [Ip,term1]:
= Kab + Kcd = 8601.624 + 70.509
= 142110656.000 mm^4

Flange Inertia Term [Ip,term2]:
= Ar • t^3(1/3 - 0.21(t/AR)(1 - 1/12(t/AR)^4))
= 180.0 • 176.0^3(1/3 - 0.21(176.0/180.0)(1 - 1/12(176.0/7.087)^4))
= 140955216.000 mm^4

Flange Inertia [Ip]:
= max(Ip,term1, Ip,term2)
= max(0.14211E+09, 0.14096E+09)
= 142110656.0 mm^4

Flange Moment of Inertia for Integral Types [I]:
= 0.0874 • L • go^2 • ho • B / V
= 0.0874 • 2.975 • 33.0^2 • 159.198 • 768.002/0.336
= 103093784.000 mm^4

Summary of Moments for Internal Pressure: (N-mm)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	4356794.	97.4990	1.0000	424955168.
Face Pressure,	Mt	510434.	112.0632	1.0000	57224104.
Gasket Load,	Mg	1025663.	101.1274	1.0000	103764680.
Gasket Seating,	Matm	7688536.	101.1274	1.0000	777837120.
Moment Load,	Moe	...	...	...	409696.

Moment Value, eq. 4.16.16, [Moe]:
= 4Me[I/(0.384 • IP + I))•(hD /(C-2•hD)) + FA • hD
= 4•555057[.10309E+09/(0.384•.14211E+09+.10309E+09))•(97.5/(1014.0-2•97.5))+
2428•97.499
= 409696.500 N-mm

Total Moment for Operation, Mo 586353664. N-mm
Total Moment for Gasket seating, Mg 777837120. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 159.198 mm
Hub Ratio, h/h0 = HL / H0 0.377
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.545

Flange Factors for Integral Flange:
Factor F per 4.16.5 0.865
Factor V per 4.16.5 0.336
Factor f per 4.16.5 1.007

Stress factor from Table 4.16.4 [K]:
= A / B
= 1128.0/768.002
= 1.469

Stress factor from Table 4.16.4 [Y]:
= (0.66845 + (5.7169(K^2 • log10(K))) / (K^2 - 1))/(K - 1)
= (0.66845 + (5.7169(1.469^2 • log10(1.469)))/(1.469^2 - 1))/(1.469 - 1)
= 5.222

Stress factor from Table 4.16.4 [T]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K^2) • (K - 1))
= (1.469^2(1 + 8.55246 • log10(1.469)) - 1)/((1.0472 + 1.9448 • 1.469^2) • (1.469 - 1))
= 1.724

Stress factor from Table 4.16.4 [U]:
= (K^2(1 + 8.55246 • log10(K)) - 1)/(1.36136(K^2 - 1) • (K - 1))
= (1.469^2(1 + 8.55246 • log10(1.469)) - 1)/(1.36136(1.469^2 - 1) • (1.469 - 1))
= 5.738

Stress factor from Table 4.16.4 [Z]:
= (K^2 + 1)/(K^2 - 1)
= (1.469^2 + 1)/(1.469^2 - 1)
= 2.728

Stress factor from Table 4.16.4 for integral flanges [d]:
= U • $g0^2$ • h0 / V
= 5.738 • 33.0^2 • 159.198/0.336
= 2962546.250 mm^3

Stress factor from Table 4.16.4 for integral flanges [e]:
= F / h0
= 0.865/159.198
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:
= (t • e + 1) / T + t^3/d
= (176.0 • 0.005 + 1)/1.724 + 176.0^3/2962546
= 2.975

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.0068•.58635E+09/768.002)/(2.9748•51.0²)
= 99.31 MPa

Longitudinal Hub Stress, Seating [SHa]:
= (f • Matm / Bcor) / (L • g1²)
= (1.0068•0.77784E+09/768.002)/(2.9748•51.0²)
= 131.74 MPa

Radial Flange Stress, Operating [SRo]:
= (β • Mop / Bcor) / (L • t²)
= (2.2751•.58635E+09/768.002)/(2.9748•176.0²)
= 18.84 MPa

Radial Flange Stress, Seating [SRa]:
= (β • Matm/Bcor) / (L • t²)
= (2.2751•.77784E+09/768.002)/(2.9748•176.0²)
= 25.00 MPa

Tangential Flange Stress, Operating [STo]:
= (Y • Mo / (t² • Bcor)) - Z • SRo
= (5.2216•.58635E+09/(176.0²•768.002))-2.7283•19
= 77.25 MPa

Tangential Flange Stress, Seating [STa]:
= (y • Matm / (t² • Bcor)) - Z • SRa
= (5.2216•.77784E+09/(176.0²•768.002))-2.7283•25
= 102.47 MPa

Average Flange Stress, Operating [SAo]:
= (SHo + max(SRo, STo)) / 2
= (99+max(19,77))/2
= 88.28 MPa

Average Flange Stress, Seating [SAa]:
= (SHa + max(SRa, STa)) / 2
= (132+max(25,102))/2
= 117.11 MPa

Bolt Stress, Operating [BSo]:
= (Wm1 + Fa + 4Me/G) / Ab
= 5898053/44705.3984
= 131.94 MPa

Bolt Stress, Seating [BSa]:
= (Wm2 / Ab)
= (1253136/44705.3984)
= 28.03 MPa

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	99.31	207.00	131.74	207.00
Radial Flange	18.84	138.00	25.00	138.00
Tangential Flange	77.25	138.00	102.47	138.00
Maximum Average	88.28	138.00	117.11	138.00
Bolting	131.94	172.00	28.03	172.00

```
Minimum Required Flange Thickness                    157.988 mm

Estimated M.A.W.P. ( Operating )                         11 MPa
Estimated Finished Weight of Flange at given Thk.     794.1 kgm
Estimated Unfinished Weight of Forging at given Thk   993.7 kgm
```

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating Actual	Operating Allowed	stress ratio	Gasket Seating Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	425.21	499.99	0.850	425.21	499.99	0.850
Tangential Flange	St	329.38	374.99	0.878	329.38	374.99	0.878
Tangential Flange	Sto	151.84	250.00	0.607	151.84	250.00	0.607
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	409.73	749.99	0.546	409.73	749.99	0.546
Combined	Sh + Sto	577.04	749.99	0.769	577.04	749.99	0.769
Radial Flange	Sr	80.35	499.99	0.161	80.35	499.99	0.161
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	f	f2	f3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c\left(\frac{g1}{g0}\right) + e\left(\frac{h}{h0}\right) + g\left(\frac{g1}{g0}\right)^2 + i\left(\frac{h}{h0}\right)^2 + k\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}{\left[1 + b\left(\frac{g1}{g0}\right) + d\left(\frac{h}{h0}\right) + f\left(\frac{g1}{g0}\right)^2 + h\left(\frac{h}{h0}\right)^2 + j\left(\frac{g1}{g0}\right)\left(\frac{h}{h0}\right)\right]}$$

Hub Stress Correction Factor [f]:
= max(1; F1num / F1den)
= max(1; 1.4045/1.3893)
= 1.0109

Hub Stress Correction Factor [f2]:
= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
= max(0.25; 0.4957/1.3795) if h/ho < 0.35, else max (0.25; 0.2738/0.7584)
= 0.3610

In this case, the hub ratio h/ho, 0.3769 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress Sbmax 507.5 MPa

Minimum permissible bolt stress	Sbmin	290.0	MPa
Max. permissible bolt stress prior to flange damage	Sfmax	552.9	MPa
Target assembly gasket stress	SgT	172.0	MPa
Maximum permissible gasket stress	Sgmax	380.0	MPa
Minimum gasket operating stress	Sgmin-O	97.0	MPa
Minimum gasket seating stress	Sgmin-S	140.0	MPa
Nut Factor	K	0.200	
Fraction of Gasket Load remaining after relaxation	Phig	0.700	
Root Area of a Single Bolt	Abs	1862.7250	mm²
Max. permissible single flange rotation for gasket	Thetagmax	1.000	deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0	MPa
Flange Material Operating Yield Stress	Syo	230.1	MPa
Single flange rotation at Sfmax	Thetafmax	0.338	deg

Gasket Contact Area [Ag]:
= π/4(GO.D.² - GI.D.²)
= π/4(826.0² - 794.0²)
= 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)
= SgT • Ag / (nb • Ab)
= 172.0 • 40714.922/(24 • 1862.725)
= 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
= min(Sbsel; Sbmax)
= min(156.647; 507.5)
= 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
= max(Sbsel; Sbmin)
= max(156.647; 290.0)
= 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
= min(Sbsel; Sfmax)
= min(290.0; 552.851)
= 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
:: Sbsel ≥ Sgmin-S(Ag / (nb • Abs))
:: 290.0 ≥ 140.0(40714.922/(24 • 1862.725))
:: 290.0 ≥ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
:: Sbsel ≥ (Sgmin-O • Ag + π/4 • Pmax • GI.D.²)/(φG • Abs • nb)
:: 290.0 ≥ (97.0 • 40714.922 + π/4 • 9.406 • 794.0²)/(0.7 • 1862.725 • 24)
:: 290.0 ≥ 275.0 MPa Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
:: Sbsel ≤ Sgmax(Ag / (nb • Abs))
:: 290.0 ≤ 380.0(40714.922/(24 • 1862.725))
:: 290.0 ≤ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
:: Sbsel ≤ Sfmax(thetagmax / thetafmax)
:: 290.0 ≤ 552.851(1.0/0.338)
:: 290.0 ≤ 1635.3 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 290.0 • 0.2 • 1862.725 • 56.0
= 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118	MPa
Bolt Allowable Stress for: SA-193 B7	Sb	171	MPa

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.77784E+09/1.0•1.•0.336/(2.409•202713•33.0²•159.198•0.3)
= 0.536 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.58635E+09/1.0•1.•0.336/(2.409•199226•33.0²•159.198•0.3)
= 0.411 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14•0.77784E+09/1.0•1.•0.336/(2.975•202713•33.0²•159.198•0.3)
= 0.434 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14•0.58635E+09/1.0•1.•0.336/(2.975•199226•33.0²•159.198•0.3)
= 0.332 (should be ≤ 1)

Minimum Design Metal Temperature Results:
Thickness Ratio = 0.779, Temperature Reduction per Fig. UCS 66.1 = 12 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B             9 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit       -8 °C
Min Metal Temp. at Required thickness (UCS 66.1)         -20 °C
```

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Longitudinal Hub	99.15	207.00	129.70	207.00
Radial Flange	20.84	138.00	27.26	138.00
Tangential Flange	73.16	138.00	95.70	138.00
Maximum Average	86.16	138.00	112.70	138.00
Bolting	131.94	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 152.629 mm

HEXAGON

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2		...	16	1.6	813	...	1
Skirt-1		...	16	1.6	813	...	2
Bottom head		9.5357	36	3	762	138	3
Shell-1		9.5335	32	3	762	138	4
Flange-1		9.5155	178	3	762	138	5
Flange-2		9.5135	178	3	762	138	6
Shell-2		9.5115	32	3	762	138	7
Shell-3		9.4844	32	3	762	138	8
Flange-3		9.4741	178	3	762	138	9
Flange-4		9.4721	178	3	762	138	10
Shell-4		9.4702	32	3	826	138	11
Shell-5		9.4431	32	3	762	138	12
Flange-5		9.4329	178	3	762	138	13
Flange-6		9.4311	178	3	762	138	14
Shell-6		9.4292	32	3	762	138	15
Flange-7		9.4075	180	3	762	138	16
Flange-8		9.4057	178	3	762	138	17
Top head		9.4051	36	3	762	138	18

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2		...	No Calc	No Calc	16	No Calc	1
Skirt-1		...	No Calc	No Calc	16	No Calc	2
Bottom head		9.4029	9.60193	10.781	30	29.4441	3
Shell-1		9.4029	9.8393	11.0343	32	30.6757	4
Flange-1		9.4029	9.85733	10.7095	176	161.468	5
Flange-2		9.4029	9.85932	10.7596	176	161.468	6
Shell-2		9.4029	9.86132	11.0343	32	30.609	7
Shell-3		9.4029	9.88837	11.0343	32	30.5271	8
Flange-3		9.4029	9.89874	11.0343	176	161.468	9
Flange-4		9.4029	9.90069	11.0343	176	161.468	10
Shell-4		9.4029	9.90264	11.0343	32	30.5852	11
Shell-5		9.4029	9.9297	11.0343	32	30.4021	12
Flange-5		9.4029	9.93987	11.0343	176	161.468	13
Flange-6		9.4029	9.94174	11.0343	176	161.468	14
Shell-6		9.4029	9.94362	11.0343	32	30.3599	15
Flange-7		9.4029	9.96526	11.0343	178	166.218	16
Flange-8		9.4029	11.2749	12.2635	176	157.988	17
Top head		9.4029	9.55396	10.6027	29.5	29.0782	18
Minimum			10	11			

MAWP: 10 MPa, limited by: Top head.

Elements Suitable for Design Internal Pressure.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Bottom head

Elliptical Head Design Information:				
Design Pressure P	9.536 MPa	Int. Design Temperature	82	°C
Material	SA-516 70	Ext. Design Temperature	82	°C
Length	50.000 mm	External Pressure Chart	CS-2	
		UNS Number	K02700	
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0	MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0	MPa
Inside Diameter Di	762.002 mm	Actual Stress	135.2	MPa
Spec. Min. Thk t,e, or sa	30.0000 mm	Spec. Nominal Thk.	36.000	mm
Surface Area	884980.8 mm²	SG of Contents	0.920	
Element Volume	80.7 ltr	Weight of Contents	727.9	N
Empty Weight	2428.3 N	Operating Weight	3156.2	N
		Comp. All. Ope. Stress	147.3	MPa
		Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.525 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.763 mm

Required Thickness due to Internal Pressure [tr]:
= (P•D•Kcor)/(2•S•E-0.2•P) Appendix 1-4(c)
= (9.536•768.002•0.99)/(2•138.0•1.0-0.2•9.536)
= 26.4441 + 3.0000 = 29.4441 mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 0.133 MPa
= (2•S•E•t)/(Kcor•D+0.2•t) per Appendix 1-4 (c)
= (2•138.0•1.0•27.0)/(0.99•768.002+0.2•27.0)
= 10 - 0 = 10 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:
= (2•S•E•t)/(K•D+0.2•t) per Appendix 1-4 (c)
= (2•138.0•1.0•30.0)/(1.0•762.002+0.2•30.0)
= 11 MPa

Actual stress at given pressure and thickness, corroded [Sact]:
= (P•(Kcor•D+0.2•t))/(2•E•t)
= (9.536•(0.99•768.002+0.2•27.0))/(2•1.0•27.0)
= 135.176 MPa

Straight Flange Required Thickness:
= (P•R)/(S•E-0.6•P) + c per UG-27 (c)(1)
= (9.536•384.001)/(138.0•1.0-0.6•9.536)+3.0
= 30.682 mm

Straight Flange Maximum Allowable Working Pressure:
Less Operating Hydrostatic Head Pressure of 0.131 MPa
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 33.0)/(384.001 + 0.6 • 33.0)
= 11 - 0 = 11 MPa

Factor K, corroded condition [Kcor]:
= (2 + (Inside Diameter/(2 • Inside Head Depth))²)/6

= (2 + (768.002/(2 • 193.501))²)/6
= 0.989704

Percent Elong. per UCS-79, VIII-1-01-57 (75*tnom/Rf)*(1-Rf/Ro) 18.300 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 30.0, tr = 26.866, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.995, Temp. Reduction = 0 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B            4 °C
Min Metal Temp. at Required thickness (UCS 66.1)          4 °C
```

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 36.0, tr = 28.14, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.853, Temp. Reduction = 8 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B            9 °C
Min Metal Temp. at Required thickness (UCS 66.1)          1 °C
```

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.687 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	36.000 mm	Basic MDMT	9 °C
Stress/Thk. Ratio	0.853	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	1 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

```
Head inner surface maximum deviation outside the specified shape,  1.25% of D: 9.525 mm
Head inner surface maximum deviation inside the specified shape,  0.625% of D: 4.763 mm
```

EXTERNAL PRESSURE RESULTS for: Bottom head

<u>Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 82 °C</u>

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
27.000	822.00	30.44	0.0045620	118.28

MAEP = B/(K0*Do/t) = 118.277/(0.9 *30.4445) = 4.3167 MPa

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.128	822.00	386.22	0.0003596	35.95

MAEP = B/(K0*Do/t) = 35.9531/(0.9 *386.2185) = 0.1034 MPa

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	9.534 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		2000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		132.0 MPa
Spec. Min. Thk t,e, or sa		32.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		5189924.0 mm²	SG of Contents		0.920
Element Volume		912.2 ltr	Weight of Contents		8224.5 N
Empty Weight		12133.1 N	Operating Weight		20357.7 N
			Comp. All. Ope. Stress		147.3 MPa

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:
= (P•R)/(S•E-0.6•P) per UG-27 (c)(1)
= (9.534•384.001)/(138.0•1.0-0.6•9.534)
= 27.6757 + 3.0000 = 30.6757 mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 0.131 MPa
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•29.0)/(384.001+0.6•29.0)
= 10 - 0 = 10 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•32.0)/(381.001+0.6•32.0)
= 11 MPa

Actual stress at given pressure and thickness, corroded [Sact]:
= (P•(R+0.6•t))/(E•t)
= (9.534•(384.001+0.6•29.0))/(1.0•29.0)
= 131.957 MPa

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 32.0, tr = 28.134, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.97, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	4 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.685 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.970	Temperature Difference	2 °C

UCS-66(c) red.	No	Computed Min. Temp.	4 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	2113.50	28.48	2.5587	0.0031987	113.02

MAEP = (4*B)/(3*(Do/t)) = (4*113.0161)/(3*28.4828) = 5.2905 MPa

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.919	826.00	2113.50	210.78	2.5587	0.0001635	16.35

MAEP = (4*B)/(3*(Do/t)) = (4*16.351)/(3*210.7801) = 0.1034 MPa

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	98941.76	28.48	50.0000	0.0013690	94.37

MAEP = (4*B)/(3*(Do/t)) = (4*94.3701)/(3*28.4828) = 4.4176 MPa

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:			
Design Pressure P	9.511 MPa	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	3000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	762.002 mm	Actual Stress	131.7 MPa
Spec. Min. Thk t,e, or sa	32.0000 mm	Spec. Nominal Thk.	32.000 mm
Surface Area	7784886.0 mm²	SG of Contents	0.920
Element Volume	1368.4 ltr	Weight of Contents	12336.8 N
Empty Weight	18199.7 N	Operating Weight	30536.5 N
		Comp. All. Ope. Stress	147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:
= (P•R)/(S•E-0.6•P) per UG-27 (c)(1)
= (9.511•384.001)/(138.0•1.0-0.6•9.511)
= 27.6090 + 3.0000 = 30.6090 mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 0.109 MPa
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•29.0)/(384.001+0.6•29.0)
= 10 - 0 = 10 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•32.0)/(381.001+0.6•32.0)
= 11 MPa

Actual stress at given pressure and thickness, corroded [Sact]:
= (P•(R+0.6•t))/(E•t)
= (9.511•(384.001+0.6•29.0))/(1.0•29.0)
= 131.652 MPa

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 32.0, tr = 28.067, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.968, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.663 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.968	Temperature Difference	2 °C

UCS-66(c) red.	No	Computed Min. Temp.	4 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	4150.00	28.48	5.0242	0.0016252	98.83

MAEP = (4*B)/(3*(Do/t)) = (4*98.8288)/(3*28.4828) = 4.6264 MPa

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.227	826.00	4150.00	158.03	5.0242	0.0001226	12.26

MAEP = (4*B)/(3*(Do/t)) = (4*12.2593)/(3*158.0346) = 0.1034 MPa

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	181399.97	28.48	50.0000	0.0013690	94.37

MAEP = (4*B)/(3*(Do/t)) = (4*94.3701)/(3*28.4828) = 4.4176 MPa

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:			
Design Pressure P	9.484 MPa	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1150.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	762.002 mm	Actual Stress	131.3 MPa
Spec. Min. Thk t,e, or sa	32.0000 mm	Spec. Nominal Thk.	32.000 mm
Surface Area	2984206.2 mm²	SG of Contents	0.920
Element Volume	524.5 ltr	Weight of Contents	4729.1 N
Empty Weight	6976.5 N	Operating Weight	11705.7 N
		Comp. All. Ope. Stress	147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:
= (P•R)/(S•E-0.6•P) per UG-27 (c)(1)
= (9.484•384.001)/(138.0•1.0-0.6•9.484)
= 27.5271 + 3.0000 = 30.5271 mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 0.082 MPa
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•29.0)/(384.001+0.6•29.0)
= 10 - 0.1 = 10 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•32.0)/(381.001+0.6•32.0)
= 11 MPa

Actual stress at given pressure and thickness, corroded [Sact]:
= (P•(R+0.6•t))/(E•t)
= (9.484•(384.001+0.6•29.0))/(1.0•29.0)
= 131.278 MPa

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 32.0, tr = 27.985, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.965, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	4 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.635 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.965	Temperature Difference	2 °C

UCS-66(c) red.	No	Computed Min. Temp.	4 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	4150.00	28.48	5.0242	0.0016252	98.83

MAEP = (4*B)/(3*(Do/t)) = (4*98.8288)/(3*28.4828) = 4.6264 MPa

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.227	826.00	4150.00	158.03	5.0242	0.0001226	12.26

MAEP = (4*B)/(3*(Do/t)) = (4*12.2593)/(3*158.0346) = 0.1034 MPa

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	181399.97	28.48	50.0000	0.0013690	94.37

MAEP = (4*B)/(3*(Do/t)) = (4*94.3701)/(3*28.4828) = 4.4176 MPa

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure P	9.470 MPa	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	3000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Outside Diameter Do	826.002 mm	Actual Stress	131.1 MPa
Spec. Min. Thk t,e, or sa	32.0000 mm	Spec. Nominal Thk.	32.000 mm
Surface Area	7784886.0 mm²	SG of Contents	0.920
Element Volume	1368.4 ltr	Weight of Contents	12336.8 N
Empty Weight	18199.7 N	Operating Weight	30536.5 N
		Comp. All. Ope. Stress	147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:
= (P•Ro) / (S•E+0.4•P) per Appendix 1-1 (a)(1)
= (9.47•413.001)/(138.0•1.0+0.4•9.47)
= 27.5852 + 3.0000 = 30.5852 mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 0.067 MPa
= (S•E•t)/(Ro-0.4•t) per Appendix 1-1 (a)(1)
= (138.0•1.0•29.0)/(413.001-0.4•29.0)
= 10 - 0.1 = 10 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:
= (S•E•t)/(Ro-0.4•t) per Appendix 1-1 (a)(1)
= (138.0•1.0•32.0)/(413.001-0.4•32.0)
= 11 MPa

Actual stress at given pressure and thickness, corroded [Sact]:
= (P•(Ro-0.4•t))/(E•t)
= (9.47•(413.001-0.4•29.0))/(1.0•29.0)
= 131.080 MPa

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 32.0, tr = 28.013, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.966, Temp. Reduction = 2 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B          6 °C
Min Metal Temp. at Required thickness (UCS 66.1)        4 °C
```

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.621 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.966	Temperature Difference	2 °C

UCS-66(c) red.	No	Computed Min. Temp.	4 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	4128.00	28.48	4.9976	0.0016289	98.89

MAEP = (4*B)/(3*(Do/t)) = (4*98.8856)/(3*28.4828) = 4.629 MPa

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.214	826.00	4128.00	158.43	4.9976	0.0001229	12.29

MAEP = (4*B)/(3*(Do/t)) = (4*12.2901)/(3*158.4315) = 0.1034 MPa

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	180490.12	28.48	50.0000	0.0013690	94.37

MAEP = (4*B)/(3*(Do/t)) = (4*94.3701)/(3*28.4828) = 4.4176 MPa

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	9.443 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		1128.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		130.7 MPa
Spec. Min. Thk t,e, or sa		32.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		2927117.0 mm²	SG of Contents		0.920
Element Volume		514.5 ltr	Weight of Contents		4638.6 N
Empty Weight		6843.1 N	Operating Weight		11481.7 N
			Comp. All. Ope. Stress		147.3 MPa

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:
= (P•R)/(S•E-0.6•P) per UG-27 (c)(1)
= (9.443•384.001)/(138.0•1.0-0.6•9.443)
= 27.4021 + 3.0000 = 30.4021 mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 0.040 MPa
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•29.0)/(384.001+0.6•29.0)
= 10 - 0 = 10 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•32.0)/(381.001+0.6•32.0)
= 11 MPa

Actual stress at given pressure and thickness, corroded [Sact]:
= (P•(R+0.6•t))/(E•t)
= (9.443•(384.001+0.6•29.0))/(1.0•29.0)
= 130.706 MPa

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 32.0, tr = 27.86, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.961, Temp. Reduction = 2 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B        6 °C
Min Metal Temp. at Required thickness (UCS 66.1)      4 °C
```

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.594 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.961	Temperature Difference	2 °C

UCS-66(c) red.	No	Computed Min. Temp.	4 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 120 to 130 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	4128.00	28.48	4.9976	0.0016289	98.89

MAEP = (4*B)/(3*(Do/t)) = (4*98.8856)/(3*28.4828) = 4.629 MPa

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.214	826.00	4128.00	158.43	4.9976	0.0001229	12.29

MAEP = (4*B)/(3*(Do/t)) = (4*12.2901)/(3*158.4315) = 0.1034 MPa

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	180490.12	28.48	50.0000	0.0013690	94.37

MAEP = (4*B)/(3*(Do/t)) = (4*94.3701)/(3*28.4828) = 4.4176 MPa

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:			
Design Pressure P	9.429 MPa	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	2400.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	762.002 mm	Actual Stress	130.5 MPa
Spec. Min. Thk t,e, or sa	32.0000 mm	Spec. Nominal Thk.	32.000 mm
Surface Area	6227909.0 mm²	SG of Contents	0.920
Element Volume	1094.7 ltr	Weight of Contents	9869.4 N
Empty Weight	14559.8 N	Operating Weight	24429.2 N
		Comp. All. Ope. Stress	147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:
= (P•R)/(S•E-0.6•P) per UG-27 (c)(1)
= (9.429•384.001)/(138.0•1.0-0.6•9.429)
= 27.3599 + 3.0000 = 30.3599 mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 0.026 MPa
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•29.0)/(384.001+0.6•29.0)
= 10 - 0 = 10 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0•1.0•32.0)/(381.001+0.6•32.0)
= 11 MPa

Actual stress at given pressure and thickness, corroded [Sact]:
= (P•(R+0.6•t))/(E•t)
= (9.429•(384.001+0.6•29.0))/(1.0•29.0)
= 130.513 MPa

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 32.0, tr = 27.817, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.959, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	3 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.580 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.959	Temperature Difference	2 °C

UCS-66(c) red.	No	Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-6

Cylindrical Shell From 150 to 160 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	2400.00	28.48	2.9056	0.0027737	110.48

MAEP = (4*B)/(3*(Do/t)) = (4*110.4818)/(3*28.4828) = 5.1719 MPa

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.145	826.00	2400.00	199.29	2.9056	0.0001546	15.46

MAEP = (4*B)/(3*(Do/t)) = (4*15.4589)/(3*199.2896) = 0.1034 MPa

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	111047.70	28.48	50.0000	0.0013690	94.37

MAEP = (4*B)/(3*(Do/t)) = (4*94.3701)/(3*28.4828) = 4.4176 MPa

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Top head

Elliptical Head Design Information:				
Design Pressure P	9.405 MPa	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	762.002 mm	Actual Stress	135.8 MPa
Spec. Min. Thk t,e, or sa	29.5000 mm	Spec. Nominal Thk.	36.000 mm
Surface Area	884980.8 mm²	SG of Contents	0.920
Element Volume	80.7 ltr	Weight of Contents	727.9 N
Empty Weight	2428.3 N	Operating Weight	3156.2 N
		Comp. All. Ope. Stress	147.3 MPa
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.525 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.763 mm

Required Thickness due to Internal Pressure [tr]:
= (P•D•Kcor)/(2•S•E-0.2•P) Appendix 1-4(c)
= (9.405•768.002•0.99)/(2•138.0•1.0-0.2•9.405)
= 26.0782 + 3.0000 = 29.0782 mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 0.002 MPa
= (2•S•E•t)/(Kcor•D+0.2•t) per Appendix 1-4 (c)
= (2•138.0•1.0•26.5)/(0.99•768.002+0.2•26.5)
= 10 - 0 = 10 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:
= (2•S•E•t)/(K•D+0.2•t) per Appendix 1-4 (c)
= (2•138.0•1.0•29.5)/(1.0•762.002+0.2•29.5)
= 11 MPa

Actual stress at given pressure and thickness, corroded [Sact]:
= (P•(Kcor•D+0.2•t))/(2•E•t)
= (9.405•(0.99•768.002+0.2•26.5))/(2•1.0•26.5)
= 135.816 MPa

Straight Flange Required Thickness:
= (P•R)/(S•E-0.6•P) + c per UG-27 (c)(1)
= (9.405•384.001)/(138.0•1.0-0.6•9.405)+3.0
= 30.286 mm

Straight Flange Maximum Allowable Working Pressure:
Less Operating Hydrostatic Head Pressure of 0.002 MPa
= (S•E•t)/(R+0.6•t) per UG-27 (c)(1)
= (138.0 • 1.0 • 33.0)/(384.001 + 0.6 • 33.0)
= 11 - 0 = 11 MPa

Factor K, corroded condition [Kcor]:
= (2 + (Inside Diameter/(2 • Inside Head Depth))²)/6

= (2 + (768.002/(2 • 193.501))²)/6
= 0.989704

Percent Elong. per UCS-79, VIII-1-01-57 (75*tnom/Rf)*(1-Rf/Ro) 18.300 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 29.5, tr = 26.501, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 1.0, Temp. Reduction = 0 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B              4 °C
```

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 36.0, tr = 27.744, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.841, Temp. Reduction = 9 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B              9 °C
Min Metal Temp. at Required thickness (UCS 66.1)            0 °C
```

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.556 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	36.000 mm	Basic MDMT	9 °C
Stress/Thk. Ratio	0.841	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	0 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

```
Head inner surface maximum deviation outside the specified shape,  1.25% of D: 9.525 mm
Head inner surface maximum deviation inside the specified shape,  0.625% of D: 4.763 mm
```

EXTERNAL PRESSURE RESULTS for: Top head

Elliptical Head From 180 to 190 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
26.500	821.00	30.98	0.0044830	118.05

```
MAEP = B/(K0*Do/t) = 118.0531/( 0.9 *30.9812 ) = 4.2339 MPa
```

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.126	821.00	386.22	0.0003596	35.95

```
MAEP = B/(K0*Do/t) = 35.9531/( 0.9 *386.2188 ) = 0.1034 MPa
```

Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	12.420 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	12.224 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13.776 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	10.509 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	13.446 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	13.446 MPa
App 27-4	= M.A.W.P.	9.554 MPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	176.4	234.0	0.754	13.78
Shell-1	172.4	234.0	0.737	13.78
Shell-2	172.4	234.0	0.737	13.78
Shell-3	172.4	234.0	0.737	13.78
Shell-4	172.4	234.0	0.737	13.78
Shell-5	172.4	234.0	0.737	13.78
Shell-6	172.4	234.0	0.737	13.78
Top head	179.4	234.0	0.767	13.78

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4_80	Nozzle	138.00	138.00	1.000
N1_150	Nozzle	138.00	138.00	1.000
K1_50	Nozzle	138.00	138.00	1.000
K2_50	Nozzle	138.00	138.00	1.000
K3_50	Nozzle	138.00	138.00	1.000
H2_200	Nozzle	138.00	138.00	1.000
H1_200	Nozzle	138.00	138.00	1.000
N2_80	Nozzle	138.00	138.00	1.000
K4_50	Nozzle	138.00	138.00	1.000
N3_150	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Flange-3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000
Flange-7	138.00	138.00	1.000
Flange-8	138.00	138.00	1.000
Top head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4_80	49.56	225.00	0.220
N1_150	115.27	225.00	0.512
K1_50	21.27	225.00	0.095
K2_50	21.27	225.00	0.095
K3_50	21.27	225.00	0.095
H2_200	88.97	225.00	0.395
H1_200	77.16	225.00	0.343
N2_80	57.43	225.00	0.255
K4_50	21.27	225.00	0.095
N3_150	115.27	225.00	0.512

Elements Suitable for Test Pressure.

Hydrostatic Test Pressure Results:

```
UG99(b)          = 1.30 * M.A.W.P. * Sa/S               12.420 MPa
UG99b(b)(35)     = 1.30 * Design Pres * Sa/S            12.224 MPa
UG99(c)          = 1.30 * M.A.P. - Head(Hyd)            13.781 MPa
UG100            = 1.10 * M.A.W.P. * Sa/S               10.509 MPa
PED              = max(1.43*DP, 1.25*DP*ratio)          13.446 MPa
PED&UG99(b)(35)  = max(1.43*DP, 1.3*DP*ratio)           13.446 MPa
App 27-4         = M.A.W.P.                              9.554 MPa
```

UG-99(b) Note 35, Test Pressure Calculation:
= Test Factor • Design Pressure • Stress Ratio
= 1.3 • 9.403 • 1.0
= 12.224 MPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	175.4	234.0	0.749	12.37
Shell-1	171.2	234.0	0.732	12.37
Shell-2	170.9	234.0	0.730	12.34
Shell-3	170.5	234.0	0.728	12.32
Shell-4	170.2	234.0	0.728	12.30
Shell-5	169.8	234.0	0.726	12.27
Shell-6	169.6	234.0	0.725	12.25
Top head	176.6	234.0	0.755	12.23

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4_80	Nozzle	138.00	138.00	1.000
N1_150	Nozzle	138.00	138.00	1.000
K1_50	Nozzle	138.00	138.00	1.000
K2_50	Nozzle	138.00	138.00	1.000
K3_50	Nozzle	138.00	138.00	1.000
H2_200	Nozzle	138.00	138.00	1.000
H1_200	Nozzle	138.00	138.00	1.000
N2_80	Nozzle	138.00	138.00	1.000
K4_50	Nozzle	138.00	138.00	1.000
N3_150	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Flange-3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000

Flange-7	138.00	138.00	1.000
Flange-8	138.00	138.00	1.000
Top head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4_80	44.48	225.00	0.198
N1_150	103.44	225.00	0.460
K1_50	19.09	225.00	0.085
K2_50	19.09	225.00	0.085
K3_50	19.09	225.00	0.085
H2_200	79.68	225.00	0.354
H1_200	68.85	225.00	0.306
N2_80	51.05	225.00	0.227
K4_50	18.91	225.00	0.084
N3_150	102.25	225.00	0.454

Elements Suitable for Test Pressure.

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	3200	...	...	No Calc	No Calc
20	30	3200	...	...	No Calc	No Calc
30	40	No Calc	822.002	27	0.004562	118.277
40	50	2113.5	826.002	29	0.0031987	113.016
50	60	No Calc	...	173	No Calc	No Calc
60	70	No Calc	...	173	No Calc	No Calc
70	80	4150	826.002	29	0.0016252	98.8288
80	90	4150	826.002	29	0.0016252	98.8288
90	100	No Calc	...	173	No Calc	No Calc
100	110	No Calc	...	173	No Calc	No Calc
110	120	4128	826.002	29	0.0016289	98.8856
120	130	4128	826.002	29	0.0016289	98.8856
130	140	No Calc	...	173	No Calc	No Calc
140	150	No Calc	...	173	No Calc	No Calc
150	160	2400	826.002	29	0.0027737	110.482
160	170	No Calc	...	175	No Calc	No Calc
170	180	No Calc	...	173	No Calc	No Calc
180	190	No Calc	821.002	26.5	0.004483	118.053

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	30	5.12833	0.10343	4.31667
40	50	32	6.91879	0.10343	5.29049
50	60	176	161.468	0.10343	No Calc
60	70	176	161.468	0.10343	No Calc
70	80	32	8.22672	0.10343	4.62636
80	90	32	8.22672	0.10343	4.62636
90	100	176	161.468	0.10343	No Calc
100	110	176	161.468	0.10343	No Calc
110	120	32	8.21362	0.10343	4.62901
120	130	32	8.21362	0.10343	4.62901
130	140	176	161.468	0.10343	No Calc
140	150	176	161.468	0.10343	No Calc
150	160	32	7.14473	0.10343	5.17185
160	170	178	166.218	0.10343	No Calc
170	180	176	157.988	0.10343	No Calc
180	190	29.5	5.12574	0.10343	4.23386
Minimum					4

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	3200	No Calc	No Calc	No Calc
20	30	3200	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	2113.5	98941.8	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

HEXAGON

70	80	4150	181400	No Calc	No Calc
80	90	4150	181400	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	4128	180490	No Calc	No Calc
120	130	4128	180490	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc
140	150	No Calc	No Calc	No Calc	No Calc
150	160	2400	111048	No Calc	No Calc
160	170	No Calc	No Calc	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc
180	190	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1321.57	...	1253.27	...	...
20	30	310.497	...	279.447	...	...
30	40	247.633	80.7336	226.997	82.4733	...
40	50	1237.32	912.242	1125.56	926.664	...
50	60	791.498	109.84	778.198	110.273	...
60	70	791.498	109.84	778.198	110.273	...
70	80	1855.98	1368.36	1688.33	1390	...
80	90	711.457	524.539	647.194	532.832	...
90	100	791.498	109.84	778.198	110.273	...
100	110	791.498	109.84	778.198	110.273	...
110	120	1855.98	1368.36	1688.33	1390	...
120	130	697.847	514.504	634.814	522.639	...
130	140	791.498	109.84	778.198	110.273	...
140	150	791.498	109.84	778.198	110.273	...
150	160	1484.78	1094.69	1350.67	1112	...
160	170	804.804	110.767	791.393	111.2	...
170	180	802.57	109.84	789.27	110.273	...
180	190	247.633	80.7336	226.997	82.4733	...
Total		16327	6823.82	15371	6922.18	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	200	...	...	...	Tailing lug
20	Insl	39.5469	...	750	...	FP
30	Liqd	74.2273	...	-95.2503	...	Liquid 30
30	Insl	34.1963	...	-70.2503	...	Ins: 30
30	Nozl	22.8393	...	-325.5	...	N4_80
30	Forc	...	...	-190	...	Guide sup later
40	Liqd	838.725	...	1000	...	Liquid 40
40	Insl	160.491	...	1000	...	Ins: 40
40	Nozl	90.0621	-481.001	1175	...	N1_150
40	Nozl	23.5956	474.663	200	...	K1_50
40	Nozl	23.5956	474.663	1200	...	K2_50
40	Nozl	23.5956	-474.663	550	...	K3_50
40	Wght	259	-1649	1175	...	N1_FY
50	Liqd	92.2626	...	110	...	Liquid 50
50	Insl	20.8233	...	103	...	Ins: 50
60	Liqd	93.0985	...	111	...	Liquid 60
60	Insl	20.8233	...	103	...	Ins: 60
70	Pack	296.59	...	1550	...	Bed 2
70	Liqd	1258.09	...	1500	...	Liquid 70
70	Insl	240.736	...	1500	...	Ins: 70
70	Nozl	221.645	481.001	343.992	...	H2_200
70	Wght	50	350	343.99	...	H2_Davit
70	Wght	10	...	50	...	Bed support 2
70	Wght	35	...	80	...	Bed sup grid 2
80	Pack	81.8179	...	400	...	Bed2
80	Liqd	482.267	...	575	...	Liquid 80
80	Insl	92.2822	...	575	...	Ins: 80
80	Wght	10	...	820	...	Bed limiter sup
80	Wght	10	...	850	...	Bed limiter
90	Liqd	90.5823	...	108	...	Liquid 90
90	Insl	20.8233	...	103	...	Ins: 90
100	Liqd	90.5823	...	108	...	Liquid 100
100	Insl	20.8233	...	103	...	Ins: 100
110	Pack	298.84	...	1539	...	Bed-1
110	Liqd	1258.09	...	1500	...	Liquid 110
110	Insl	240.736	...	1500	...	Ins: 110

110	Nozl	221.93	481.001	673.241	...	H1_200
110	Wght	50	350	673.239	...	H1_Davit
110	Wght	400	...	664.998	...	Clip weight
110	Wght	10	...	49.9987	...	Bed support
110	Wght	35	...	59.9987	...	Bed sup grid
110	Wght	1750	...	606.958	...	Misc wt 10%
120	Pack	79.5679	...	389	...	Bed1
120	Liqd	473.041	...	564	...	Liquid 120
120	Insl	90.5168	...	564	...	Ins: 120
120	Wght	10	...	799.998	...	Bed limiter sup
120	Wght	10	...	819.998	...	Bed limiter 1
130	Liqd	87.2274	...	104	...	Liquid 130
130	Insl	20.8233	...	103	...	Ins: 130
140	Liqd	87.2274	...	104	...	Liquid 140
140	Insl	20.8233	...	103	...	Ins: 140
150	Liqd	1006.47	...	1200	...	Liquid 150
150	Insl	192.589	...	1200	...	Ins: 150
150	Nozl	34.9242	-481.001	744.71	...	N2_80
150	Nozl	23.6567	474.663	1506.51	...	K4_50
150	Wght	5	...	744.7	...	N2_Pipe
150	Wght	10	...	2250	...	Demister sup
150	Wght	20	...	2300	...	Demister
150	Wght	150	...	2300	...	Lifting lug1
150	Wght	150	...	2300	...	Lifting lug 2
160	Liqd	87.2274	...	104	...	Liquid 160
160	Insl	20.9609	...	103	...	Ins: 160
170	Liqd	27.4805	...	32.7646	...	Liquid 170
170	Insl	21.1146	...	97	...	Ins: 170
180	Liqd	74.2272	...	145.251	...	Liquid 180
180	Insl	34.7592	...	120.25	...	Ins: 180
180	Nozl	89.804	...	325.75	...	N3_150
180	Wght	320	1649	...	...	N3_Wt
180	Forc	...	...	170	...	N3_F/M

Total Weight of Each Detail Type:

Packing	756.8
Liquid	6120.8
Insulation	1292.9
Nozzles	775.6
Weights	3494.0
Sum of the Detail Weights	12440.2 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	16327.1	16327.1	16327.1	16327.1	16327.1	16327.1
Nozzles	775.6	775.6	775.6	775.6	775.6	775.6
Wld Weights	2800.0	2800.0	2800.0	2800.0	2800.0	2800.0
Packing	...	...	756.8	756.8	756.8	756.8
Insulation	...	...	1292.9	1292.9	1292.9	1292.9
Ope Weights	...	...	...	...	...	694.0
Ope. Liquid	...	...	...	...	...	6120.8
Test Liquid	...	6819.7	...	...	...	...
Totals	19902.7	26722.4	21952.4	21952.4	21952.4	28767.2

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	19902.7 kgm
Shop Test Wt.	- Fabricated weight + water (full)	26722.4 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	21952.4 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	21952.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	21952.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	28767.2 kgm

Field Test Wt. - Empty weight + test medium (full) 28015.2 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	10161.3 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	9621.5 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	7715.1 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm²	Inside Surface Area mm²
10	20	5619052	5397884
20	30	2554114	2453583
30	40	884981	749108
40	50	5189924	4787800
50	60	1331141	569748
60	70	1331141	569748
70	80	7784886	7181699
80	90	2984206	2752985
90	100	1331141	569748
100	110	1331141	569748
110	120	7784886	7181699
120	130	2927117	2700319
130	140	1331141	569748
140	150	1331141	569748
150	160	6227909	5745360
160	170	1342887	574536
170	180	1338062	569748
180	190	884981	749108

Total	53509852.0	44262324.0	mm²
Total	53.5	44.3	m²

Element and Detail Weights:

From	To To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	1521.57	1521.57	1521.57	...	1521.57
20	30	350.044	350.044	350.044	...	350.044
30	40	304.668	378.895	385.352	...	304.668
40	50	1558.66	2656.38	2470.34	4504846	1817.66
50	60	812.321	904.584	922.095	...	812.321
60	70	812.321	905.42	922.095	...	812.321
70	80	2674.95	3968.04	3745.89	1217530	2709.95
80	90	895.557	1387.82	1337.96	...	905.557
90	100	812.321	902.904	922.095	...	812.321
100	110	812.321	902.904	922.095	...	812.321
110	120	4827.48	6120.57	5896.17	1218875	4862.48
120	130	877.932	1360.97	1312.55	...	887.932
130	140	812.321	899.549	922.095	...	812.321
140	150	812.321	899.549	922.095	...	812.321
150	160	2045.95	3077.42	3139.97	54637.4	2070.95
160	170	825.765	912.992	936.465	...	825.765
170	180	823.684	851.165	933.458	...	823.684
180	190	372.196	766.423	452.88	5176526	692.196

Empty Support Force + the Sum of the Y forces	230109.86	N
Operating Support Force + the Sum of the Y forces	296935.94	N
Hydro Support Force + the Sum of the Y forces	289562.00	N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid	Cumulative Oper. Wgt.	Cumulative Hydro. Wgt.

		kgm	kgm	kgm
10	20	22646.4	28767.2	28015.2
20	30	21124.8	27245.6	26493.7
30	40	20774.8	26895.6	26143.6
40	50	20470.1	26516.7	25758.3
50	60	18652.4	23860.3	23287.9
60	70	17840.1	22955.7	22365.8
70	80	17027.8	22050.3	21443.7
80	90	14317.9	18082.3	17697.8
90	100	13412.3	16694.4	16359.9
100	110	12600	15791.5	15437.8
110	120	11787.7	14888.6	14515.7
120	130	6925.17	8768.07	8619.52
130	140	6037.24	7407.1	7306.97
140	150	5224.92	6507.55	6384.87
150	160	4412.6	5608	5462.78
160	170	2341.65	2530.58	2322.8
170	180	1515.88	1617.59	1386.34
180	190	692.196	766.423	452.88

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	12172417	12172417	12172417
20	30	12172417	12172417	12172417
30	40	12172417	12172417	12172417
40	50	12172417	12172417	12172417
50	60	7667570	7667570	7667570
60	70	7667570	7667570	7667570
70	80	7667570	7667570	7667570
80	90	6450039	6450039	6450039
90	100	6450039	6450039	6450039
100	110	6450039	6450039	6450039
110	120	6450039	6450039	6450039
120	130	5231163	5231163	5231163
130	140	5231163	5231163	5231163
140	150	5231163	5231163	5231163
150	160	5231163	5231163	5231163
160	170	5176526	5176526	5176526
170	180	5176526	5176526	5176526
180	190	5176526	5176526	5176526

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure UG-44	Max Pressure 50%	Max Pressure DNV
N4_80	14.34	15.31	82	900	GR 1.1	...	...	...	...
N1_150	14.34	15.31	82	900	GR 1.1	...	...	...	...
K1_50	14.34	15.31	82	900	GR 1.1	...	...	...	...
K2_50	14.34	15.31	82	900	GR 1.1	...	...	...	...
K3_50	14.34	15.31	82	900	GR 1.1	...	...	...	...
H2_200	14.34	15.31	82	900	GR 1.1	...	...	...	...
H1_200	14.34	15.31	82	900	GR 1.1	...	...	...	...
N2_80	14.34	15.31	82	900	GR 1.1	...	...	...	...
K4_50	14.34	15.31	82	900	GR 1.1	...	...	...	...
N3_150	14.34	15.31	82	900	GR 1.1	...	...	...	...
Min Rating	14.342	15.311, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:
There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/ eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.1925 Hz.

The Natural Frequency for the Vessel (Ope...) is 1.06419 Hz.

The Natural Frequency for the Vessel (Filled) is 1.08531 Hz.

Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	25228.9	-4795438
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	15161.8	9579076

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	25228.9	-4795438
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	15161.8	9579076

User Input Forces and Moments:

From	Distance	- - - -	Forces - - - -		- - - -	Moments - - - -	
Node	From	Fx	Fy	Fz	Mx	My	Mz
30	-190.00	-20791.		14291.			
180	170.00	10721.	-14845.	10721.	0.495E+07		0.495E+07

Wind Load Calculation Results:

Input Values:

Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		300	mm
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm
Distance Upwind of Crest	Lh	0	mm
Distance from Crest to the Vessel	x	0	mm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
= min(0.85, 0.925((1 + 1.7 • gQ • Izbar • Q)/(1 + 1.7 • gV • Izbar)))
= min(0.85,0.925((1+1.7•3.4•0.198•0.926)/(1+1.7•3.4•0.198)))
= min(0.85, 0.888)
= 0.850

Natural Frequency of Vessel (Operating)	1.064 Hz
Natural Frequency of Vessel (Empty)	1.192 Hz
Natural Frequency of Vessel (Test)	1.085 Hz

User Entered Importance Factor is		1.000
Force Coefficient	[Cf]	0.668
Structure Height to Diameter ratio		19.269
Height to top of Structure		18185.998 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
Exposure Category: C from Table 26.11-1
α = 9.5: Zg = 274320.0 mm

Effective Height [z]:
= Centroid Height + Vessel Base Elevation
= 1100.0 + 300.0 = 1400.0 mm
= 4.593 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
Because z (4.593 ft.) < 15 ft.
= 2.01(15 / Zg)2/α
= 2.01(15/900.0)2/9.5
= 0.849

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
= (1 + K1 • K2 • K3)²
= (1 + 0.0• 0.0• 0.0)²
= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:
= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)
= max(16, 0.00256 • 0.849 • 1.0 • 1.0 • 1.0 • 60.399²)
= 16.0 psf [0.766] kPa

Force on the first element [F]:
= qz • G • Cf • WindArea
= 16.0 • 0.85 • 0.668 • 31.574
= 286.9 lbs. [1276.2] N

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	1400.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-1	3000.0	0.000	0.000	0.000	0.849	1.000	0.766
Bottom head	3525.0	0.000	0.000	0.000	0.849	1.000	0.766
Shell-1	4550.0	0.000	0.000	0.000	0.849	1.000	0.766
Flange-1	5672.5	0.000	0.000	0.000	0.888	1.000	0.766
Flange-2	5908.5	0.000	0.000	0.000	0.896	1.000	0.766
Shell-2	7526.5	0.000	0.000	0.000	0.943	1.000	0.766
Shell-3	9601.5	0.000	0.000	0.000	0.992	1.000	0.766
Flange-3	10299.0	0.000	0.000	0.000	1.007	1.000	0.766
Flange-4	10535.0	0.000	0.000	0.000	1.012	1.000	0.766
Shell-4	12153.0	0.000	0.000	0.000	1.043	1.000	0.766
Shell-5	14217.0	0.000	0.000	0.000	1.078	1.000	0.766
Flange-5	14903.5	0.000	0.000	0.000	1.089	1.000	0.766
Flange-6	15139.5	0.000	0.000	0.000	1.092	1.000	0.766
Shell-6	16457.5	0.000	0.000	0.000	1.112	1.000	0.766
Flange-7	17781.0	0.000	0.000	0.000	1.130	1.000	0.766
Flange-8	18018.0	0.000	0.000	0.000	1.133	1.000	0.766
Top head	18286.7	0.000	0.000	0.000	1.137	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

```
Cf  - Correction factor for natural frequency
 D  - Average outer insulated diameter of vessel, mm
Df  - Damping Factor < 0.75 Unstable,  > 0.95 Stable
Dr  - Average outer insulated of top half of vessel, mm
 f  - Natural frequency of vibration (Hertz)
f1  - Natural frequency of bare vessel based on a unit value of (D/L²)(104)
 L  - Total height of structure mm
Lc  - Total length of conical section(s) of vessel, mm
tb  - Uncorroded plate thickness at bottom of vessel, mm
V30 - Design Wind Speed provided by user, km/hr
Vc  - Critical wind velocity, km/hr
Vw  - Maximum wind speed at top of structure, km/hr
W   - Total corroded weight of structure, N
Ws  - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z  - Maximum amplitude of vibration at top of vessel, mm
Dl  - Logarithmic decrement ( taken as 0.03 for Welded Structures )
Vp  - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
P30 - wind pressure 30 feet above the base
```

Check other Conditions and Basic Assumptions:
:: Total Cone Length / Total Length < 0.5
:: 0.0/17885.998 = 0.0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)

:: [$3.12/58.68^2$] • 10^4 = 9.046 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.000004 no chance. [Vp]:
= W / (L • Dr^2)
= 272721/(17886.0 • 978.482^2)
= 0.15926E-04

Since Vp is > 0.000004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	300.00	0.77	0.00
N1FY	0.00	4725.00	0.77	0.00
H2Davit	0.00	6370.49	0.77	0.00
Bed support 2	0.00	6076.50	0.77	0.00
Bed sup grid	0.00	6106.50	0.77	0.00
Bed limiter s	0.00	9846.50	0.77	0.00
Bed limiter	0.00	9876.50	0.77	0.00
H1Davit	0.00	11326.24	0.77	0.00
Clip weight	0.00	11318.00	0.77	0.00
Bed support	0.00	10703.00	0.77	0.00
Bed sup grid	0.00	10713.00	0.77	0.00
Misc wt 10%	0.00	11259.96	0.77	0.00
Bed limiter s	0.00	14453.00	0.77	0.00
Bed limiter 1	0.00	14473.00	0.77	0.00
N2Pipe	0.00	16002.20	0.77	0.00
Demister sup	0.00	17507.50	0.77	0.00
Demister	0.00	17557.50	0.77	0.00
Lifting lug1	0.00	17557.50	0.77	0.00
Lifting lug 2	0.00	17557.50	0.77	0.00
N3Wt	0.00	18136.00	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 1.06419 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm²	Wind Pressure kPa	Element Wind Load N
10	20	1400	1333.32	2933303	0.76606	1403.79
20	30	3000	1464.52	1464519	0.76606	700.876
30	40	3525	1610.48	80524.2	0.76606	38.5365
40	50	4550	1617.04	3234086	0.76606	1547.74
50	60	5672.5	1478.73	348980	0.76606	167.011
60	70	5908.5	1478.73	348980	0.76606	167.011
70	80	7526.5	1617.04	4851130	0.76606	2321.61
80	90	9601.5	1617.04	1859599	0.76606	889.95
90	100	10299	1478.73	348980	0.76606	167.011
100	110	10535	1478.73	348980	0.76606	167.011
110	120	12153	1617.04	4851130	0.76606	2321.61
120	130	14217	1617.04	1824024	0.76606	872.925
130	140	14903.5	1478.73	348980	0.76606	167.011
140	150	15139.5	1478.73	348980	0.76606	167.011
150	160	16457.5	1617.04	3880904	0.76606	1857.29
160	170	17781	1476.8	351479	0.76606	168.208
170	180	18018	1465.38	345831	0.76606	165.504
180	190	18286.7	1608.84	459518	0.76606	219.912

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		2.000
Table Value Fa		1.300
Table Value Fv		1.500
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		2.000
Site Class		C
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	2.500
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

```
SMS = Fa * Ss = 1.3 * 0.147 = 0.191
SM1 = Fv * S1 = 1.5 * 0.065 = 0.097
SDS = 2/3 * Sms = 2/3 * 0.191 = 0.127
SD1 = 2/3 * Sm1 = 2/3 * 0.097 = 0.065
```

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= Ct • hnx where Ct = 0.020, x = 0.75 and hn = Structural Height (ft.)
= 0.020(59.30610.75)
= 0.427 seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/1.064)
= 0.940

Check the Value of T which is the smaller of Cu•Ta and T:
= Minimum Value of (1.7 • 0.427, 0.94) per 12.8.2
= 0.727

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= Sds / (R / I)
= 0.127/(2.0/2.0)
= 0.127

Maximum value of Cs per equation 12.8-3 [Cs]:
= Sd1 / (T(R / Ie))
= 0.065/(0.727(2.0/2.0))
= 0.089

Minimum value of Cs per equation 15.4-1 [Cs]:
= max((0.044 • SDS • Ie), 0.030)
= max((0.044 • 0.1274 • 2.0), 0.030)
= 0.030

Total Base Shear [V]:
= Cs • W (Equation 12.8-1):
= 0.0895 • 282091.1
= 25234.770 N

Vertical load per 12.4-4, [Ev]:
= 0.2 • Sds • W
= 0.2 • 0.127 • 282091

= 7187.7 N

Distribute the Base shear force to each element according to the equations
Fx = Cvx * V (eqn. 12.8-11) and the vertical distribution factor
Cvx = Wx*hxk/(Sum of Wi*hik) and k is an exponent which is related
to the period of Vibration.

In this case, the value of k was 1.2198

The Natural Frequency for the Vessel (Ope...) is 1.06419 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1100	14920.6	84.1525	95.6153	-7657.12
20	30	2700	3432.53	57.8894	62.8559	-8037.29
30	40	3225	3715.45	77.8261	67.341	-8124.76
40	50	4250	26048.5	764.022	475.722	-8219.42
50	60	5372.5	8870.35	346.281	326.099	-8883.14
60	70	5608.5	8878.54	365.262	342.91	-9109.16
70	80	7226.5	38910.5	2180.78	1518.74	-9335.38
80	90	9301.5	13609	1037.76	683.04	-10326.8
90	100	9999	8853.87	737.411	674.218	-10673.6
100	110	10235	8853.87	758.696	692.862	-10899.2
110	120	11853	60018.3	6151.37	4888.44	-11124.8
120	130	13917	13345.7	1663.69	1072.58	-12654
130	140	14603.5	8820.97	1166.16	1049.9	-12994.1
140	150	14839.5	8820.97	1189.19	1069.77	-13218.8
150	160	16157.5	30177.2	4513.27	2976.24	-13443.6
160	170	17481	8952.8	1473.94	1317.07	-14212.5
170	180	17718	8346.52	1396.89	1334.6	-14440.6
180	190	17861	7515.54	1270.21	608.759	-14653.3

Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1100	40390.7	40390.7	355082560	355082560
20	30	2700	40390.7	40390.7	266872384	266872384
30	40	3225	40390.7	40390.7	226863376	226863376
40	50	4250	15161.8	15161.8	230818496	230818496
50	60	5372.5	15161.8	15161.8	200414352	200414352
60	70	5608.5	15161.8	15161.8	196834720	196834720
70	80	7226.5	15161.8	15161.8	193255104	193255104
80	90	9301.5	15161.8	15161.8	147751312	147751312
90	100	9999	15161.8	15161.8	130239944	130239944
100	110	10235	15161.8	15161.8	126660304	126660304
110	120	11853	15161.8	15161.8	123080688	123080688
120	130	13917	15161.8	15161.8	77576896	77576896
130	140	14603.5	15161.8	15161.8	60399216	60399216
140	150	14839.5	15161.8	15161.8	56819584	56819584
150	160	16157.5	15161.8	15161.8	53239964	53239964
160	170	17481	15161.8	15161.8	16768661	16768661
170	180	17718	15161.8	15161.8	13158700	13158700
180	190	17861	15161.8	15161.8	9579076	9579076

Note:
The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1100	13510	25234.8	124878792	325366912
20	30	2700	12106.2	25150.6	96689520	269920480
30	40	3225	11405.3	25092.7	84928960	244788608
40	50	4250	11366.8	25014.9	84359432	243535440
50	60	5372.5	9819.07	24250.9	63120756	194140512
60	70	5608.5	9652.06	23904.6	60822232	188455856
70	80	7226.5	9485.05	23539.3	58563152	182855200
80	90	9301.5	7163.44	21358.6	33580292	115481064
90	100	9999	6273.49	20320.8	25822684	91414232
100	110	10235	6106.48	19583.4	24361258	86703632
110	120	11853	5939.47	18824.7	22939264	82169656
120	130	13917	3617.86	12673.3	8597463	34903448
130	140	14603.5	2744.93	11009.6	4995039	21491276
140	150	14839.5	2577.92	9843.48	4366689	19029614
150	160	16157.5	2410.91	8654.3	3777769	16845996
160	170	17481	553.624	4141.03	216393	1466727
170	180	17718	385.416	2667.09	104603	656235
180	190	17861	219.912	1270.21	33145.2	191446

Note:
The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	40390.7	0.3381	1.29362	0.0011226
20	30	40390.7	1.9083	2.62422	0.0015276
30	40	40390.7	2.66247	2.70086	0.0015383
40	50	15161.8	4.33601	6.15619	0.0019084
50	60	15161.8	6.3815	6.60703	0.0019123
60	70	15161.8	6.8328	7.0588	0.0019162
70	80	15161.8	10.1124	13.4953	0.0023552
80	90	15161.8	14.8701	16.2843	0.0024924
90	100	15161.8	16.5785	16.8728	0.0024949
100	110	15161.8	17.1673	17.4619	0.0024974
110	120	15161.8	21.3196	25.371	0.0027558
120	130	15161.8	26.9354	28.5187	0.0028225
130	140	15161.8	28.8518	29.185	0.0028237
140	150	15161.8	29.5182	29.8515	0.0028248
150	160	15161.8	33.2703	36.7324	0.0028968
160	170	15161.8	37.0772	37.4219	0.0028971
170	180	15161.8	37.7638	38.1057	0.0028973
180	190	15161.8	38.1781	38.2505	0.0028974

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
Allowable deflection : 89.430 Actual deflection : 38.251 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	13510	0.1205	0.46285	0.00040362
20	30	12106.2	0.68431	0.94341	0.00055425
30	40	11405.3	0.9573	0.97123	0.00055824
40	50	11366.8	1.56449	2.22154	0.00068593
50	60	9819.07	2.30251	2.38357	0.00068719
60	70	9652.06	2.46469	2.54589	0.0006884
70	80	9485.05	3.63125	4.80529	0.00080701
80	90	7163.44	5.27395	5.75111	0.0008366
90	100	6273.49	5.84984	5.94861	0.00083711
100	110	6106.48	6.0474	6.14622	0.00083759
110	120	5939.47	7.42227	8.72775	0.00087719
120	130	3617.86	9.22358	9.72126	0.00088377
130	140	2744.93	9.82554	9.92984	0.00088386
140	150	2577.92	10.0341	10.1384	0.00088394
150	160	2410.91	11.201	12.2655	0.00088733
160	170	553.624	12.371	12.4766	0.00088734
170	180	385.416	12.5813	12.6861	0.00088734
180	190	219.912	12.7082	12.7304	0.00088734

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	25.4716	159.197
20	30	27.978	174.862

30	40	30.7665	192.29
40	50	30.8918	193.074
50	60	28.2494	176.559
60	70	28.2494	176.559
70	80	30.8918	193.074
80	90	30.8918	193.074
90	100	28.2494	176.559
100	110	28.2494	176.559
110	120	30.8918	193.074
120	130	30.8918	193.074
130	140	28.2494	176.559
140	150	28.2494	176.559
150	160	30.8918	193.074
160	170	28.2126	176.329
170	180	27.9945	174.966
180	190	30.7351	192.095

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 89.430 Actual deflection : 12.730 mm

Total Deflection in the Operating Condition + Applied Forces:
Allowable deflection : 89.430 Actual deflection : 50.981 mm

Longitudinal Stress Constants:

From	To	Metal Area New mm²	Metal Area Corroded mm²	Section Modulus New mm³	Section Modulus Corroded mm³
10	20	40061.6	36055.4	7828332	7058852
20	30	40061.6	36055.4	7828332	7058852
30	40	74644.4	67434.4	14260668	12977350
40	50	79821.7	72611.7	15255559	13978457
50	60	79821.7	72611.7	15255559	13978457
60	70	79821.7	72611.7	15255559	13978457
70	80	79821.7	72611.7	15255559	13978457
80	90	79821.7	72611.7	15255559	13978457
90	100	79821.7	72611.7	15255559	13978457
100	110	79821.8	72611.8	15255572	13978465
110	120	79821.8	72611.8	15255572	13978465
120	130	79821.7	72611.7	15255559	13978457
130	140	79821.7	72611.7	15255559	13978457
140	150	79821.7	72611.7	15255559	13978457
150	160	79821.7	72611.7	15255559	13978457
160	170	79821.7	72611.7	15255559	13978457
170	180	73354	66144	14012844	12727963
180	190	73354	66144	14012844	12727963

Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	115.918	196.556	-141.502	-141.502
20	30	82.7985	140.397	-141.502	-141.502
30	40	165.597	280.795	-147.277	-147.277
40	50	165.597	280.795	-147.277	-147.277
50	60	165.597	269.995	-147.277	-147.277
60	70	165.597	269.995	-147.277	-147.277
70	80	165.597	280.795	-147.277	-147.277
80	90	165.597	280.795	-147.277	-147.277
90	100	165.597	269.995	-147.277	-147.277
100	110	165.597	269.995	-147.277	-147.277
110	120	165.597	280.795	-147.277	-147.277
120	130	165.597	280.795	-147.277	-147.277
130	140	165.597	269.995	-147.277	-147.277
140	150	165.597	269.995	-147.277	-147.277
150	160	165.597	280.795	-147.277	-147.277
160	170	165.597	269.995	-147.277	-147.277
170	180	165.597	269.995	-147.277	-147.277
180	190	165.597	280.795	-147.277	-147.277

Longitudinal Stress Report

Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	64.9847	-0.81392	84.4801
40	50	60.3733	-0.76326	78.4853
50	60	...	...	...
60	70	...	...	...
70	80	60.3733	-0.76326	78.4853
80	90	60.3733	-0.76326	78.4853
90	100	...	...	...
100	110	...	...	...
110	120	60.3733	-0.76326	78.4852
120	130	60.3733	-0.76326	78.4853
130	140	...	...	...
140	150	...	...	...
150	160	60.3733	-0.76326	78.4853
160	170	...	...	...
170	180	...	...	...
180	190	66.2463	-0.82778	86.1202

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-6.15966	-7.82448	-7.61995	1.72387	1.72387
20	30	-5.7458	-7.41063	-7.20609	1.72387	1.72387
30	40	-3.02123	-3.02123	-3.02123	0.93767	0.93767
40	50	-2.76466	-2.76466	-2.76466	0.87052	0.87052
50	60	-2.51917	-2.51917	-2.51917	0.54835	0.54835
60	70	-2.40946	-2.40946	-2.40946	0.54835	0.54835
70	80	-2.29975	-2.29975	-2.29975	0.54835	0.54835
80	90	-1.93375	-1.93375	-1.93375	0.46128	0.46128
90	100	-1.81144	-1.81144	-1.81144	0.46128	0.46128
100	110	-1.70173	-1.70173	-1.70173	0.46128	0.46128
110	120	-1.59202	-1.59202	-1.59202	0.46128	0.46128
120	130	-0.9353	-0.9353	-0.9353	0.37411	0.37411
130	140	-0.81538	-0.81538	-0.81538	0.37411	0.37411
140	150	-0.70567	-0.70567	-0.70567	0.37411	0.37411
150	160	-0.59596	-0.59596	-0.59596	0.37411	0.37411
160	170	-0.31626	-0.31626	-0.31626	0.3702	0.3702
170	180	-0.22475	-0.22475	-0.22475	0.40657	0.40657
180	190	-0.10263	-0.10263	-0.10263	0.40657	0.40657

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	1.72387	17.6854	46.0787	8.84271	...
20	30	1.72387	13.6932	38.2263	6.84662	...
30	40	0.93767	6.5423	18.8567	3.27115	...

40	50	0.87052	6.03303	17.4166	3.01651	...
50	60	0.54835	4.51413	13.8841	2.25706	...
60	70	0.54835	4.34975	13.4776	2.17487	...
70	80	0.54835	4.18819	13.077	2.09409	...
80	90	0.46128	2.40152	8.25871	1.20076	...
90	100	0.46128	1.84673	6.53756	0.92336	...
100	110	0.46128	1.74221	6.20067	0.87111	...
110	120	0.46128	1.64052	5.87642	0.82026	...
120	130	0.37411	0.61485	2.49615	0.30743	...
130	140	0.37411	0.35722	1.53696	0.17861	...
140	150	0.37411	0.31229	1.36092	0.15614	...
150	160	0.37411	0.27017	1.20475	0.13508	...
160	170	0.3702	0.015476	0.10489	0.0077378	...
170	180	0.40657	0.0082157	0.051542	0.0041079	...
180	190	0.40657	0.0026033	0.015037	0.0013016	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	35.0137
20	30	...	...	...	29.0264
30	40	...	...	...	14.3143
40	50	...	...	...	13.221
50	60	...	...	...	10.5259
60	70	...	...	...	10.2154
70	80	...	...	...	9.91046
80	90	...	...	...	6.23445
90	100	...	...	...	4.91084
100	110	...	...	...	4.65171
110	120	...	...	...	4.40412
120	130	...	...	...	1.85573
130	140	...	...	...	1.13539
140	150	...	...	...	1.00316
150	160	...	...	...	0.88882
160	170	...	...	...	0.072412
170	180	...	...	...	0.030869
180	190	...	...	...	0.0072064

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.41176	-0.21239
20	30	-0.41176	-0.22293
30	40	-0.22016	-0.12049
40	50	-0.20446	-0.11321
50	60	-0.20446	-0.12235
60	70	-0.20446	-0.12546
70	80	-0.20446	-0.12858
80	90	-0.20446	-0.14223
90	100	-0.20446	-0.14701
100	110	-0.20446	-0.15011
110	120	-0.20446	-0.15322
120	130	-0.20446	-0.17428
130	140	-0.20446	-0.17897
140	150	-0.20446	-0.18206
150	160	-0.20446	-0.18516
160	170	-0.20446	-0.19575
170	180	-0.22445	-0.21834
180	190	-0.22445	-0.22156

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	50.287	50.287	45.3441	45.3441
20	30	37.7947	37.7947	34.0797	34.0797
30	40	17.4759	17.4759	15.9032	15.9032
40	50	16.5072	16.5072	15.1253	15.1253
50	60	14.3328	14.3328	13.1329	13.1329
60	70	14.0768	14.0768	12.8984	12.8984
70	80	13.8208	13.8208	12.6638	12.6638
80	90	10.5665	10.5665	9.68198	9.68198
90	100	9.31421	9.31421	8.53448	8.53448
100	110	9.0582	9.0582	8.2999	8.2999
110	120	8.8022	8.8022	8.06533	8.06533
120	130	5.54797	5.54797	5.08353	5.08353
130	140	4.31949	4.31949	3.95789	3.95789
140	150	4.06349	4.06349	3.72332	3.72332
150	160	3.8075	3.8075	3.48875	3.48875
160	170	1.19922	1.19922	1.09883	1.09883
170	180	1.03351	1.03351	0.93874	0.93874
180	190	0.75236	0.75236	0.68337	0.68337

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

```
IP = Longitudinal Stress due to Internal  Pressure
EP = Longitudinal Stress due to External  Pressure
HP = Longitudinal Stress due to Hydrotest Pressure
NP = No Pressure
EW = Longitudinal Stress due to Weight (No Liquid)
OW = Longitudinal Stress due to Weight (Operating)
HW = Longitudinal Stress due to Weight (Hydrotest)
WI = Bending Stress due to Wind Moment (Operating)
EQ = Bending Stress due to Earthquake Moment (Operating)
EE = Bending Stress due to Earthquake Moment (Empty)
HI = Bending Stress due to Wind Moment (Hydrotest)
HE = Bending Stress due to Earthquake Moment (Hydrotest)
WE = Bending Stress due to Wind Moment (Empty) (no CA)
WF = Bending Stress due to Wind Moment (Filled) (no CA)
CW = Longitudinal Stress due to Weight (Empty) (no CA)
VO = Bending Stress due to Vortex Shedding Loads ( Ope )
VE = Bending Stress due to Vortex Shedding Loads ( Emp )
VF = Bending Stress due to Vortex Shedding Loads ( Test No CA. )
FW = Axial Stress due to Vertical Forces for the Wind Case
FS = Axial Stress due to Vertical Forces for the Seismic Case
BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
BL = Bending Stress due to Blast Loads, Corroded
```

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	63.12	115.92	-76.27	141.50	0.5446	0.5390
20	47.05	82.80	-59.37	141.50	0.5683	0.4196
30	21.71	165.60	-28.20	147.28	0.1311	0.1915
40	20.44	165.60	-26.38	147.28	0.1234	0.1791
70	16.05	165.60	-21.06	147.28	0.0969	0.1430
80	11.29	165.60	-15.57	147.28	0.0682	0.1057
110	9.11	165.60	-12.70	147.28	0.0550	0.0862
120	5.40	165.60	-7.68	147.28	0.0326	0.0521
150	3.65	165.60	-5.25	147.28	0.0220	0.0357
180	0.83	165.60	-1.49	147.28	0.0050	0.0101

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	80.65	115.92	-93.40	141.50	0.6958	0.6600
20	62.58	82.80	-74.51	141.50	0.7558	0.5266
30	29.59	165.60	-35.87	147.28	0.1787	0.2436
40	27.72	165.60	-33.48	147.28	0.1674	0.2273
70	21.85	165.60	-26.71	147.28	0.1320	0.1813
80	15.19	165.60	-19.34	147.28	0.0917	0.1313
110	11.92	165.60	-15.41	147.28	0.0720	0.1047
120	6.67	165.60	-8.89	147.28	0.0403	0.0603
150	4.29	165.60	-5.85	147.28	0.0259	0.0397

180	0.84	165.60	-1.49	147.28	0.0051	0.0101

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.46	115.92	-77.93	141.50	0.5302	0.5508
20	45.39	82.80	-61.03	141.50	0.5482	0.4313
30	21.71	165.60	-28.20	147.28	0.1311	0.1915
40	20.44	165.60	-26.38	147.28	0.1234	0.1791
70	16.05	165.60	-21.06	147.28	0.0969	0.1430
80	11.29	165.60	-15.57	147.28	0.0682	0.1057
110	9.11	165.60	-12.70	147.28	0.0550	0.0862
120	5.40	165.60	-7.68	147.28	0.0326	0.0521
150	3.65	165.60	-5.25	147.28	0.0220	0.0357
180	0.83	165.60	-1.49	147.28	0.0050	0.0101

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	90.05	115.92	-106.13	141.50	0.7769	0.7500
20	70.11	82.80	-85.38	141.50	0.8468	0.6034
30	34.13	165.60	-40.41	147.28	0.2061	0.2744
40	31.92	165.60	-37.67	147.28	0.1927	0.2558
70	25.02	165.60	-29.87	147.28	0.1511	0.2028
80	17.21	165.60	-21.36	147.28	0.1039	0.1450
110	13.39	165.60	-16.89	147.28	0.0809	0.1146
120	7.31	165.60	-9.53	147.28	0.0441	0.0647
150	4.61	165.60	-6.17	147.28	0.0278	0.0419
180	0.85	165.60	-1.50	147.28	0.0051	0.0102

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.95	196.56	-18.19	141.50	0.0150	0.1285
20	1.36	140.40	-15.78	141.50	0.0097	0.1115
30	1.19	280.79	-7.23	147.28	0.0042	0.0491
40	1.12	280.79	-6.65	147.28	0.0040	0.0452
70	0.34	280.79	-4.94	147.28	0.0012	0.0336
80		280.79	-3.60	147.28		0.0244
110		280.79	-2.87	147.28		0.0195
120		280.79	-1.62	147.28		0.0110
150		280.79	-1.11	147.28		0.0075
180	0.31	280.79	-0.51	147.28	0.0011	0.0035

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-9.34	141.50		0.0660
20		140.40	-8.93	141.50		0.0631
30		280.79	-3.96	147.28		0.0269
40		280.79	-3.64	147.28		0.0247
70		280.79	-2.85	147.28		0.0193
80		280.79	-2.40	147.28		0.0163
110		280.79	-2.05	147.28		0.0139
120		280.79	-1.31	147.28		0.0089
150		280.79	-0.97	147.28		0.0066
180	0.30	280.79	-0.51	147.28	0.0011	0.0035

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.46	115.92	-77.93	141.50	0.5302	0.5508
20	45.39	82.80	-61.03	141.50	0.5482	0.4313
30	86.70	165.60		147.28	0.5236	
40	80.81	165.60		147.28	0.4880	

70	76.43	165.60		147.28	0.4615	
80	71.66	165.60		147.28	0.4328	
110	69.48	165.60		147.28	0.4196	
120	65.77	165.60		147.28	0.3972	
150	64.02	165.60		147.28	0.3866	
180	67.08	165.60		147.28	0.4051	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	90.05	115.92	-106.13	141.50	0.7769	0.7500
20	70.11	82.80	-85.38	141.50	0.8468	0.6034
30	99.11	165.60		147.28	0.5985	
40	92.29	165.60		147.28	0.5573	
70	85.39	165.60		147.28	0.5157	
80	77.58	165.60		147.28	0.4685	
110	73.77	165.60		147.28	0.4455	
120	67.68	165.60		147.28	0.4087	
150	64.98	165.60		147.28	0.3924	
180	67.10	165.60		147.28	0.4052	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.46	115.92	-77.93	141.50	0.5302	0.5508
20	45.39	82.80	-61.03	141.50	0.5482	0.4313
30	20.90	165.60	-29.01	147.28	0.1262	0.1970
40	19.68	165.60	-27.14	147.28	0.1188	0.1843
70	15.29	165.60	-21.82	147.28	0.0923	0.1482
80	10.53	165.60	-16.33	147.28	0.0636	0.1109
110	8.34	165.60	-13.46	147.28	0.0504	0.0914
120	4.63	165.60	-8.44	147.28	0.0280	0.0573
150	2.89	165.60	-6.02	147.28	0.0174	0.0408
180	0.01	165.60	-2.32	147.28	0.0000	0.0157

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	90.05	115.92	-106.13	141.50	0.7769	0.7500
20	70.11	82.80	-85.38	141.50	0.8468	0.6034
30	33.31	165.60	-41.23	147.28	0.2012	0.2799
40	31.15	165.60	-38.44	147.28	0.1881	0.2610
70	24.25	165.60	-30.64	147.28	0.1465	0.2080
80	16.45	165.60	-22.13	147.28	0.0993	0.1502
110	12.63	165.60	-17.65	147.28	0.0763	0.1198
120	6.55	165.60	-10.29	147.28	0.0395	0.0699
150	3.84	165.60	-6.93	147.28	0.0232	0.0471
180	0.02	165.60	-2.33	147.28	0.0001	0.0158

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.95	196.56	-18.19	141.50	0.0150	0.1285
20	1.36	140.40	-15.78	141.50	0.0097	0.1115
30	85.67	280.79		147.28	0.3051	
40	79.61	280.79		147.28	0.2835	
70	78.83	280.79		147.28	0.2807	
80	78.21	280.79		147.28	0.2785	
110	78.17	280.79		147.28	0.2784	
120	78.23	280.79		147.28	0.2786	
150	78.40	280.79		147.28	0.2792	
180	86.43	280.79		147.28	0.3078	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio

10		196.56	-9.34	141.50		0.0660
20		140.40	-8.93	141.50		0.0631
30	82.40	280.79		147.28	0.2934	
40	76.59	280.79		147.28	0.2728	
70	76.73	280.79		147.28	0.2733	
80	77.01	280.79		147.28	0.2743	
110	77.35	280.79		147.28	0.2755	
120	77.92	280.79		147.28	0.2775	
150	78.26	280.79		147.28	0.2787	
180	86.42	280.79		147.28	0.3078	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.88	141.50		0.0557
20		82.80	-7.47	141.50		0.0528
30	62.90	165.60		147.28	0.3798	
40	58.48	165.60		147.28	0.3531	
70	58.62	165.60		147.28	0.3540	
80	58.90	165.60		147.28	0.3557	
110	59.24	165.60		147.28	0.3578	
120	59.81	165.60		147.28	0.3612	
150	60.15	165.60		147.28	0.3632	
180	66.55	165.60		147.28	0.4019	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.04	141.50		0.0498
20		82.80	-6.67	141.50		0.0471
30	62.26	165.60		147.28	0.3759	
40	57.86	165.60		147.28	0.3494	
70	58.28	165.60		147.28	0.3519	
80	58.61	165.60		147.28	0.3540	
110	58.93	165.60		147.28	0.3558	
120	59.52	165.60		147.28	0.3594	
150	59.83	165.60		147.28	0.3613	
180	66.15	165.60		147.28	0.3995	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-9.55	141.50		0.0675
20		82.80	-9.13	141.50		0.0646
30	62.90	165.60		147.28	0.3798	
40	58.48	165.60		147.28	0.3531	
70	58.62	165.60		147.28	0.3540	
80	58.90	165.60		147.28	0.3557	
110	59.24	165.60		147.28	0.3578	
120	59.81	165.60		147.28	0.3612	
150	60.15	165.60		147.28	0.3632	
180	66.55	165.60		147.28	0.4019	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.88	141.50		0.0557
20		82.80	-7.47	141.50		0.0528
30	62.90	165.60		147.28	0.3798	
40	58.48	165.60		147.28	0.3531	
70	58.62	165.60		147.28	0.3540	
80	58.90	165.60		147.28	0.3557	
110	59.24	165.60		147.28	0.3578	
120	59.81	165.60		147.28	0.3612	
150	60.15	165.60		147.28	0.3632	

180	66.55	165.60		147.28	0.4019	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-9.55	141.50		0.0675
20		82.80	-9.13	141.50		0.0646
30		165.60	-3.96	147.28		0.0269
40		165.60	-3.64	147.28		0.0247
70		165.60	-2.85	147.28		0.0193
80		165.60	-2.40	147.28		0.0163
110		165.60	-2.05	147.28		0.0139
120		165.60	-1.31	147.28		0.0089
150		165.60	-0.97	147.28		0.0066
180	0.30	165.60	-0.51	147.28	0.0018	0.0035

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.97	96.60	-60.05	117.92	0.4552	0.5092
20	31.88	69.00	-47.15	117.92	0.4621	0.3999
30	80.26	138.00		122.73	0.5816	
40	74.87	138.00		122.73	0.5426	
70	72.31	138.00		122.73	0.5240	
80	69.33	138.00		122.73	0.5024	
110	67.89	138.00		122.73	0.4920	
120	65.19	138.00		122.73	0.4724	
150	63.77	138.00		122.73	0.4621	
180	67.08	138.00		122.73	0.4861	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.97	96.60	-60.05	117.92	0.4552	0.5092
20	31.88	69.00	-47.15	117.92	0.4621	0.3999
30	14.46	138.00	-22.37	122.73	0.1048	0.1823
40	13.74	138.00	-21.02	122.73	0.0995	0.1713
70	11.18	138.00	-17.56	122.73	0.0810	0.1431
80	8.19	138.00	-13.87	122.73	0.0593	0.1130
110	6.75	138.00	-11.77	122.73	0.0490	0.0959
120	4.05	138.00	-7.79	122.73	0.0293	0.0635
150	2.64	138.00	-5.73	122.73	0.0191	0.0467
180	0.01	138.00	-2.31	122.73	0.0001	0.0188

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-9.55	141.50		0.0675
20		82.80	-9.13	141.50		0.0646
30	62.90	165.60		147.28	0.3798	
40	58.48	165.60		147.28	0.3531	
70	58.62	165.60		147.28	0.3540	
80	58.90	165.60		147.28	0.3557	
110	59.24	165.60		147.28	0.3578	
120	59.81	165.60		147.28	0.3612	
150	60.15	165.60		147.28	0.3632	
180	66.55	165.60		147.28	0.4019	

Absolute Maximum of the all of the Stress Ratio's 0.8468

Governing Element: Skirt-1
Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side

Skirt-2	...	-8.0	1.7	96.4	90.1	+	Tens
Skirt-2	...	-8.0	-1.7	-96.4	-106.1	-	Comp
Skirt-1	...	-7.6	1.7	76.0	70.1	+	Tens
Skirt-1	...	-7.6	-1.7	-76.0	-85.4	-	Comp
Bottom head	...	-3.1	0.9	36.3	34.1	+	Tens
Bottom head	...	-3.1	-0.9	-36.3	-40.4	-	Comp
Shell-1	...	-2.9	0.9	33.9	31.9	+	Tens
Shell-1	...	-2.9	-0.9	-33.9	-37.7	-	Comp
Shell-2	...	-2.4	0.5	26.9	25.0	+	Tens
Shell-2	...	-2.4	-0.5	-26.9	-29.9	-	Comp
Shell-3	...	-2.1	0.5	18.8	17.2	+	Tens
Shell-3	...	-2.1	-0.5	-18.8	-21.4	-	Comp
Shell-4	...	-1.7	0.5	14.7	13.4	+	Tens
Shell-4	...	-1.7	-0.5	-14.7	-16.9	-	Comp
Shell-5	...	-1.1	0.4	8.0	7.3	+	Tens
Shell-5	...	-1.1	-0.4	-8.0	-9.5	-	Comp
Shell-6	...	-0.8	0.4	5.0	4.6	+	Tens
Shell-6	...	-0.8	-0.4	-5.0	-6.2	-	Comp
Top head	...	-0.3	0.4	0.8	0.8	+	Tens
Top head	...	-0.3	-0.4	-0.8	-1.5	-	Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

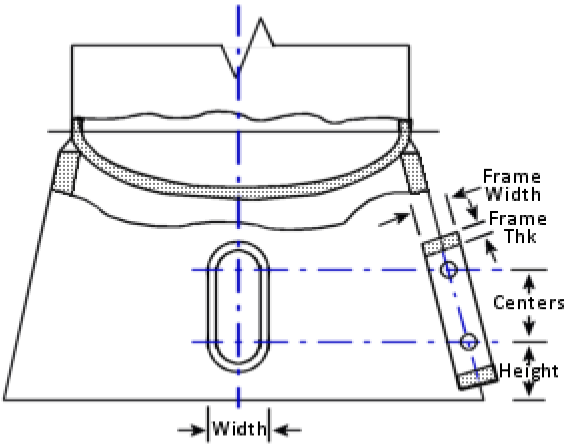
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	206066.328	N
Weight of vessel contents (liquid)	Ff	60020.727	N
Vessel Gross Weight	Fg + Ff	266087.062	N
Skirt Mean Diameter	Dz	797.000	mm
Skirt Analysis Thickness (Corroded)	ez	14.400	mm
Skirt Height	H	2200.000	mm
Additional Axial Force at section 1-1	F1	-14844.800	N
Moment at section 1-1	M1	536792864.000	N-mm
Applied Moment - Bottom of Skirt	Mb	680449472.000	N-mm
Weight at Bottom of Skirt	W	282091.125	N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	300.0000	0.0	700.00	0.00	80.000	18.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-22.1116	22.1116

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-337.888	-22.112

Height of Weakest Section	H	700.030	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	-19.076	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	2886281216.000	mm4
Principle Moment of Inertia	Iyy	2574906112.000	mm4
Distance to Extreme Fiber	ybar	405.700	mm
Distance to Extreme Fiber	xbar	424.776	mm
Minimum Section Modulus	W4	6061800.500	mm3
Metal Area at the Cutting Plane	A4	34506.309	mm^2

Half Opening Width Subtended Angle at Weakest Section [δ]:
= sin-1(Largest Opening Width / Dz)
= sin-1(300.0/797.0)
= 0.3859 rad (22.1116 deg)

Skirt Mean Radius at Section 4-4 [r]:
= Dz / 2
= 797.0/2
= 398.5 mm

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:
= $\delta \cdot \sqrt{r / ez}$
= $0.386 \cdot \sqrt{398.5/14.4}$
= 2.030

WARNING: ratio Exceeds 2 - See EN 13445 16.12.76(a).
This equation is a precaution against radial bucking at the opening.
The frame (if there is one), may prevent this. The code does not
say what size this frame should be. Your judgement is required.

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:
= ex(new) - ci - cext
= 16.0 - 0.8 - 0.8
= 14.4 mm

Find the moment acting at the Critical Height [M4]:
= (Mbottom - Mtop)•(H - CutHeight)/H + Mtop
= (.68045E+09 - 0.53679E+09)•(2200.0 - 700.03)/2200.0 + 0.53679E+09
= 0.63474E+09 N-mm

Compute the section modulus [Zxx and Zyy]:
Zxx = Ixx/ybar = 0.28863E+10/405.7 = 7114331 mm^3
Zyy = Iyy/xbar = 0.25749E+10/424.776 = 6061801 mm^3
Z = min(Zxx, Zyy) = min(7114331,6061801) = 6061801 mm^3

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:
= $\sqrt{\min(Ixx, Iyy) / Area(A4)}$
= $\sqrt{\min(0.28863E+10, 0.25749E+10)/34506.309}$
= 273.169 mm

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:
= rg / L
= 273.169/2200.0
= 0.124

Location of the Skirt Centroid at critical height [e]:
= $\sqrt{ex^2 + ey^2}$
= $\sqrt{-19.076^2 + 0.0^2}$
= 24.625 mm

Where ex = -19.076 : ey = 0.0

Moment Increment at Centroid [ΔM4]:
= abs(F1 - Ff - Fg) • e = F4 • e
= abs(-14844.8 - 206066.33 - 60020.73) • 19.076
= 5361283.000 N-mm

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:
= abs((lt - ea3))/2
= abs((80.0 - 14.4))/2
= 32.800 mm

Mean Diameter of Opening, Figure 16.12-5 [d]:
= opening width - et
= 300.0 - 18.0
= 282.000 mm

Angle, 16.12-72 [δ]:
= sin-1(d / D3)
= sin-1(282.0/797.0)
= 0.362 radians

Area in the Frame [At]:
= 2 • lt • et
= 2 • 80.0 • 18.0
= 2880.000 mm²

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:
= (π - δ) • D3 • ea3
= (π - 0.362) • 797.0 • 14.4
= 31904.738 mm²

Combined Area at Opening, 16.12-73 [A4]:
= AS + AT
= 31904.738 + 2880.0
= 34784.738 mm²

Computed Offset, 16.12-74(b) [yt]:
= 0.5 • D3 • cos(δ) + ht - 0.5 • lt
= 0.5 • 797.0 • cos(0.362) + 32.8 - 0.5 • 80.0
= 365.521 mm

Distance to Neutral Axis, 16.12-74 [yG]:
= (0.5 • D3 • ea3 • d - 2 • lt • eat • yt)/A4
= (0.5 • 797.0 • 14.4 • 282.0 - 2 • 80.0 • 18.0 • 365.521)/34784.738
= 16.258 mm

Computed Offset [ymax]:
= max[abs(0.5 • d3 • cos(δ) + ht + yG) ; abs(0.5 • D3 - yG)]
= max[abs(0.5 • 797.0 • cos(0.362) + 32.8 + 16.258) ; abs(0.5 • 797.0 - 16.258)]
= 421.779 mm

Inertia Value, 16.12-75(b) [IS]:
= (π - δ - sin(δ) • cos(δ)) • ea3 • (D3/2)³
= (π - 0.362 - sin(0.362) • cos(0.362)) • 14.4 • (797.0/2)³
= 2231695616.0 mm⁴

Section Modulus [W4]:
= (Is + At(yt² + lt²/12) - A4 • yG²)/ymax
= (0.22317E+10 + 2880.0(365.521² + 80.0²/12) - 34784.738 • 16.258²)/421.779
= 6185283.000 mm³

Weakening Factor, 16.12-70(b) [psi1]:
= min(1; A4/(π • D3 • ea3)
= min(1; 34784.738/(π • 797.0 • 14.4))
= 0.965

Weakening Factor, 16.12-70(c) [psi2]:

= min(1; 4 • W4 /(π • D²3/(π • D3 • ea3)
= min(1; 4.0 • 6185283/(π • 797.0² • 14.4))
= 0.861

Maximum Bending Moment [Mmax]:
= π/4 • d3² • ea3 • σcall
= π/4 • 797.0² • 14.4 • 141.5
= 1016884352.0 N-mm

Maximum Compressive Force [Fc,max]:
= π • d3 • ea3 • σcall
= π • 797.0 • 14.4 • 141.5
= 5101493.0 N

Unity Check of Weakened Section, 16.12-70 [UC1]:
= F4/(psi1 • Fc,max) + (abs(M4) + F4 • abs(yg))/(psi2 • Mmax))
= 280931.8/(0.965 • 5101493) + (abs(0.63474E+09) + 280931.8 • abs(16.258))/
(0.861 • 0.10169E+10)
= 0.787 must be ≤ to 1, Passed

The stresses are satisfactory in the region with the opening(s).

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt-1

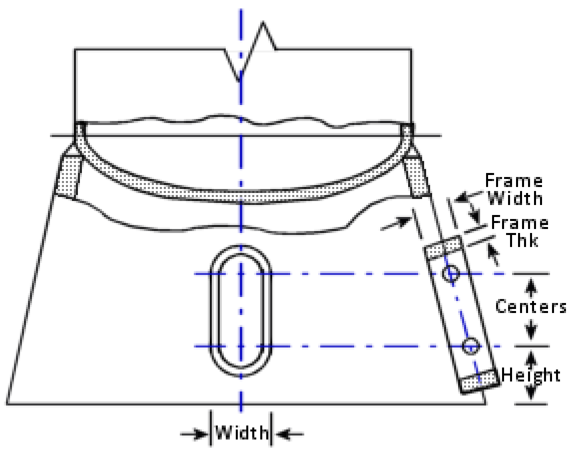
Total vessel (empty) weight	Fg	206066.328	N
Head Weight below section 2-2	DeltaFg	2987.574	N
Weight of vessel contents (liquid)	Ff	60020.727	N
Vessel Gross Weight	Fg + Ff	266087.062	N
Skirt Mean Diameter	Dz	797.000	mm
Skirt Analysis Thickness (Corroded)	ez	14.400	mm
Skirt Height	H	1000.000	mm
Additional Axial Force at section 1-1	F1	-14844.800	N
Moment at section 1-1	M1	471651968.000	N-mm
Applied Moment - Bottom of Skirt	Mb	536792864.000	N-mm
Weight at Bottom of Skirt	W	267170.594	N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical	
Head Mean Diameter at Skirt Top	Db	795.002	mm
Head Corroded Thickness	eb	27.000	mm

Per EN 13445 Section 7.5.4 [R, r]:

```
hi = 193.501 mm
K = Di/(2*hi) = 768.002/(2*193.501 ) = 1.9845
R = Di(0.44*K+0.02) = 768.002*(0.44*1.984+0.02) = 685.9628 mm
r = Di((0.5/K)-0.08) = 768.002 * ((0.5/1.984 - 0.08) = 132.0603 mm
```



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	150.0000	0.0	450.00	0.00	80.000	16.000
2	80.0000	0.0	850.00	0.00	80.000	16.000
3	80.0000	180.0	850.00	0.00	80.000	16.000

Interference check:

Access Access Pitch Ligament

		mm	mm
1	2	400.0000	285.0000
2	3	1251.9036	1171.9037
1	3	1314.2535	1199.2537

Interference Check: No Overlap Detected between Openings

```
Minimum Pitch between openings     is:    400.0000 mm
Minimum Ligament between Openings is:    285.0000 mm
```

Recommended minimum Ligament [lMin]:
= $2 \cdot \sqrt{(D \cdot t)}$
= $2 \cdot \sqrt{(797.0 \cdot 14.4)}$mm
= 214.260 mm

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 3

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
2	-5.7608	5.7608
3	174.2392	185.7608

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
0.000	-5.761
5.761	174.239
-174.239	0.000

Height of Weakest Section	H	849.967	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	24.625	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	2867176960.000	mm4
Principle Moment of Inertia	Iyy	3687956736.000	mm4
Distance to Extreme Fiber	ybar	405.700	mm
Distance to Extreme Fiber	xbar	430.325	mm
Minimum Section Modulus	W4	7067241.000	mm3
Metal Area at the Cutting Plane	A4	41427.543	mm^2

Half Opening Width Subtended Angle at Weakest Section [δ]:
= sin-1(Largest Opening Width / Dz)
= sin-1(150.0/797.0)
= 0.1893 rad (10.8481 deg)

Skirt Mean Radius at Section 4-4 [r]:
= Dz / 2
= 797.0/2
= 398.5 mm

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
2560.0	1938.8	...	...
2560.0	922.6	...	...
2560.0	922.6	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:
= $\delta \cdot \sqrt{(r / ez)}$
= $0.189 \cdot \sqrt{(398.5/14.4)}$
= 0.996

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:
= ex(new) - ci - cext
= 16.0 - 0.8 - 0.8
= 14.4 mm

Find the moment acting at the Critical Height [M4]:
= (Mbottom - Mtop)•(H - CutHeight)/H + Mtop
= (.53679E+09 - 0.47165E+09)•(1000.0 - 849.967)/1000.0 + 0.47165E+09
= 0.48143E+09 N-mm

Compute the section modulus [Zxx and Zyy]:
Zxx = Ixx/ybar = 0.28672E+10/405.7 = 7067241 mm^3
Zyy = Iyy/xbar = 0.36880E+10/430.325 = 8570171 mm^3
Z = min(Zxx, Zyy) = min(7067241,8570171) = 7067241 mm^3

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:
= $\sqrt{(\min(Ixx, Iyy) / Area(A4))}$
= $\sqrt{(\min(0.28672E+10, 0.36880E+10)/41427.543)}$
= 263.077 mm

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:
= rg / L
= 263.077/1000.0
= 0.263

Location of the Skirt Centroid at critical height [e]:
= $\sqrt{(ex^2 + ey^2)}$
= $\sqrt{(24.625^2 + -0.0^2)}$
= 19.076 mm

Where ex = 24.625 : ey = -0.0

Moment Increment at Centroid [ΔM4]:
= abs(F1 - Ff - Fg) • e = F4 • e
= abs(-14844.8 - 206066.33 - 60020.73) • 24.625
= 6920799.500 N-mm

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
= F1 - Fg - Ff + 4 • M1 / Dz
= -14844.8 - 206066.328 - 60020.727 + 4 • 0.47165E+09/797.0
= 2085246 N

Force at the joint, 16.12-2 [Fzq]:
= F1 - Fg - Ff - 4 • M1 / Dz
= -14844.8 - 206066.328 - 60020.727 - 4 • 0.47165E+09/797.0
= -2647109.2 N

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
= (Fzp + ΔFg) + Ff)/(π • Db • eb) + P • Db /(4 • eb)
= (2085246 + 2987.574 + 60020.727)/(π • 795.002 • 27.0) + 9.403 • 795.002/(4 • 27.0)

= 101.08 MPa

Stress [$\sigma1q^m$]:
= (Fzq + ΔFg + Ff)/(π • Db • eb) + P • Db / (4 • eb)
= (-0.3E+07 + 2987.574 + 60020.727)/(π • 795.002 • 27.0) + 9.403 • 795.002/(4 • 27.0)
= 30.89 MPa

Membrane stress at 2-2, 16.12-9:

Stress [$\sigma2p^m$ & $\sigma2q^m$]:
= (Ff + ΔFg)/(π • Db • eb) + P • Db / (4 • eb)
= (60020.727 + 2987.574)/(π • 795.002 • 27.0) + 9.403 • 795.002/(4 • 27.0)
= 70.15 MPa

Membrane stress at 3-3, 16.12 12 & 13:

Stress [$\sigma3p^m$]:
= Fzp / (π • Dz • ez)
= 2085246/(π • 797.0 • 14.4)
= 57.84 MPa

Stress [$\sigma3q^m$]:
= Fzq / (π • Dz • ez)
= -2647109.250/(π • 797.0 • 14.4)
= -73.42 MPa

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:
= 1 - 0.5(Db + eb - Dz + ez)/(r + eb)
= 1 - 0.5(795.0 + 27.0 - 797.0 + 14.4)/(132.06 + 27.0)
= 0.8761

gamma = acos(cos(gamma)) = acos(0.876) = 28.82 deg

Dimension [a]:
= $0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$
= $0.5 \cdot \sqrt{(27.0^2 + 14.4^2 + 2 \cdot 27.0 \cdot 14.4 \cdot 0.876)}$
= 20.11 mm

Bending moment at verification points p and q [Mp & Mq]:

```
Mp         = a * Fzp = 20.11 *    2085245.8 =  41.9513x106 N-mm
Mq         = a * Fzq = 20.11 *   -2647109.2 = -53.2549x106 N-mm
```

Correction factor [C]:
= 0.63 - 0.057(eb/ez)²
= 0.63 - 0.057(27.0/14.4)²
= 0.430

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma1p^b$(o) & $\sigma2p^b$(o)]:
= C • 6 • Mp / (π • Db • eb²)
= 0.43 • 6 • 41951280/(π • 795.002 • 27.0²)
= 59.37 MPa

Stress [$\sigma1q^b$(o) & $\sigma2q^b$(o)]:
= C • 6 • Mq / (π • Db • eb²)
= 0.43 • 6 • -0.5E+08/(π • 795.002 • 27.0²)
= -75.37 MPa

Stress [$\sigma3p^b$(o)]:
= C • 6 • Mp /(π • Dz • ez²)
= 0.43 • 6 • 41951280/(π • 797.0 • 14.4²)
= 208.21 MPa

Stress [$\sigma3q^b$(o)]:
= C • 6 • Mq / (π • Dz • ez²)
= 0.43 • 6 • -0.5E+08/(π • 797.0 • 14.4²)
= -264.31 MPa

Geometry variables, 16.12-32 to 36:
eb/Db = 1.063/31.299 = 0.034
y = 125 * eb/Db = 125 * 27.0/795.002 = 4.2453

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:
= 4.2 - 0.2 • y = 4.2 - 0.2•4.245 = 3.3509

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [σ1b(p) & σ2b(p)]:
= (P + Ph) • Db((γ / γA) • α - 1)/(4 • eb)
= (9.403 + 0.0) • 795.0((28.82/40.0) • 3.351 - 1)/(4 • 27.0)
= 97.89 MPa

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [$\sigma1pi^{tot}$]:
= $\sigma1p^m$ - $\sigma1p^{b(o)}$ + $\sigma1^{b(p)}$
= 101.076 - 59.373 + 97.894
= 139.60 MPa

Stress [$\sigma1po^{tot}$]:
= $\sigma1p^m$ + $\sigma1p^{b(o)}$ - $\sigma1^{b(p)}$
= 101.076 + 59.373 - 97.894
= 62.55 MPa

Stress [$\sigma1qi^{tot}$]:
= $\sigma1q^m$ - $\sigma1q^{b(o)}$ + $\sigma1^{b(p)}$
= 30.893 - -75.37 + 97.894
= 204.16 MPa

Stress [$\sigma1qo^{tot}$]:
= $\sigma1q^m$ + $\sigma1q^{b(o)}$ - $\sigma1^{b(p)}$
= 30.893 + -75.37 - 97.894
= -142.37 MPa

Stress [$\sigma2pi^{tot}$]:
= $\sigma2p^m$ + $\sigma2p^{b(o)}$ + $\sigma2^{b(p)}$
= 70.15 + 59.373 + 97.894
= 227.42 MPa

Stress [$\sigma2po^{tot}$]:
= $\sigma2p^m$ - $\sigma2p^{b(o)}$ - $\sigma2^{b(p)}$
= 70.15 - 59.373 - 97.894
= -87.12 MPa

Stress [$\sigma2qi^{tot}$]:
= $\sigma2q^m$ + $\sigma2q^{b(o)}$ + $\sigma2^{b(p)}$
= 70.15 + -75.37 + 97.894
= 92.67 MPa

Stress [$\sigma2qo^{tot}$]:
= $\sigma2p^m$ - $\sigma2p^{b(o)}$ - $\sigma2^{b(p)}$
= 70.15 - -75.37 - 97.894
= 47.63 MPa

Stress [$\sigma3pi^{tot}$]:
= $\sigma3p^m$ - $\sigma3p^b$
= 57.839 - 208.208
= -150.37 MPa

Stress [$\sigma3po^{tot}$]:
= $\sigma3p^m$ + $\sigma3p^b$
= 57.839 + 208.208
= 266.05 MPa

Stress [$\sigma3qi^{tot}$]:

= $\sigma 3q^m$ - $\sigma 3q^b$
= -73.424 - -264.309
= 190.89 MPa

Stress [$\sigma 3qo^{tot}$]:
= $\sigma 3q^m$ + $\sigma 3q^b$
= -73.424 + -264.309
= -337.73 MPa

Allowable stresses:

For $\sigma 1pi^{tot}$ and $\sigma 1po^{tot}$, the stress must not exceed:
= fs(3 - (1 / 1.5) • ($\sigma 1p^m$ / f)²)
= 137.997(3 - (1/1.5) • (101.076/137.997)²)
= 364.64 MPa

For $\sigma 1qi^{tot}$ and $\sigma 1qo^{tot}$, the stress must not exceed:
= fs(3 - (1 / 1.5) • ($\sigma 1q^m$ / f)²)
= 137.997(3 - (1/1.5) • (30.893/137.997)²)
= 409.38 MPa

For $\sigma 2pi^{tot}$ and $\sigma 2po^{tot}$, the stress must not exceed:
= fs(3 - (1 / 1.5) • ($\sigma 2p^m$ / f)²)'
= 137.997(3 - (1/1.5) • (70.15/137.997)²)
= 390.22 MPa

For $\sigma 2qi^{tot}$ and $\sigma 2qo^{tot}$, the stress must not exceed:
= fs(3 - (1 / 1.5) • ($\sigma 2q^m$ / f)²)
= 137.997(3 - (1/1.5) • (70.15/137.997)²)
= 390.22 MPa

For $\sigma 3pi^{tot}$ & $\sigma 3po^{tot}$, the stress must not exceed:
= fs(3 - (1 / 1.5) • ($\sigma 3p^m$ / f)²)
= 137.997(3 - (1/1.5) • (57.839/137.997)²)
= 397.83 MPa

For $\sigma 3qi^{tot}$ & $\sigma 3qo^{tot}$, the stress must not exceed:
= fs(3 - (1 / 1.5) • ($\sigma 3q^m$ / f)²)
= 137.997(3 - (1/1.5) • (-73.424/137.997)²)
= 387.95 MPa

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:
= min(1; A4/(π • D3 • ea3)
= min(1; 41427.543/(π • 797.0 • 14.4))
= 1.000

Weakening Factor, 16.12-70(c) [psi2]:
= min(1; 4 • W4 /(π • D²3/(π • D3 • ea3)
= min(1; 4.0 • 8570171/(π • 797.0² • 14.4))
= 1.000

Maximum Bending Moment [M_{max}]:
= π/4 • d3² • ea3 • σcall
= π/4 • 797.0² • 14.4 • 141.5
= 1016884352.0 N-mm

Maximum Compressive Force [$F_{c,max}$]:
= π • d3 • ea3 • σcall
= π • 797.0 • 14.4 • 141.5
= 5101493.0 N

Unity Check of Weakened Section, 16.12-70 [UC1]:
= F4/(psi1 • $F_{c,max}$) + (abs(M4) + F4 • abs(yg))/(psi2 • M_{max}))
= 280931.8/(1.0 • 5101493) + (abs(0.48143E+09) + 280931.8 • abs(24.625))/
(1.0 • 0.10169E+10)
= 0.535 must be ≤ to 1, Passed

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	139.60	364.64	Pass	46
Stot1po	62.55	364.64	Pass	47
Stot1qi	204.16	409.38	Pass	48
Stot1qo	-142.37	409.38	Pass	49
Stot2pi	227.42	390.22	Pass	50
Stot2po	-87.12	390.22	Pass	51
Stot2qi	92.67	390.22	Pass	52
Stot2qo	47.63	390.22	Pass	53
Stot3pi	-150.37	397.83	Pass	54
Stot3po	266.05	397.83	Pass	55
Stot3qi	190.89	387.95	Pass	56
Stot3qo	-337.73	387.95	Pass	57

The stresses are satisfactory in the region without openings.
The stresses are satisfactory in the region with the opening(s).

Shop/Field Installation Options :

Packing is installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	9978.749 mm
Center of Gravity of the Liquid	10437.472 mm
Center of Gravity of the Insulation	10328.857 mm
Center of Gravity of the Nozzles	9155.775 mm
Center of Gravity of the Added Weights (Operating)	10953.780 mm
Center of Gravity of the Added Weights (Empty)	10782.169 mm
Center of Gravity of Bare Shell New and Cold	10093.839 mm
Center of Gravity of Bare Shell Corroded	10123.719 mm
Vessel CG in the Operating Condition	10275.646 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	10158.363 mm
Vessel CG in the Test Condition	10250.071 mm

Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	813.0000	mm
Skirt Thickness	STHK	16.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basering Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basering	TBA, ea4	54.0000	mm	
Design Temperature of the Basering		38.00	°C	
Basering Material		SA-516 70		
Basering Ope. All. Stress	BASOPE, f4	138.00	MPa	
Basering Yield Stress		262.00	MPa	
Inside Diameter of Basering	DI, D4	629.0000	mm	
Outside Diameter of Basering	DOU, D5	1149.0000	mm	
Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm	
Bolt Corrosion Allowance	BCA	1.5000	mm	
Bolt Material		SA-193 B7		
Bolt Material Class/Thickness Range		<= 64		
Bolt Operating Allowable Stress	SA, fB	138.00	MPa	
Number of Bolts	RN, nb	16		
Diameter of Bolt Circle	DC, DBC	981.0000	mm	
Bolt Allowable Shear Stress		82.800	MPa	
Ultimate Comp. Strength of Concrete	FPC	41.0	MPa	
Allowable Comp. Strength of Concrete	FC	22.5	MPa	
Modular ratio Steel/Concrete		9.833		
Thickness of Gusset Plates	TGA, e7	16.0000	mm	
Width of Gussets at Top Plate	TWDT	158.0000	mm	
Width of Gussets at Base Plate	BWDT	158.0000	mm	
Gusset Plate Elastic Modulus	E	201746.0	MPa	
Gusset Plate Yield Stress	SY	262.0	MPa	
Height of Gussets	HG, h1	252.0000	mm	
Distance between Gussets	RG, b6	96.0000	mm	
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm	
Number of Gussets per bolt	NG	2		
Thickness of Top Plate or Ring	TTA, e5	44.0000	mm	
Radial Width of the Top Plate	TOPWTH, b5	168.0000	mm	
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm	
External Corrosion Allowance	CA	3.0000	mm	
Dead Weight of Vessel	DW	230109.9	N	
Operating Weight of Vessel	ROW	296935.9	N	
Earthquake Moment on Basering	EQMOM	326696992.0	N-mm	
Wind Moment on Basering	WIMOM	126208888.0	N-mm	
User Moment at the Base	[EarthQuake Case]	355082560.0	N-mm	(UEQMOM)
User Moment at the Base	[WindLoad Case]	355082560.0	N-mm	(UWIMOM)
Percent Bolt Preload	ppl	100.0		
Use AISC A5.2 Increase in Fc and Bolt Stress		No		
Use Allowable Weld Stress per AISC J2.5		No		
Factor for Increase of Allowables	Fact	1.0000		

Results for Brownell and Young Basering Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:
= ((4 • M/DC) - W) / nb per Jawad & Farr, Eq. 12.3

= ((4 • 0.68178E+09/981.0) - 289748)/16
= 155566.4 N

Bolt Load, Windload Empty Case:
= ((4 • M/DC) - W) / nb per Jawad & Farr, Eq. 12.3
= ((4 • 0.48129E+09/981.0) - 230110)/16
= 108221.7 N

Bolt Load, Windload Vortex Shedding Case:
= ((4 • M/DC) - W) / nb per Jawad & Farr, Eq. 12.3
= ((4 • 0/981.0) - 296936)/16
= -18558.5 N

Governing Bolt Load Condition, Earthquake + Operating Condition:

```
 Area Available in 1 Bolt (corroded)      Abss :       1155.1653  mm²
 Area Available in all the Bolts      Abss * RN :     18482.6445  mm²
```

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.150	1.511	2.441	0.438	0.781	726998.4	1016746.6	65318.0
4	0.225	0.263	1.297	2.605	0.454	0.777	724754.2	1014502.5	61021.6
6	0.282	0.291	1.459	2.482	0.442	0.780	726294.7	1016042.9	64181.0
8	0.296	0.298	1.498	2.451	0.439	0.781	726813.8	1016562.1	65029.5
10	0.299	0.299	1.507	2.444	0.438	0.781	726951.6	1016699.9	65245.6

The Actual Stress in a Single Bolt [Sbolt]:
= 2 • Ft / (T1 • Dc • Ct)
= 2 • 726952/(5.997 • 981.0 • 2.444)
= 101.139 MPa, Should be ≤ 138.0 MPa

Thickness of the Band of Bolting Steel [T1]
= RN • Bolt Area / (π • Dc)
= 16 • 1155.165/(π • 981.0)
= 5.997 mm

Check the Bearing Stress in the Concrete [fc(max)]
= fc'[(2kd + t3) / (2kd)]
= 4.394[(2•0.2993•981.0+260.0)/(2•0.2993•981.0)]
= 6.34 MPa, Should be ≤ 22.5 MPa

Values for table 10.3, l = 168.0, b = 96.309, l/b = 1.744377

Maximum Moment per unit width [Mmax]:
= max(Mx, My) = max(7462.164, 22273.053) = 22273.053 N

Reqd Thickness of Basering, Brownell & Young Method [T]:
= √(6 • Mmax / fallow) + Ca
= √(6 • 22273.053/172.9) + 3.0
= 30.801 mm

Nomenclature:

```
 a   = ( Dc-Ds )/2         Skirt Distance to Bolt Circle
 P   = Sa * Abss           Maximum Load on one Bolt
 l                         Radial Top Plate Width
 lavg                      Average Gusset Plate Width
 g1  = Gamma 1             Constant Term f( b/l )
 g2  = Gamma 2             Constant Term f( b/l )
 g   = Flat distance / 2   Nut 1/2 Dimension (from Tema)
 Fb                        Allowable Bending Stress
```

Values for table 10.6, l = 168.0, b = 96.0, b/l = 0.571429
As b/l (0.571) is less than 1, inverting b/l = 1.75.

Moment Term, based on geometry [Mo]:
= P/(4 • π) [1.3(ln((2lsin(π•a/l)/(π•g))) + 1] - [(0.7-g2)P/(4π)]
= 159399.25/(4 • π) [1.3(ln((2•168.0•sin(π • 84.0/168.0))/
(π • 40.0))) + 1] - [(0.7 - 0.044)• 159399.25/(4 • π)]
= 20587.5703 N

Required Thickness of Continuous Top Ring [Tc]:

$= \sqrt{(6 \cdot Abs(Mo) / Fb) + Ca}$
$= \sqrt{(6 \cdot Abs(20587.57)/174.66) + 3.0}$
= 29.5947 mm

Required Gusset Plate Thickness [tg]:
= P / (Stress Term • lavg) + Ca
= 159399.25/(124.11 • 158.0) + 3.0
= 12.525 (not less than 9.525 + 3.0) mm

Bolt spacing [m]:
= π • Bolt Circle Diameter / number of Bolts
= π • 981.0/16 = 192.619 mm

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:
$= 1.76(P \cdot a/(m \cdot (h + tba) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$
$= 1.76(159399 \cdot 84.0/(192.62 \cdot 306.0 \cdot 207))^{2/3} \cdot 406.5^{1/3}+Ca$
= 13.872 + 1.6 = 15.472 mm

Shear Stress in a Single Bolt [τb]:
= Shear Force/(Bolt Area • Number of Bolts)
= 65625.5/(1155.165 • 16)
= 3.6 MPa. Must be less than 82.8 MPa.

Summary of Basering Thickness Calculations

Required Basering Thickness (tension)	30.8013	mm
Actual Basering Thickness as entered by user	54.0000	mm
Required Thickness of Chair Cap	29.5947	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	12.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	15.4722	mm
Given Thickness of Skirt	16.0000	mm
Required Gusset Height to meet local stress	283.3070	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
= SkirtStress • (SkirtThickness - CA)
= 106.126 • (16.0 - 1.6)
= 1528.09 N/mm

Results for Computed Minimum Basering Weld Size [BWeld]:
= W / [(0.4 • Yield) • 2 • 0.707]
= 1528/[(0.4 • 243) • 2 • 0.707]
= 11.138 mm

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
= Bolt Load / (Cmwth + 2 • (Hg + Tta))
= 159399.2/(156.0 + 2 • (252.0 + 44.0))
= 213.101 N/mm

Horizontal Plate Load [Wh]:
$= Bolt Load \cdot e / (Cmwth \cdot (Hg+Tta) + 0.6667 \cdot (Hg+Tta)^2)$
$= 159399.2 \cdot 84.0/(156.0 \cdot (296.0) + 0.6667 \cdot (296.0)^2)$
= 128.023 N/mm

Resultant Weld Load [Wr]:
$= \sqrt{(Wv^2 + Wh^2)}$
$= \sqrt{(213.1^2 + 128.02^2)}$
= 248.6 N/mm

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 248.6/[(0.4 • 243) • 2 • 0.707]

= 1.812 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= 159399.2/(2 • 158.0)
= 504.428 N/mm

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= 159399.2 • 84.0/(2 • 296.0 • 158.0)
= 143.148 N/mm

Resultant Weld Load [Wr]:
= $\sqrt{(Wv^2 + Wh^2)}$
= $\sqrt{(504.43^2 + 143.15^2)}$
= 524.346 N/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 524.35/[(0.4 • 243) • 2 • 0.707]
= 3.822 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basering to Skirt Double Fillet Weld Size	11.1376	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	3.8217	mm

Review notes about nozzle loadings and supports in the Warnings report.

Input, Nozzle Desc: N480 From: 30

Pressure for Reinforcement Calculations	P	9.5357	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	762.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	30.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		80.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	13.8750	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		25.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		600.0	mm
Centerline Distance to Tangent		600.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N480

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	88.900	mm.
Actual Thickness Used in Calculation	11.125	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= P•D•K1/(2•Sv•E-0.2•P) per Appendix 1-4(c)
= 9.54•768.002•0.8933/(2•138.0•1.0-0.2•9.54)
= 23.8694 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= P•Ro/(Sn•E+0.4•P) per Appendix 1-1 (a)(1)
= 9.54•44.45/(138•1.0+0.4•9.54)
= 2.9888 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3989 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 13.875/0.5773
= 24.0319 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	145.2992	mm
Parallel to Vessel Wall, opening length	d	72.6496	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	44.3449	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1734.099	77.311	NA
Area in Shell	A1	227.439	1806.916	NA
Area in Nozzle Wall	A2	455.542	685.240	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1230.552	1230.552	NA
TOTAL AREA AVAILABLE	Atot	2013.534	3822.709	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (72.6496•23.8694•1.0+2•22.0•23.8694•1.0•(1-1.0))

= 1734.099 mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 72.65(1.0 • 27.0 - 1.0 • 23.869) - 2 • 22.0
(1.0 • 27.0 - 1.0 • 23.8694) • (1 - 1.0)
= 227.439 mm²

Area Available in Nozzle Projecting Outward [A2]:
= (2 • tlnp)(tn - trn)fr2
= (2 • 44.34)(8.13 - 2.99)1.0
= 455.542 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:
= Wo² • fr2 + (Wi-can/0.707)² • fr2
= 10.0² • 1.0 + (0.0)² • 1.0
= 100.000 mm²

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(44.3,200.0,50.0)) • (25.0 - 11.1) • 1.0)
= 1230.552 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

```
Wall Thickness for Internal/External pressures       ta = 5.9888 mm
Wall Thickness per UG16(b),                        tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure      trb1 = 29.4441 mm
Wall Thickness                  tb1 = max(trb1, tr16b) = 29.4441 mm
Wall Thickness, shell/head, external pressure      trb2 = 3.2849 mm
Wall Thickness                  tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45                      tb3 = 7.8000 mm
```

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(29.4441, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5.9888, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 0.875 * 11.125 = 9.735 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

```
Sustained  :     49.4,  Allowable :    138.0 MPa   Passed
Expansion  :      0.0,  Allowable :    295.6 MPa   Passed
Occasional :     19.2,  Allowable :    183.5 MPa   Passed
Shear      :     25.1,  Allowable :     96.6 MPa   Passed
```

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.735, tr = 2.989, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.444, Temp. Reduction = 41 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve A            -8 °C
Min Metal Temp. at Required thickness (UCS 66.1)         -48 °C
Min Metal Temp. w/o impact per UG-20(f)                  -29 °C
```

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 25.0, tr = 2.989, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.136, Temp. Reduction = 78 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B              -1 °C
Min Metal Temp. at Required thickness (UCS 66.1)          -104 °C
Min Metal Temp. w/o impact per UG-20(f)                    -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly :   -48 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)     -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)       -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 9.54/15.31 = 0.623

Weld Size Calculations, Description: N480

```
Intermediate Calc. for nozzle/shell Welds   Tmin    19.0000  mm
```

Results Per UW-16.1:

```
                     Required Thickness    Actual Thickness
Nozzle Weld        6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
```

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (1734.0995 - 227.4395 + 2 • 22.0 • 1.0 •
(1.0 • 27.0 - 23.8694))138)
= 226904.92 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (1686.0946 + 100.0 - 0.0 • 1.0) • 138
= 246455.64 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1686.0946 + 0.0 + 100.0 + (1188.0)) • 138
= 410382.69 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1686.0946 + 0.0 + 100.0 + 0.0 + (1188.0)) • 138
= 410382.69 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 88.9 • 10.0 • 0.49 • 138
= 94417. N

Shear, Nozzle Wall [Snw]:
= (π •(Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 40.3874) • (25.0 - 3.0) • 0.7 • 138
= 269624. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 88.9 • (30.0 - 3.0) • 0.74 • 138
= 384999. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (94417 + 269624) = 364041 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (94417 + 0 + 384999 + 0) = 479416 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (94417 + 384999 + 0) = 479416 N

Summary of Failure Path Calculations:

```
Path 1-1 = 364041   N  , must exceed W = 226904 N     or W1 = 246455 N
Path 2-2 = 479416   N  , must exceed W = 226904 N     or W2 = 410382 N
Path 3-3 = 479416   N  , must exceed W = 226904 N     or W3 = 410382 N
```

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.3931 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 231.3931 mm

Input Echo, WRC107/537 Item 1, Description: N480 :

```
Diameter Basis for Vessel                  Vbasis      ID
Cylindrical or Spherical Vessel            Cylsph      Spherical
Internal Corrosion Allowance                  Cas      3.0000  mm
Vessel Diameter                                Dv    1371.604  mm
Vessel Thickness                               Tv      30.000  mm

Design Temperature                             T1        82.0  C
Vessel Material                                     SA-516 70
Vessel UNS Number                                      K02700
Vessel Cold S.I. Allowable                    Smc      138.00  MPa
Vessel Hot  S.I. Allowable                    Smh      138.00  MPa
```

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

```
Attachment Type                              Type      Round
WRC107 Attachment Classification           Holsol      Hollow

Diameter Basis for Nozzle                  Nbasis      OD
Corrosion Allowance for Nozzle                Can      3.0000  mm
Nozzle Diameter                                Dn     116.650  mm
Nozzle Thickness                               Tn      25.000  mm
Nozzle Material                                        SA-105
Nozzle UNS Number                                      K03504
Nozzle Cold S.I. Allowable                   SNmc      138.00  MPa
Nozzle Hot  S.I. Allowable                   SNmh      138.00  MPa

Design Internal Pressure                       Dp       9.536  MPa
Include Pressure Thrust                                    No
```

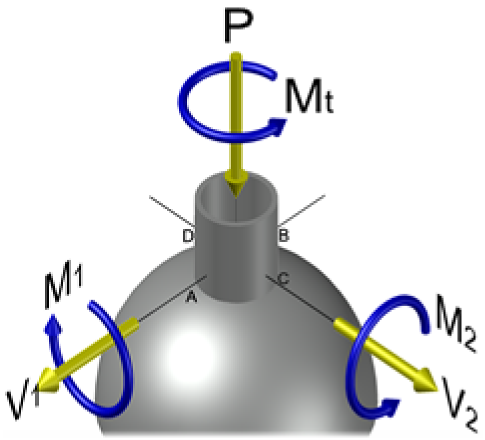
External Forces and Moments in WRC 107/537 Convention:

```
Radial Load            (SUS)               P     -5997.7   N
Longitudinal Shear     (SUS)          (Vl) V1     3998.5   N
Circumferential Shear  (SUS)          (Vc) V2     3998.5   N
Circumferential Moment (SUS)          (Mc) M1   896046.4   N-mm
Longitudinal Moment    (SUS)          (Ml) M2   896046.4   N-mm
Torsional Moment       (SUS)               Mt  1280070.6   N-mm
```

```
Use Interactive Control                              No
WRC107 Version                               Version March  1979

Include Pressure Stress Indices per Div. 2                   No
Compute Pressure Stress per WRC-368                          No
Local Loads applied at end of Nozzle/Attachment              No
```

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
= 116.65 + 2 • 1.65 • √(702.302 (30.0 - 3.0))
= 571.070 mm

WRC 107 Stress Calculation for SUStained loads:

```
Radial Load                                 P      -5997.7   N
Circumferential Shear                 (VC)  V2      3998.5   N
Longitudinal Shear                    (VL)  V1      3998.5   N
Circumferential Moment                (MC)  M1    896046.4   N-mm
Longitudinal Moment                   (ML)  M2    896046.4   N-mm
Torsional Moment                            MT   1280070.6   N-mm

Dimensionless Param: U =    0.42 TAU =    5.00 (   2.15) RHO =    1.23
```

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 2	0.08312	(A,B,C,D)
M(x) / P	SP 2	0.09214	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 2	0.17564	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 2	0.25353	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 2	0.17564	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 2	0.25353	(A,B,C,D)
N(y) * T / P	SP 2	0.16411	(A,B,C,D)
M(y) / P	SP 2	0.04349	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 2	0.08855	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 2	0.14753	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 2	0.08855	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 2	0.14753	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

	Stress Intensity Values at

Type of Stress Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Rad. Bend. P	4.5	-4.5	4.5	-4.5	4.5	-4.5	4.5	-4.5
Rad. Memb. MC	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Rad. Bend. MC	0.0	0.0	0.0	0.0	-13.6	13.6	13.6	-13.6
Rad. Memb. ML	-1.6	-1.6	1.6	1.6	0.0	0.0	0.0	0.0
Rad. Bend. ML	-13.6	13.6	13.6	-13.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-9.9	8.1	20.4	-15.9	-9.9	8.1	20.4	-15.9
Tang. Memb. P	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Tang. Bend. P	2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-7.9	7.9	7.9	-7.9
Tang. Memb. ML	-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Tang. Bend. ML	-7.9	7.9	7.9	-7.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-5.2	6.3	12.2	-7.9	-5.2	6.3	12.2	-7.9
Shear VC	0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Tot. Shear	3.0	3.0	1.4	1.4	1.4	1.4	3.0	3.0
Str. Int.	11.4	10.4	20.6	16.1	10.3	8.9	21.4	16.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress Load	Stress Intensity Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)	121.8	121.8	121.8	121.8	121.8	121.8	121.8	121.8
Rad. Pl (SUS)	-0.9	-0.9	2.3	2.3	-0.9	-0.9	2.3	2.3
Rad. Q (SUS)	-9.0	9.0	18.1	-18.1	-9.0	9.0	18.1	-18.1
Long. Pm (SUS)	121.8	121.8	121.8	121.8	121.8	121.8	121.8	121.8
Long. Pl (SUS)	0.6	0.6	2.1	2.1	0.6	0.6	2.1	2.1
Long. Q (SUS)	-5.8	5.8	10.0	-10.0	-5.8	5.8	10.0	-10.0
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear Q (SUS)	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Pm (SUS)	121.8	121.8	121.8	121.8	121.8	121.8	121.8	121.8
Pm+Pl (SUS)	122.7	122.7	124.8	124.8	122.7	122.7	124.8	124.8
Pm+Pl+Q (Total)	118.0	132.1	142.4	114.1	117.0	130.7	143.1	114.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.76	138.00	Passed
Pm+Pl (SUS)	124.76	207.00	Passed
Pm+Pl+Q (TOTAL)	143.13	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

Input, Nozzle Desc: N1150 **From: 40**

Pressure for Reinforcement Calculations	P	9.5229	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2113.5002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1225.00	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105 +IT	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	37.5095	mm
Hub Thickness of Integral Nozzle	(tn or x+tp)	50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.8000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	168.275	mm.
Actual Thickness Used in Calculation	9.601	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= P•R/(Sv•E-0.6•P) per UG-27 (c)(1)

= 9.52•384.001/(138•1.0-0.6•9.52)
= 27.6436 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= P•Ro/(Sn•E+0.4•P) per Appendix 1-1 (a)(1)
= 9.52•84.1375/(138•1.0+0.4•9.52)
= 5.6501 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5805 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 40.399/0.5773
= 69.9728 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	310.1452	mm
Parallel to Vessel Wall, opening length	d	155.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	72.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4286.771	303.848	NA
Area in Shell	A1	210.333	3889.408	NA
Area in Nozzle Wall	A2	137.912	873.005	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	5857.826	5857.826	NA
TOTAL AREA AVAILABLE	Atot	6306.071	10720.238	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (155.0726•27.6436•1.0+2•47.0•27.6436•1.0•(1-1.0))
= 4286.771 mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 155.073(1.0 • 29.0 - 1.0 • 27.644) - 2 • 47.0
(1.0 • 29.0 - 1.0 • 27.6436) • (1 - 1.0)
= 210.333 mm²

Area Available in Nozzle Projecting Outward [A2]:
= (2 • tlnp)(tn - trn)fr2
= (2 • 72.5)(6.6 - 5.65)1.0
= 137.912 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:
= Wo² • fr2 + (Wi-can/0.707)² • fr2
= 10.0² • 1.0 + (0.0)² • 1.0
= 100.000 mm²

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(72.5,200.0,100.0)) • (50.0 - 9.6) • 1.0)
= 5857.826 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

```
Wall Thickness for Internal/External pressures      ta = 8.6501 mm
Wall Thickness per UG16(b),                      tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure     trb1 = 30.6436 mm
Wall Thickness                 tb1 = max(trb1, tr16b) = 30.6436 mm
Wall Thickness, shell/head, external pressure     trb2 = 3.2879 mm
Wall Thickness                 tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45                     tb3 = 9.2200 mm
```

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[9.22, max(30.6436, 4.5)]
= 9.2200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(8.6501, 9.22)
= 9.2200 mm

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

```
Sustained   :     84.4,  Allowable :    138.0 MPa  Passed
Expansion   :      0.0,  Allowable :    260.6 MPa  Passed
Occasional  :     53.6,  Allowable :    183.5 MPa  Passed
Shear       :     24.6,  Allowable :     96.6 MPa  Passed
```

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 32.0, tr = 27.644, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.953, Temp. Reduction = 3 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B            6 °C
Min Metal Temp. at Required thickness (UCS 66.1)          3 °C

Gov. MDMT of the nozzle to shell joint welded assembly :  3 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
MDMT of ASME B16.5/47 flange per Matl. Specification       -18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)           -39 °C
```

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

```
Design Pressure/Ambient Rating = 9.52/15.31 = 0.622
```

Weld Size Calculations, Description: N1150

```
Intermediate Calc. for nozzle/shell Welds   Tmin     19.0000  mm
```

Results Per UW-16.1:

```
                     Required Thickness    Actual Thickness
Nozzle Weld         6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
```

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (4286.7715 - 210.3335 + 2 • 47.0 • 1.0 •
(1.0 • 29.0 - 27.6436))138)
= 580083.25 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (5995.7378 + 100.0 - 0.0 • 1.0) • 138
= 841125.06 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (5995.7378 + 0.0 + 100.0 + (2726.0)) • 138
= 1217274.25 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (5995.7378 + 0.0 + 100.0 + 0.0 + (2726.0)) • 138
= 1217274.25 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 168.275 • 10.0 • 0.49 • 138
= 178719. N

Shear, Nozzle Wall [Snw]:
= (π •(Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 80.8369) • (50.0 - 3.0) • 0.7 • 138
= 1152916. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 168.275 • (32.0 - 3.0) • 0.74 • 138
= 782729. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (178719 + 1152916) = 1331634 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (178719 + 0 + 782729 + 0) = 961447 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (178719 + 782729 + 0) = 961447 N

Summary of Failure Path Calculations:

```
Path 1-1 = 1331634 N  , must exceed W = 580083 N    or W1 = 841125 N
Path 2-2 = 961447  N  , must exceed W = 580083 N    or W2 = 1217274 N
Path 3-3 = 961447  N  , must exceed W = 580083 N    or W3 = 1217274 N
```

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 9.4063 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.4062 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 155.07 mm

Input Echo, WRC297 Item 1, Description: N1150 :

```
Diameter Basis for Cylindrical Shell                 ID
Shell Internal Corrosion Allowance               3.0000  mm
Shell External Corrosion Allowance               0.0000  mm
Shell Diameter                                  762.002  mm
Shell Thickness                                 32.0000  mm
Shell Stress Concentration Factor                 1.000
Vessel Material                               SA-516 70
Vessel Cold S.I. Allowable             Smc       138.00  MPa
Vessel Hot  S.I. Allowable             Smh       138.00  MPa
```

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

```
Diameter Basis for Nozzle                            OD
Nozzle Corrosion Allowance                       3.0000  mm
Nozzle Diameter                                 249.073  mm
Nozzle Thickness                                50.0000  mm
Nozzle Stress Concentration Factor                1.000
Nozzle Material                                  SA-105
Nozzle Cold S.I. Allowable            SNmc       138.00  MPa
Nozzle Hot  S.I. Allowable            SNmh       138.00  MPa
```

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.

```
Design Internal Pressure                Dp         9.52  MPa
Radial Load                              P    -11245.70  N
Circumferential Shear                   Vc      7497.10  N
Longitudinal Shear                      Vl      7497.10  N
Circumferential Moment                  Mc    0.315E+07  N-mm
Longitudinal Moment                     Ml    0.315E+07  N-mm
Torsional Moment                        Mt    0.450E+07  N-mm

Include Axial Pressure Thrust                          No
Include Pressure Stress Indices per Div. 2             No
Local Loads applied at end of Nozzle/Attachment        No
```

Warning - The ratio, Dn/Tn (4.981) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • $\sqrt{(\text{Rmean}(t - ca))}$
= 249.073 + 2 • 1.65 • $\sqrt{(398.501(32.0 - 3.0))}$
= 603.827 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	797.002	mm
Nozzle Outside Diameter	(d) =	249.073	mm
Vessel Thickness used	(T) =	29.000	mm
Nozzle Thickness used	(t) =	47.000	mm
T / t	=	0.617	
d / t	=	10.000	

$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.638$

Nr/P	=	0.082
Mr/P	=	0.156
M0/P	=	0.047
N0/P	=	0.107
MrD/Mc	=	0.285
NrDL/Mc	=	0.109
M0d/Mc	=	0.092
N0DL/Mc	=	0.059
MrD/Ml	=	0.134
NrDL/Ml	=	0.076
M0D/Ml	=	0.045
N0DL/Ml	=	0.131

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	121	131	121	131
Outplane Stress Summary	120	130	132	126

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-12	12	12	-12
Inplane Pressure Stress	60	60	60	60
Inplane Stress Summary	61	60	87	38

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	120	130	132	126

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	60	60	60	60
Outplane Stress Summary	56	65	75	51

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-25	25	25	-25
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	121	131	121	131
Inplane Stress Summary	107	143	162	95

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	107	143	162	95

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	25	14	25
Hoop Stress Summary	13	25	17	29

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	5	4	16	3

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	14	25	18	29

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	25	14	25
Hoop Stress Summary	15	26	16	28

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-7	7	7	-7
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	0	9	21	-1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	15	26	21	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	121	131	121	131	121	131	121	131
Circ. Pl (SUS)	0	0	2	2	0	0	2	2
Circ. Q (SUS)	-1	1	7	-7	-13	13	37	-37
Long. Pm (SUS)	60	60	60	60	60	60	60	60
Long. Pl (SUS)	0	0	2	2	1	1	1	1
Long. Q (SUS)	0	0	24	-24	-5	5	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	121.0	131.0	121.0	131.0	121.0	131.0	121.0	131.0
Pm+Pl (SUS)	121.0	131.0	123.0	133.0	121.0	131.0	123.0	133.0
Pm+Pl+Q (Total)	120.0	132.0	130.0	126.0	108.0	144.0	160.0	96.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	133.00	207.00	Passed
Pm+Pl+Q (TOTAL)	160.01	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	25	14	25	14	25	14	25
Circ. Pl (SUS)	0	0	2	2	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-4	4	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	14.0	25.0	14.0	25.0	14.0	25.0	14.0	25.0
Pm+Pl (SUS)	14.0	25.0	16.0	27.0	15.0	26.0	15.0	26.0
Pm+Pl+Q (Total)	14.0	25.0	16.0	27.0	15.0	26.0	19.0	27.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.00	138.00	Passed
Pm+Pl (SUS)	27.00	207.00	Passed
Pm+Pl+Q (TOTAL)	27.00	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

Input, Nozzle Desc: K150 **From: 40**

Pressure for Reinforcement Calculations	P	9.5317	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2113.5002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		250.00	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	187.3250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.8000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	26.924	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= P•R/(Sv•E-0.6•P) per UG-27 (c)(1)
= 9.53•384.001/(138•1.0-0.6•9.53)
= 27.6703 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= R(exp([P/(Sn•E)] - 1) per Appendix 1-2 (a)(1)
= 28.4(exp([9.53/(138.0•1.0]-1)
= 2.0309 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4278 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

```
Parallel to Vessel Wall (Diameter Limit)            Dl    162.6480  mm
Parallel to Vessel Wall                        Rn+tn+t     81.3240  mm
Normal to Vessel Wall (Thickness Limit), no pad   Tlnp     59.8100  mm
```

Taking a UG-36(c)(3)(a) exemption for nozzle: K150.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

```
Wall Thickness for Internal/External pressures      ta = 5.0309 mm
Wall Thickness per UG16(b),                       tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure      trb1 = 30.6703 mm
Wall Thickness                tb1 = max(trb1, tr16b) = 30.6703 mm
Wall Thickness, shell/head, external pressure      trb2 = 3.2879 mm
Wall Thickness                tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45                       tb3 = 8.2578 mm
```

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[8.258, max(30.6703, 4.5)]
= 8.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5.0309, 8.2578)
= 8.2578 mm

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 26.924, tr = 2.031, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.085, Temp. Reduction = 78 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B              1 °C
Min Metal Temp. at Required thickness (UCS 66.1)         -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly :  -104 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)      -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)        -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 9.53/15.31 = 0.623

Weld Size Calculations, Description: K150

```
Intermediate Calc. for nozzle/shell Welds    Tmin     19.0000  mm
```

Results Per UW-16.1:

```
                       Required Thickness    Actual Thickness
Nozzle Weld           6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
```

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6100 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.9350 mm

Input, Nozzle Desc: K250 **From: 40**

Pressure for Reinforcement Calculations	P	9.5227	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2113.5002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1250.00	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	187.3250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.8000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	26.924	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= P•R/(Sv•E-0.6•P) per UG-27 (c)(1)
= 9.52•384.001/(138•1.0-0.6•9.52)
= 27.6430 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= R(exp([P/(Sn•E)] - 1) per Appendix 1-2 (a)(1)
= 28.4(exp([9.52/(138.0•1.0]-1)
= 2.0289 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4278 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	162.6480	mm
Parallel to Vessel Wall	Rn+tn+t	81.3240	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K250.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

```
Wall Thickness for Internal/External pressures      ta = 5.0289 mm
Wall Thickness per UG16(b),                       tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure      trb1 = 30.6430 mm
Wall Thickness              tb1 = max(trb1, tr16b) = 30.6430 mm
Wall Thickness, shell/head, external pressure      trb2 = 3.2879 mm
Wall Thickness              tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45                      tb3 = 8.2578 mm
```

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[8.258, max(30.643, 4.5)]
= 8.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5.0289, 8.2578)
= 8.2578 mm

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 26.924, tr = 2.029, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.085, Temp. Reduction = 78 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B          1 °C
Min Metal Temp. at Required thickness (UCS 66.1)     -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly :  -104 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)      -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)        -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 9.52/15.31 = 0.622

Weld Size Calculations, Description: K250

```
Intermediate Calc. for nozzle/shell Welds    Tmin     19.0000  mm
```

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6100 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.9350 mm

Input, Nozzle Desc: K350 From: 40

Pressure for Reinforcement Calculations	P	9.5285	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2113.5002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		600.00	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	187.3250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.8000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	26.924	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= P•R/(Sv•E-0.6•P) per UG-27 (c)(1)
= 9.53•384.001/(138•1.0-0.6•9.53)
= 27.6607 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= R(exp([P/(Sn•E)] - 1) per Appendix 1-2 (a)(1)
= 28.4(exp([9.53/(138.0•1.0]-1)
= 2.0302 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4278 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

```
Parallel to Vessel Wall (Diameter Limit)            Dl    162.6480  mm
Parallel to Vessel Wall                        Rn+tn+t     81.3240  mm
Normal to Vessel Wall (Thickness Limit), no pad   Tlnp     59.8100  mm
```

Taking a UG-36(c)(3)(a) exemption for nozzle: K350.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

```
Wall Thickness for Internal/External pressures      ta = 5.0302 mm
Wall Thickness per UG16(b),                       tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure      trb1 = 30.6607 mm
Wall Thickness                 tb1 = max(trb1, tr16b) = 30.6607 mm
Wall Thickness, shell/head, external pressure      trb2 = 3.2879 mm
Wall Thickness                 tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45                      tb3 = 8.2578 mm
```

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[8.258, max(30.6607, 4.5)]
= 8.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5.0302, 8.2578)
= 8.2578 mm

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 26.924, tr = 2.03, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.085, Temp. Reduction = 78 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B              1 °C
Min Metal Temp. at Required thickness (UCS 66.1)         -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly :  -104 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)      -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)        -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 9.53/15.31 = 0.622

Weld Size Calculations, Description: K350

```
Intermediate Calc. for nozzle/shell Welds   Tmin     19.0000  mm
```

Results Per UW-16.1:

```
                     Required Thickness     Actual Thickness
Nozzle Weld         6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
```

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6100 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.9350 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 81.32 mm

Input, Nozzle Desc: H2200 **From: 70**

Pressure for Reinforcement Calculations	P	9.5084	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	4150.0000	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2870.49	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	34.0000	mm
Hub Thickness of Integral Nozzle	(tn or x+tp)	50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H2200

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	219.075	mm.
Actual Thickness Used in Calculation	15.980	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= P•R/(Sv•E-0.6•P) per UG-27 (c)(1)
= 9.51•384.001/(138•1.0-0.6•9.51)
= 27.5996 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= P•Ro/(Sn•E+0.4•P) per Appendix 1-1 (a)(1)
= 9.51•109.5375/(138•1.0+0.4•9.51)
= 7.3448 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6751 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 34.02/0.5773
= 58.9248 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	386.2309	mm
Parallel to Vessel Wall, opening length	d	193.1154	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	72.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5329.918	504.680	NA
Area in Shell	A1	270.429	4590.988	NA
Area in Nozzle Wall	A2	817.065	1784.177	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	4932.932	4932.932	NA
TOTAL AREA AVAILABLE	Atot	6120.426	11408.097	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (193.1154•27.5996•1.0+2•47.0•27.5996•1.0•(1-1.0))
= 5329.918 mm²

Reinforcement Areas per Figure UG-37.1

HEXAGON

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 193.115(1.0 • 29.0 - 1.0 • 27.6) - 2 • 47.0
(1.0 • 29.0 - 1.0 • 27.5996) • (1 - 1.0)
= 270.429 mm²

Area Available in Nozzle Projecting Outward [A2]:
= (2 • tlnp)(tn - trn)fr2
= (2 • 72.5)(12.98 - 7.34)1.0
= 817.065 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:
= Wo² • fr2 + (Wi-can/0.707)² • fr2
= 10.0² • 1.0 + (0.0)² • 1.0
= 100.000 mm²

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(72.5,200.0,100.0)) • (50.0 - 16.0) • 1.0)
= 4932.932 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

```
Wall Thickness for Internal/External pressures      ta = 10.3448 mm
```

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 10.3448 mm

Available Nozzle Neck Thickness = 15.9798 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 15.98, tr = 7.345, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.566, Temp. Reduction = 25 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve A            6 °C
Min Metal Temp. at Required thickness (UCS 66.1)        -20 °C
```

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 32.0, tr = 27.6, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.952, Temp. Reduction = 3 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B            6 °C
Min Metal Temp. at Required thickness (UCS 66.1)          3 °C

Gov. MDMT of the nozzle to shell joint welded assembly :  3 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)  -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)    -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

```
Design Pressure/Ambient Rating = 9.51/15.31 = 0.621
```

Weld Size Calculations, Description: H2200

```
Intermediate Calc. for nozzle/shell Welds   Tmin     19.0000  mm
```

Results Per UW-16.1:

```
                         Required Thickness     Actual Thickness
Nozzle Weld            6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
```

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)

= max(0, (5329.9185 - 270.4287 + 2 • 47.0 • 1.0 •
(1.0 • 29.0 - 27.5996))138)
= 716301.06 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (5749.9966 + 100.0 - 0.0 • 1.0) • 138
= 807216.31 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (5749.9966 + 0.0 + 100.0 + (2726.0)) • 138
= 1183365.50 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (5749.9966 + 0.0 + 100.0 + 0.0 + (2726.0)) • 138
= 1183365.50 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 219.075 • 10.0 • 0.49 • 138
= 232671. N

Shear, Nozzle Wall [Snw]:
= (π •(Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 103.0476) • (50.0 - 3.0) • 0.7 • 138
= 1469690. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 219.075 • (32.0 - 3.0) • 0.74 • 138
= 1019024. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (232671 + 1469691) = 1702362 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (232671 + 0 + 1019024 + 0) = 1251696 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (232671 + 1019024 + 0) = 1251696 N

Summary of Failure Path Calculations:

```
Path 1-1 = 1702361 N  , must exceed W = 716301 N     or W1 = 807216 N
Path 2-2 = 1251695 N  , must exceed W = 716301 N     or W2 = 1183365 N
Path 3-3 = 1251695 N  , must exceed W = 716301 N     or W3 = 1183365 N
```

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 16.0855 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 248.0855 mm

Input, Nozzle Desc: H1200 From: 110

Pressure for Reinforcement Calculations	P	9.4641	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	4128.0000	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7826.24	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	32.7400	mm
Hub Thickness of Integral Nozzle	(tn or x+tp)	50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H1200

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	219.075	mm.
Actual Thickness Used in Calculation	18.263	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= P•R/(Sv•E-0.6•P) per UG-27 (c)(1)
= 9.46•384.001/(138•1.0-0.6•9.46)
= 27.4656 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= P•Ro/(Sn•E+0.4•P) per Appendix 1-1 (a)(1)
= 9.46•109.5375/(138•1.0+0.4•9.46)
= 7.3115 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6751 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 31.737/0.5773
= 54.9708 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	377.0996	mm
Parallel to Vessel Wall, opening length	d	188.5498	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	72.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5178.628	491.514	NA
Area in Shell	A1	289.316	4484.917	NA
Area in Nozzle Wall	A2	1152.902	2115.187	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	4601.923	4601.923	NA
TOTAL AREA AVAILABLE	Atot	6144.141	11302.026	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (188.5498•27.4656•1.0+2•47.0•27.4656•1.0•(1-1.0))
= 5178.628 mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 188.55(1.0 • 29.0 - 1.0 • 27.466) - 2 • 47.0
(1.0 • 29.0 - 1.0 • 27.4656) • (1 - 1.0)
= 289.316 mm²

Area Available in Nozzle Projecting Outward [A2]:
= (2 • tlnp)(tn - trn)fr2
= (2 • 72.5)(15.26 - 7.31)1.0
= 1152.902 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:
= Wo² • fr2 + (Wi-can/0.707)² • fr2
= 10.0² • 1.0 + (0.0)² • 1.0
= 100.000 mm²

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(72.5,200.0,100.0)) • (50.0 - 18.3) • 1.0)
= 4601.923 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

```
Wall Thickness for Internal/External pressures      ta = 10.3115 mm
```

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 10.3115 mm

Available Nozzle Neck Thickness = 0.875 * 18.263 = 15.980 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 15.98, tr = 7.312, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.563, Temp. Reduction = 26 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve A              6 °C
Min Metal Temp. at Required thickness (UCS 66.1)          -20 °C
```

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 32.0, tr = 27.466, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.947, Temp. Reduction = 3 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B              6 °C
Min Metal Temp. at Required thickness (UCS 66.1)            3 °C

Gov. MDMT of the nozzle to shell joint welded assembly :    3 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)      -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

```
Design Pressure/Ambient Rating = 9.46/15.31 = 0.618
```

Weld Size Calculations, Description: H1200

```
Intermediate Calc. for nozzle/shell Welds    Tmin     19.0000  mm
```

Results Per UW-16.1:

```
                      Required Thickness    Actual Thickness
Nozzle Weld          6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
```

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)

= max(0, (5178.6279 - 289.3158 + 2 • 47.0 • 1.0 •
(1.0 • 29.0 - 27.4656))138)
= 694558.06 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (5754.8252 + 100.0 - 0.0 • 1.0) • 138
= 807882.50 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (5754.8252 + 0.0 + 100.0 + (2726.0)) • 138
= 1184031.75 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (5754.8252 + 0.0 + 100.0 + 0.0 + (2726.0)) • 138
= 1184031.75 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 219.075 • 10.0 • 0.49 • 138
= 232671. N

Shear, Nozzle Wall [Snw]:
= (π •(Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 101.9062) • (50.0 - 3.0) • 0.7 • 138
= 1453411. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 219.075 • (32.0 - 3.0) • 0.74 • 138
= 1019024. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (232671 + 1453411) = 1686083 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (232671 + 0 + 1019024 + 0) = 1251696 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (232671 + 1019024 + 0) = 1251696 N

Summary of Failure Path Calculations:

```
Path 1-1 = 1686082 N  , must exceed W = 694558 N    or W1 = 807882 N
Path 2-2 = 1251695 N  , must exceed W = 694558 N    or W2 = 1184031 N
Path 3-3 = 1251695 N  , must exceed W = 694558 N    or W3 = 1184031 N
```

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 16.0855 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 248.0855 mm

Input, Nozzle Desc: N280 **From: 150**

Pressure for Reinforcement Calculations	P	9.4225	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2400.0002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12502.21	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		80.0000	mm
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	82.1444	mm
Height of Beveled Transition	L'	22.4444	mm
Hub Thickness of Integral Nozzle	(tn or x+tp)	50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N280

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	88.900	mm.
Actual Thickness Used in Calculation	9.735	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= P•R/(Sv•E-0.6•P) per UG-27 (c)(1)

= 9.42•384.001/(138•1.0-0.6•9.42)
= 27.3396 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= P•Ro/(Sn•E+0.4•P) per Appendix 1-1 (a)(1)
= 9.42•44.45/(138•1.0+0.4•9.42)
= 2.9543 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3989 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 30.981/0.5773
= 53.6605 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	150.8618	mm
Parallel to Vessel Wall, opening length	d	75.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	70.4969	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2062.252	156.320	NA
Area in Shell	A1	125.244	1874.855	NA
Area in Nozzle Wall	A2	532.991	893.282	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	-0.000	-0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	4368.113	4368.113	NA
TOTAL AREA AVAILABLE	Atot	5026.348	7136.250	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (75.4309•27.3396•1.0+2•37.7154•27.3396•1.0•(1-1.0))
= 2062.252 mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 75.431(1.0 • 29.0 - 1.0 • 27.34) - 2 • 37.715
(1.0 • 29.0 - 1.0 • 27.3396) • (1 - 1.0)
= 125.244 mm²

Area Available in Nozzle Projecting Outward [A2]:
= (2 • tlnp)(tn - trn)fr2
= (2 • 70.5)(6.73 - 2.95)1.0
= 532.991 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:
= (Wo² - Area Lost)•fr2 + ((Wi-can/0.707)² - Area Lost)•fr2
= (10.0² - 100.0) • 1.0 + (0.0² -0.0) • 1.0
= -0.000 mm²

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(70.5,200.0,82.1)) • (40.7 - 9.7) • 1.0)
= 4368.113 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

```
Wall Thickness for Internal/External pressures       ta = 5.9543 mm
Wall Thickness per UG16(b),                        tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure      trb1 = 30.3396 mm
Wall Thickness                 tb1 = max(trb1, tr16b) = 30.3396 mm
Wall Thickness, shell/head, external pressure      trb2 = 3.2879 mm
Wall Thickness                 tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45                       tb3 = 7.8000 mm
```

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(30.3396, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5.9543, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 9.7346 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

```
Sustained  :     58.9,  Allowable :    138.0 MPa   Passed
Expansion  :      0.0,  Allowable :    286.1 MPa   Passed
Occasional :     24.2,  Allowable :    183.5 MPa   Passed
Shear      :     28.3,  Allowable :     96.6 MPa   Passed
```

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.735, tr = 2.954, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.439, Temp. Reduction = 42 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve A              -8 °C
Min Metal Temp. at Required thickness (UCS 66.1)           -48 °C
Min Metal Temp. w/o impact per UG-20(f)                    -29 °C
```

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 32.0, tr = 27.34, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.943, Temp. Reduction = 3 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B               6 °C
Min Metal Temp. at Required thickness (UCS 66.1)             3 °C

Gov. MDMT of the nozzle to shell joint welded assembly :     3 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)     -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)       -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
```
Design Pressure/Ambient Rating = 9.42/15.31 = 0.615
```

Weld Size Calculations, Description: N280

```
Intermediate Calc. for nozzle/shell Welds   Tmin     19.0000  mm
```

Results Per UW-16.1:

```
                      Required Thickness    Actual Thickness
Nozzle Weld          6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
```

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (2062.2522 - 125.2436 + 2 • 37.7154 • 1.0 •
(1.0 • 29.0 - 27.3396))138)
= 284561.47 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (4901.1045 + -0.0 - 0.0 • 1.0) • 138
= 676282.69 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (4901.1045 + 0.0 + -0.0 + (2187.4958)) • 138
= 978125.94 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (4901.1045 + 0.0 + -0.0 + 0.0 + (2187.4958)) • 138
= 978125.94 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 88.9 • 10.0 • 0.49 • 138
= 94417. N

Shear, Nozzle Wall [Snw]:
= (π •(Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 41.0827) • (40.7154 - 3.0) • 0.7 • 138
= 470185. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 88.9 • (32.0 - 3.0) • 0.74 • 138
= 413517. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SONW + SNW) = (94417 + 470185) = 564602 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (94417 + 0 + 413517 + 0) = 507935 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (94417 + 413517 + 0) = 507935 N

Summary of Failure Path Calculations:

```
Path 1-1 = 564602  N  , must exceed W = 284561 N    or W1 = 676282 N
Path 2-2 = 507934  N  , must exceed W = 284561 N    or W2 = 978125 N
Path 3-3 = 507934  N  , must exceed W = 284561 N    or W3 = 978125 N
```

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 2.6018 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 234.6018 mm

Input Echo, WRC297 Item 1, Description: N280 :

Diameter Basis for Cylindrical Shell		ID	
Shell Internal Corrosion Allowance		3.0000	mm
Shell External Corrosion Allowance		0.0000	mm
Shell Diameter		762.002	mm
Shell Thickness		32.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		169.431	mm
Nozzle Thickness		50.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	9.42	MPa
Radial Load	P	-5997.68	N
Circumferential Shear	Vc	3998.47	N
Longitudinal Shear	Vl	3998.47	N
Circumferential Moment	Mc	896046.44	N-mm
Longitudinal Moment	Ml	896046.44	N-mm
Torsional Moment	Mt	0.128E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (3.389) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

= NozzleOD + 2 • 1.65 • $\sqrt{(\text{Rmean}(t - ca))}$

= 169.431 + 2 • 1.65 • $\sqrt{(398.501(32.0 - 3.0))}$

= 524.185 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	797.002	mm
Nozzle Outside Diameter	(d) =	169.431	mm
Vessel Thickness used	(T) =	29.000	mm
Nozzle Thickness used	(t) =	47.000	mm
T / t	=	0.617	
d / t	=	10.000	

$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.114$

Nr/P	=	0.104
Mr/P	=	0.201
M0/P	=	0.061
N0/P	=	0.125
MrD/Mc	=	0.299
NrDL/Mc	=	0.105
M0d/Mc	=	0.098
N0DL/Mc	=	0.043
MrD/Ml	=	0.183
NrDL/Ml	=	0.090
M0D/Ml	=	0.061
N0DL/Ml	=	0.125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	120	129	120	129
Outplane Stress Summary	120	129	126	126

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	60	60	60	60
Inplane Stress Summary	61	58	76	45

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	120	129	126	126

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	60	60	60	60
Outplane Stress Summary	59	61	67	54

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-11	11	11	-11
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	120	129	120	129
Inplane Stress Summary	117	132	141	111

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	1	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	117	132	141	111

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	16	5	16

Hoop Stress Summary	5	17	7	18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	1	0	9	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	5	17	9	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	16	5	16
Hoop Stress Summary	6	17	6	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	0	2	11	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0

Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6	17	11	19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	120	129	120	129	120	129	120	129
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	4	-4	-3	3	19	-19
Long. Pm (SUS)	60	60	60	60	60	60	60	60
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	2	-2	14	-14	-1	1	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	120.0	129.0	120.0	129.0	120.0	129.0	120.0	129.0
Pm+Pl (SUS)	120.0	129.0	120.0	129.0	120.0	129.0	120.0	129.0
Pm+Pl+Q (Total)	120.0	129.0	124.0	125.0	117.0	132.0	139.0	110.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	129.00	138.00	Passed
Pm+Pl (SUS)	129.00	207.00	Passed
Pm+Pl+Q (TOTAL)	139.00	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	16	5	16	5	16	5	16
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	1	-1	3	-3	-1	1	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	5.0	16.0	5.0	16.0	5.0	16.0	5.0	16.0
Pm+Pl (SUS)	5.0	16.0	5.0	16.0	5.0	16.0	5.0	16.0

Pm+Pl+Q (Total)	5.0	16.0	6.0	16.0	5.0	16.0	8.0	18.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	138.00	Passed
Pm+Pl (SUS)	16.00	207.00	Passed
Pm+Pl+Q (TOTAL)	18.00	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

Input, Nozzle Desc: K450 **From: 150**

Pressure for Reinforcement Calculations	P	9.4156	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2400.0002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		13264.01	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	187.3250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K450

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	26.924	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= P•R/(Sv•E-0.6•P) per UG-27 (c)(1)
= 9.42•384.001/(138•1.0-0.6•9.42)
= 27.3188 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= R(exp([P/(Sn•E)] - 1) per Appendix 1-2 (a)(1)
= 28.4(exp([9.42/(138.0•1.0]-1)
= 2.0053 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4278 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	162.6480	mm
Parallel to Vessel Wall	Rn+tn+t	81.3240	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K450.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

```
Wall Thickness for Internal/External pressures      ta = 5.0053 mm
Wall Thickness per UG16(b),                       tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure      trb1 = 30.3188 mm
Wall Thickness                 tb1 = max(trb1, tr16b) = 30.3188 mm
Wall Thickness, shell/head, external pressure      trb2 = 3.2879 mm
Wall Thickness                 tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45                      tb3 = 8.2578 mm
```

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[8.258, max(30.3188, 4.5)]
= 8.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5.0053, 8.2578)
= 8.2578 mm

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 26.924, tr = 2.005, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.084, Temp. Reduction = 78 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B              1 °C
Min Metal Temp. at Required thickness (UCS 66.1)         -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly :  -104 °C
```

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)     -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)       -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 9.42/15.31 = 0.615

Weld Size Calculations, Description: K450

```
Intermediate Calc. for nozzle/shell Welds    Tmin     19.0000   mm
```

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6100 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.9350 mm

Input, Nozzle Desc: N3150 **From: 180**

```
Pressure for Reinforcement Calculations         P       9.4029  MPa
Temperature for Internal Pressure            Temp           82  °C
Design External Pressure                     Pext         0.10  MPa
Temperature for External Pressure          Tempex           82  °C

Parent Material                                     SA-516 70
Parent Allowable Stress at Temperature         Sv       138.00  MPa
Parent Allowable Stress At Ambient            Sva       138.00  MPa

Inside Diameter of Elliptical Head              D       762.00  mm
Aspect Ratio of Elliptical Head                Ar         2.00
Head Finished (Minimum) Thickness               t      29.5000  mm
Head Internal Corrosion Allowance               c       3.0000  mm
Head External Corrosion Allowance              co       0.0000  mm

Distance from Head Centerline                  L1       0.0000  mm

User Entered Minimum Design Metal Temperature           -28.00  °C
```

Type of Element Connected to the Parent : Nozzle

```
Material                                              SA-105 +IT
Material UNS Number                                     K03504
Material Specification/Type                           Forgings
Allowable Stress at Temperature                Sn       138.00  MPa
Allowable Stress At Ambient                   Sna       138.00  MPa

Diameter Basis (for tr calc only)                      Outside
Layout Angle                                              0.00  deg
Diameter                                              150.0000  mm

Size and Thickness Basis                               Minimum
Nominal Thickness                                           80

Flange Material                                         SA-105

Hub Height of Integral Nozzle                   h     100.0000  mm
Height of Beveled Transition                   L'      41.4000  mm
Hub Thickness of Integral Nozzle  ( tn or x+tp )       50.0000  mm

Corrosion Allowance                           can       3.0000  mm
Joint Efficiency of Shell Seam at Nozzle       E1         1.00
Joint Efficiency of Nozzle Neck                En         1.00

Outside Projection                             ho     200.0000  mm
Weld leg size between Nozzle and Pad/Shell     Wo      10.0000  mm
Groove weld depth between Nozzle and Vessel Wgnv      29.4000  mm
Inside Projection                               h       0.0000  mm
Weld leg size, Inside Element to Shell         Wi       0.0000  mm

Flange Class                                               900
Flange Grade                                            GR 1.1
```

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N3150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

```
Actual Outside Diameter Used in Calculation            168.275  mm.
Actual Thickness Used in Calculation                     9.601  mm
```

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= P•D•K1/(2•Sv•E-0.2•P) per Appendix 1-4(c)

= 9.4•768.002•0.8933/(2•138.0•1.0-0.2•9.4)
= 23.5347 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= P•Ro/(Sn•E+0.4•P) per Appendix 1-1 (a)(1)
= 9.4•84.1375/(138•1.0+0.4•9.4)
= 5.5808 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5805 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 40.399/0.5773
= 69.9728 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	310.1452	mm
Parallel to Vessel Wall, opening length	d	155.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	66.2500	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3649.593	164.822	NA
Area in Shell	A1	459.831	3779.779	NA
Area in Nozzle Wall	A2	135.207	797.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	5352.841	5352.841	NA
TOTAL AREA AVAILABLE	Atot	6047.879	10030.365	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (155.0726•23.5347•1.0+2•47.0•23.5347•1.0•(1-1.0))
= 3649.593 mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 155.073(1.0 • 26.5 - 1.0 • 23.535) - 2 • 47.0
(1.0 • 26.5 - 1.0 • 23.5347) • (1 - 1.0)
= 459.831 mm²

Area Available in Nozzle Projecting Outward [A2]:
= (2 • tlnp)(tn - trn)fr2
= (2 • 66.25)(6.6 - 5.58)1.0
= 135.207 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:
= Wo² • fr2 + (Wi-can/0.707)² • fr2
= 10.0² • 1.0 + (0.0)² • 1.0
= 100.000 mm²

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(66.2,200.0,100.0)) • (50.0 - 9.6) • 1.0)
= 5352.841 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

```
Wall Thickness for Internal/External pressures       ta = 8.5808 mm
Wall Thickness per UG16(b),                        tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure      trb1 = 29.0734 mm
Wall Thickness                  tb1 = max(trb1, tr16b) = 29.0734 mm
Wall Thickness, shell/head, external pressure      trb2 = 3.2849 mm
Wall Thickness                  tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45                       tb3 = 9.2200 mm
```

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[9.22, max(29.0734, 4.5)]
= 9.2200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(8.5808, 9.22)
= 9.2200 mm

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

```
Sustained  :     83.8,  Allowable :    138.0 MPa   Passed
Expansion  :      0.0,  Allowable :    261.2 MPa   Passed
Occasional :     53.0,  Allowable :    183.5 MPa   Passed
Shear      :     24.6,  Allowable :     96.6 MPa   Passed
```

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.601, tr = 5.581, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.845, Temp. Reduction = 9 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve A            -8 °C
Min Metal Temp. at Required thickness (UCS 66.1)         -16 °C
Min Metal Temp. w/o impact per UG-20(f)                  -29 °C
```

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 29.5, tr = 23.535, c = 3.0 mm, E∗ = 1.0
Thickness Ratio = tr * E∗/(tg - c) = 0.888, Temp. Reduction = 6 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B             4 °C
Min Metal Temp. at Required thickness (UCS 66.1)          -3 °C
```

Gov. MDMT of the nozzle to shell joint welded assembly : -3 °C

Flange MDMT including Temperature reduction per UCS-66.1:

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)      -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)        -39 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 9.40/15.31 = 0.614

Weld Size Calculations, Description: N3150

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

```
                     Required Thickness     Actual Thickness
Nozzle Weld         6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
```

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (3649.5925 - 459.8309 + 2 • 47.0 • 1.0 •
(1.0 • 26.5 - 23.5347))138)
= 478603.12 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (5488.0479 + 100.0 - 0.0 • 1.0) • 138
= 771071.12 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (5488.0479 + 0.0 + 100.0 + (2491.0)) • 138
= 1114793.75 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (5488.0479 + 0.0 + 100.0 + 0.0 + (2491.0)) • 138
= 1114793.75 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 168.275 • 10.0 • 0.49 • 138
= 178719. N

Shear, Nozzle Wall [Snw]:
= (π •(Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 80.8369) • (50.0 - 3.0) • 0.7 • 138
= 1152916. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 168.275 • (29.5 - 3.0) • 0.74 • 138
= 715252. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (178719 + 1152916) = 1331634 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (178719 + 0 + 715252 + 0) = 893971 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (178719 + 715252 + 0) = 893971 N

Summary of Failure Path Calculations:

```
Path 1-1 = 1331634 N  , must exceed W = 478603 N    or W1 = 771071 N
Path 2-2 = 893970  N  , must exceed W = 478603 N    or W2 = 1114793 N
Path 3-3 = 893970  N  , must exceed W = 478603 N    or W3 = 1114793 N
```

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 5.0072 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 234.5072 mm

Input Echo, WRC107/537 Item 1, Description: N3150 :

```
Diameter Basis for Vessel                   Vbasis      ID
Cylindrical or Spherical Vessel             Cylsph      Spherical
Internal Corrosion Allowance                   Cas      3.0000  mm
Vessel Diameter                                 Dv    1371.604  mm
Vessel Thickness                                Tv      29.500  mm

Design Temperature                              T1        82.0  C
Vessel Material                                      SA-516 70
Vessel UNS Number                                       K02700
Vessel Cold S.I. Allowable                     Smc      138.00  MPa
Vessel Hot  S.I. Allowable                     Smh      138.00  MPa
```

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

```
Attachment Type                               Type      Round
WRC107 Attachment Classification            Holsol      Hollow

Diameter Basis for Nozzle                   Nbasis      OD
Corrosion Allowance for Nozzle                 Can      3.0000  mm
Nozzle Diameter                                 Dn     249.073  mm
Nozzle Thickness                                Tn      50.000  mm
Nozzle Material                                         SA-105
Nozzle UNS Number                                       K03504
Nozzle Cold S.I. Allowable                    SNmc      138.00  MPa
Nozzle Hot  S.I. Allowable                    SNmh      138.00  MPa

Design Internal Pressure                        Dp       9.403  MPa
Include Pressure Thrust                                     No
```

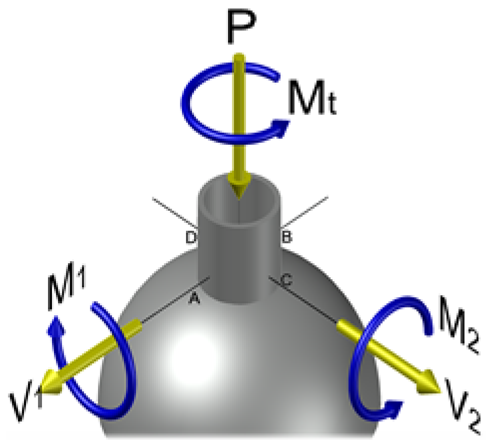
External Forces and Moments in WRC 107/537 Convention:

```
Radial Load             (SUS)               P    -11245.7   N
Longitudinal Shear      (SUS)          (Vl) V1      7497.1   N
Circumferential Shear   (SUS)          (Vc) V2      7497.1   N
Circumferential Moment  (SUS)          (Mc) M1   3150156.0   N-mm
Longitudinal Moment     (SUS)          (Ml) M2   3150156.0   N-mm
Torsional Moment        (SUS)               Mt   4500244.5   N-mm


Use Interactive Control                           No
WRC107 Version                          Version   March  1979

Include Pressure Stress Indices per Div. 2                  No
Compute Pressure Stress per WRC-368                         No
Local Loads applied at end of Nozzle/Attachment             No
```

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
$= NozzleOD + 2 \cdot 1.65 \cdot \sqrt{Rmean(t - ca)}$
$= 249.073 + 2 \cdot 1.65 \cdot \sqrt{702.052 (29.5 - 3.0)}$
= 699.185 mm

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	-11245.7	N
Circumferential Shear	(VC)	V2	7497.1	N
Longitudinal Shear	(VL)	V1	7497.1	N
Circumferential Moment	(MC)	M1	3150156.0	N-mm
Longitudinal Moment	(ML)	M2	3150156.0	N-mm
Torsional Moment		MT	4500244.5	N-mm

Dimensionless Param: U =   0.91 TAU =   5.00 (2.15) RHO =   0.56

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.04456	(A,B,C,D)
M(x) / P	SP 1	0.04720	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.05655	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.09040	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.05655	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.09040	(A,B,C,D)
N(y) * T / P	SP 1	0.07488	(A,B,C,D)
M(y) / P	SP 1	0.01436	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.07449	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.03124	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.07449	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.03124	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Rad. Bend.	P	4.5	-4.5	4.5	-4.5	4.5	-4.5	4.5	-4.5
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-1.9	-1.9	1.9	1.9

Rad. Bend. MC	0.0	0.0	0.0	0.0	-17.8	17.8	17.8	-17.8
Rad. Memb. ML	-1.9	-1.9	1.9	1.9	0.0	0.0	0.0	0.0
Rad. Bend. ML	-17.8	17.8	17.8	-17.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-14.4	12.2	24.9	-19.8	-14.4	12.2	24.9	-19.8
Tang. Memb. P	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Tang. Bend. P	1.4	-1.4	1.4	-1.4	1.4	-1.4	1.4	-1.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-2.4	-2.4	2.4	2.4
Tang. Bend. MC	0.0	0.0	0.0	0.0	-6.2	6.2	6.2	-6.2
Tang. Memb. ML	-2.4	-2.4	2.4	2.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.2	6.2	6.2	-6.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-6.0	3.5	11.2	-3.9	-6.0	3.5	11.2	-3.9
Shear VC	0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Tot. Shear	2.5	2.5	1.0	1.0	1.0	1.0	2.5	2.5
Str. Int.	15.1	12.8	25.0	19.9	14.6	12.3	25.4	20.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress Load	Stress Intensity Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)	122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
Rad. Pl (SUS)	-1.1	-1.1	2.6	2.6	-1.1	-1.1	2.6	2.6
Rad. Q (SUS)	-13.3	13.3	22.4	-22.4	-13.3	13.3	22.4	-22.4
Long. Pm (SUS)	122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
Long. Pl (SUS)	-1.2	-1.2	3.6	3.6	-1.2	-1.2	3.6	3.6
Long. Q (SUS)	-4.8	4.8	7.5	-7.5	-4.8	4.8	7.5	-7.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear Q (SUS)	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Pm (SUS)	122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
Pm+Pl (SUS)	121.8	121.8	126.3	126.3	121.8	121.8	126.3	126.3
Pm+Pl+Q (Total)	117.0	135.1	147.3	118.5	116.4	134.6	147.7	118.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	122.32	138.00	Passed
Pm+Pl (SUS)	126.33	207.00	Passed
Pm+Pl+Q (TOTAL)	147.69	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*